OnBuch+

SIGNED COURSE • PREMIUM

PURE MATHEMATICS WITH MECHANICS • UPPER SIXTH

UPPER SIXTH — PURE MATHEMATICS WITH MECHANICS

CHAPTER 3

VECTORS

SUGGESTED TIME: 11 PERIODS

IN THIS CHAPTER

This chapter extends vector methods to the geometry of planes in three-dimensional space. Learners will derive vector, parametric and Cartesian equations of planes, compute distances and angles, and study the intersection of lines and planes — essential tools for the GCE Advanced Level examination.

- By the end of this chapter I can write the vector equation of a plane given a point and a normal vector, or a point and two direction vectors.
- By the end of this chapter I can convert between the vector, parametric and Cartesian forms of the equation of a plane.
- By the end of this chapter I can find the Cartesian equation of a plane through three non-collinear points.
- By the end of this chapter I can calculate the perpendicular distance from a point to a plane.
- By the end of this chapter I can find the angle between two planes and the angle between a line and a plane.
- By the end of this chapter I can determine the point of intersection of a line and a plane and discuss existence and uniqueness.

DOWNLOAD ONBUCH

Free • Google Play
AI tutor, past papers, results

GO ONBUCH+

All signed courses, unlimited AI tutor, PDF sheets — OnBuch+ tab in the app

THIS SIGNED COURSE CONTAINS

COURSE — complete, illustrated, step by step

METHODS — and worked examples

EXERCISES — exam style + solutions

SUMMARY SHEET — the essentials to remember

CONTENTS

1. Equations of a Plane: Vector, Parametric and Cartesian Forms
2. Plane Through Three Points and Special Planes
3. Perpendicular Distance from a Point to a Plane
4. Angles: Between Two Planes and Between a Line and a Plane
5. Intersection of a Line with a Plane and of Two Planes
6. Pencils of Planes and Synthesis Problems
7. Integration activity
8. Methods and worked examples
9. Exercises
10. Solutions to the exercises
11. Summary sheet

CHAPTER OBJECTIVES

- By the end of this chapter I can write the vector equation of a plane given a point and a normal vector, or a point and two direction vectors.
- By the end of this chapter I can convert between the vector, parametric and Cartesian forms of the equation of a plane.
- By the end of this chapter I can find the Cartesian equation of a plane through three non-collinear points.
- By the end of this chapter I can calculate the perpendicular distance from a point to a plane.
- By the end of this chapter I can find the angle between two planes and the angle between a line and a plane.
- By the end of this chapter I can determine the point of intersection of a line and a plane and discuss existence and uniqueness.
- By the end of this chapter I can find the line of intersection of two planes and recognise parallel or coincident planes.
- By the end of this chapter I can write the equation of a plane through the line of intersection of two given planes (pencil of planes).
- By the end of this chapter I can solve GCE-style problems combining several of these skills.

PREREQUISITES

- Scalar (dot) product of two vectors and its link with the angle between them (Lower Sixth).
- Vector and parametric equations of a straight line in 3D (Lower Sixth).
- Vector (cross) product and its geometric meaning as area and perpendicular direction.
- Position vectors, magnitude of a vector, and basic vector algebra.
- Solving simultaneous linear equations in two and three unknowns.

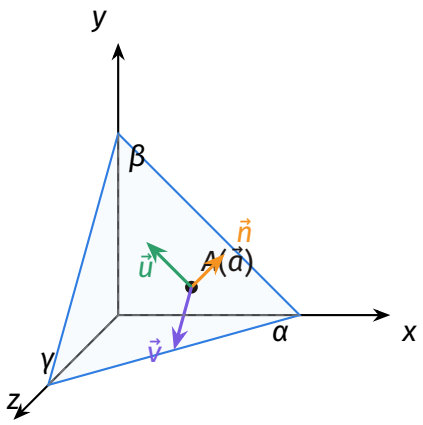
1 EQUATIONS OF A PLANE: VECTOR, PARAMETRIC AND CARTESIAN FORMS

When engineers designed the roof of the Ahmadou Ahidjo Stadium in Yaoundé, every steel panel had to be positioned in space with millimetre precision: each panel is a piece of a plane, and the angle between two panels determines how rainwater flows off. Similarly, when an aircraft lands at Douala International Airport, the pilot must control the angle between the aircraft's path (a line) and the runway (a plane). How can we describe a flat surface in space with an equation, measure distances from it, and find where lines and planes meet? Vectors give us a powerful and elegant language to answer all these questions.

1.1 The Concept of a Plane in Space

A plane in three-dimensional space is a flat, two-dimensional surface that extends infinitely in all directions. Unlike a line, which is one-dimensional and requires only one direction vector, a plane is **two-dimensional**. To fix a plane in space we need either:

- A **point** on the plane and a **normal vector** (a vector perpendicular to the plane), or
- A **point** on the plane and **two non-parallel direction vectors** lying in the plane.



A plane defined by point A, normal vector $\vec{n}$, and two direction vectors $\vec{u}, \vec{v}$ lying in the plane. The intercepts on the axes are α, β, γ.

DEFINITION **Normal Vector to a Plane**

A vector $\vec{n}$ is a normal vector to a plane Π if $\vec{n}$ is perpendicular to every vector lying in Π. If $\vec{n}$ is a normal vector, then any non-zero scalar multiple $k\vec{n}$ ($k \neq 0$) is also a normal vector.

DEFINITION **Vector Equation of a Plane (Normal Form)**

Let Π be a plane with normal vector $\vec{n}$. Let A be a fixed point on Π with position vector $\vec{a}$, and let P be any point on Π with position vector $\vec{r}$. Then the vector $\vec{AP} = \vec{r} - \vec{a}$ lies in the plane, so it is perpendicular to $\vec{n}$.

$$(\vec{r} - \vec{a}) \cdot \vec{n} = 0 \quad \Longrightarrow \quad \vec{r} \cdot \vec{n} = \vec{a} \cdot \vec{n}.$$

The quantity $\vec{a} \cdot \vec{n}$ is a constant, denoted by p (or d). The equation is

$$\boxed{\vec{r} \cdot \vec{n} = p}$$

where $\vec{n}$ is a normal vector and p is the perpendicular distance from the origin to the plane multiplied by $|\vec{n}|$.

KEY POINTS **Key Points: Vector Equation $\vec{r} \cdot \vec{n} = p$**

- $\vec{n} = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$ is a normal vector to the plane.
- $p = \vec{a} \cdot \vec{n}$ is a scalar constant.
- If $\vec{r} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$, the equation expands to the Cartesian form: $ax + by + cz = p$.
- The coefficients of x, y, z in the Cartesian equation give the components of a normal vector directly.

WORKED EXAMPLE 1 **Finding the Vector Equation Given a Point and a Normal**

Find the vector equation of the plane Π that passes through the point $A(2, -1, 3)$ and has normal vector $\vec{n} = 2\vec{i} - 3\vec{j} + 4\vec{k}$.

SOLUTION

Solution: The position vector of A is $\vec{a} = \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}$. The normal vector is $\vec{n} = \begin{pmatrix} 2 \\ -3 \\ 4 \end{pmatrix}$. The constant

$p = \vec{a} \cdot \vec{n} = (2)(2) + (-1)(-3) + (3)(4) = 4 + 3 + 12 = 19$. The vector equation is:

$$\vec{r} \cdot \begin{pmatrix} 2 \\ -3 \\ 4 \end{pmatrix} = 19$$

In Cartesian form: $2x - 3y + 4z = 19$.

1.2 Parametric Vector Form of a Plane

If we know a point $A(\vec{a})$ on the plane and two non-parallel vectors $\vec{u}$ and $\vec{v}$ parallel to the plane (direction vectors), then any point $P(\vec{r})$ on the plane can be reached by starting at A and moving some distance along $\vec{u}$ and some distance along $\vec{v}$.

DEFINITION **Parametric (Two-Parameter) Vector Equation**

Let $\vec{a}$ be the position vector of a point A on the plane Π, and let $\vec{u}, \vec{v}$ be two non-parallel vectors parallel to Π. Then the position vector $\vec{r}$ of any point P on Π is given by:

$$\boxed{\vec{r} = \vec{a} + \lambda\vec{u} + \mu\vec{v} \qquad (\lambda, \mu \in \mathbb{R})}$$

The parameters λ and μ are real numbers. Varying λ and μ independently generates the whole plane.

WARNING **Vectors Must Not Be Parallel**

In the equation $\vec{r} = \vec{a} + \lambda\vec{u} + \mu\vec{v}$, the vectors $\vec{u}$ and $\vec{v}$ must not be parallel (i.e., $\vec{u} \neq k\vec{v}$ for any scalar k). If they are parallel, the equation describes only a line, not a plane. Always check that $\vec{u} \times \vec{v} \neq \vec{0}$.

1.3 Converting Parametric Form to Cartesian Form

Given $\vec{r} = \vec{a} + \lambda\vec{u} + \mu\vec{v}$, we write the components:

$$\begin{cases} x = a_1 + \lambda u_1 + \mu v_1 \\ y = a_2 + \lambda u_2 + \mu v_2 \\ z = a_3 + \lambda u_3 + \mu v_3 \end{cases}$$

We eliminate the parameters λ and μ to obtain a single equation in x, y, z of the form $ax + by + cz = d$.

METHOD **Converting Parametric → Cartesian**

1. Write the parametric equations for x, y, z in terms of λ, μ.
2. Treat two of the equations as a linear system in λ, μ (e.g., solve for λ, μ in terms of x, y).
3. Substitute the expressions for λ, μ into the third equation.
4. Simplify to get the Cartesian form $ax + by + cz = d$.
5. Alternative (faster for exams): Compute the normal vector $\vec{n} = \vec{u} \times \vec{v}$. Then use $\vec{r} \cdot \vec{n} = \vec{a} \cdot \vec{n}$.

WORKED EXAMPLE 2 **Parametric to Cartesian via Cross Product**

A plane Π has parametric equation:

$$\vec{r} = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} -1 \\ 0 \\ 3 \end{pmatrix}.$$

Find its Cartesian equation.

SOLUTION

Method 1: Cross Product (Recommended) Direction vectors: $\vec{u} = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}, \vec{v} = \begin{pmatrix} -1 \\ 0 \\ 3 \end{pmatrix}$. Normal vector $\vec{n} = \vec{u} \times \vec{v} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & 1 & 0 \\ -1 & 0 & 3 \end{vmatrix} = \vec{i}(3-0) - \vec{j}(6-0) + \vec{k}(0-(-1)) = \begin{pmatrix} 3 \\ -6 \\ 1 \end{pmatrix}$. Point $A(1, 2, -1)$, so $\vec{a} = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$.

Constant $p = \vec{a} \cdot \vec{n} = 3(1) - 6(2) + 1(-1) = 3 - 12 - 1 = -10$. Cartesian equation: $3x - 6y + z = -10$ or $\boxed{3x - 6y + z + 10 = 0}$.

Method 2: Elimination of Parameters $x = 1 + 2\lambda - \mu$ (1)

$y = 2 + \lambda \Rightarrow \lambda = y - 2$ (2)

$z = -1 + 3\mu \Rightarrow \mu = \frac{z+1}{3}$ (3) Substitute (2) and (3) into (1): $x = 1 + 2(y-2) - \frac{z+1}{3}$ $3x = 3 + 6y - 12 - z - 1$

$3x - 6y + z = -10$. ✓

1.4 Cartesian Equation and the Normal Vector

The general Cartesian equation of a plane is

$$ax + by + cz = d \quad \text{or} \quad ax + by + cz + d = 0 \quad (d \text{ sign convention varies}).$$

PROPERTY **Normal Vector from Cartesian Equation**

For the plane $ax + by + cz = d$, a normal vector is

$$\vec{n} = \begin{pmatrix} a \\ b \\ c \end{pmatrix} = a\vec{i} + b\vec{j} + c\vec{k}.$$

WARNING **Coefficients Give the Normal, Not a Direction**

A very common mistake is to think that $\begin{pmatrix} a \\ b \\ c \end{pmatrix}$ is a vector in the plane. It is perpendicular to the plane. Direction vectors in the plane satisfy $au_1 + bu_2 + cu_3 = 0$ (they are orthogonal to $\vec{n}$).

WORKED EXAMPLE 3 **Reading the Normal and Finding a Point**

For the plane $\Pi : 2x - y + 3z = 6$:

1. State a normal vector.
2. Find the coordinates of three points on Π.

SOLUTION

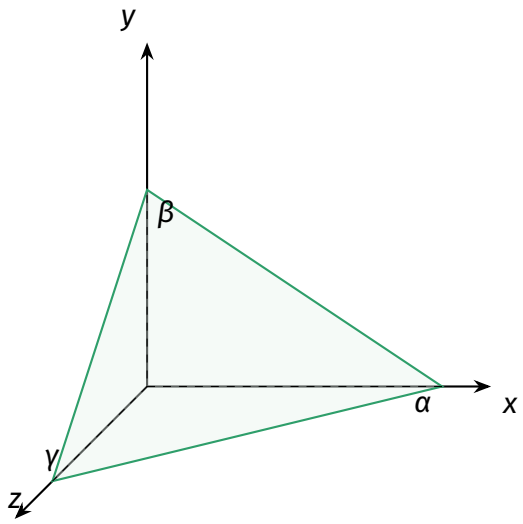
Solution:

1. Normal vector $\vec{n} = \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}$.
2. Choose arbitrary values for two variables and solve for the third.
   - Let $x = 0, y = 0 \Rightarrow 3z = 6 \Rightarrow z = 2$. Point $A(0, 0, 2)$.
   - Let $x = 0, z = 0 \Rightarrow -y = 6 \Rightarrow y = -6$. Point $B(0, -6, 0)$.
   - Let $y = 0, z = 0 \Rightarrow 2x = 6 \Rightarrow x = 3$. Point $C(3, 0, 0)$.

   These are exactly the intercepts on the axes.

1.5 Intercepts and the Intercept Form

If a plane cuts the x-, y-, and z-axes at $(\alpha, 0, 0)$, $(0, \beta, 0)$, and $(0, 0, \gamma)$ respectively, then α, β, γ are the **intercepts**. The plane does not pass through the origin, so $\alpha, \beta, \gamma \neq 0$.



A plane cutting the coordinate axes at intercepts α, β, γ.

Substituting these points into $ax + by + cz = d$ gives $a\alpha = d$, $b\beta = d$, $c\gamma = d$. Thus $a = d/\alpha$, $b = d/\beta$, $c = d/\gamma$. The equation becomes:

$$\frac{d}{\alpha}x + \frac{d}{\beta}y + \frac{d}{\gamma}z = d \quad \Longrightarrow \quad \boxed{\frac{x}{\alpha} + \frac{y}{\beta} + \frac{z}{\gamma} = 1}$$

This is the **intercept form**.

KEY POINTS **GCE Exam Tip: Intercept Form**

The intercept form $\frac{x}{\alpha} + \frac{y}{\beta} + \frac{z}{\gamma} = 1$ is extremely useful in multiple-choice questions for quickly identifying the intercepts or checking if a plane passes through a given point on an axis. If a plane is parallel to an axis, the corresponding intercept is infinite (the term is absent).

DID YOU KNOW? **Syllabus Note: Three Points Define a Plane**

The official syllabus mentions "Lesson 43: Cartesian equation of a plane through three collinear points". This is a known typo: three collinear points lie on infinitely many planes (they only define a line). A unique plane is defined by three non-collinear points. In the exam, you will be given three non-collinear points (or asked to verify they are non-collinear first). The method is: find two direction vectors $\vec{AB}$ and $\vec{AC}$, then $\vec{n} = \vec{AB} \times \vec{AC}$, and use $\vec{r} \cdot \vec{n} = \vec{a} \cdot \vec{n}$.

WORKED EXAMPLE 4 **Using Intercept Form**

Find the equation of the plane that cuts the x-axis at 4, the y-axis at -2, and is parallel to the z-axis.

SOLUTION

Solution: Intercepts: $\alpha = 4, \beta = -2$. Parallel to z-axis $\Rightarrow$ no z-intercept ($\gamma \to \infty$, term $\frac{z}{\gamma} \to 0$). Equation: $\frac{x}{4} + \frac{y}{-2} = 1$. Multiply by 4: $x - 2y = 4$ or $x - 2y - 4 = 0$. (Check: Normal vector $\vec{n} = (1, -2, 0)$ has zero z-component, so plane is vertical/parallel to z-axis.)

1.6 Checking Whether a Point Lies on a Plane

Given a plane in any form (vector, parametric, or Cartesian) and a point $P(x_0, y_0, z_0)$, we substitute the coordinates into the equation.

- **Cartesian** $ax + by + cz = d$: Check if $ax_0 + by_0 + cz_0 = d$.
- **Vector** $\vec{r} \cdot \vec{n} = p$: Check if $\vec{OP} \cdot \vec{n} = p$.
- **Parametric** $\vec{r} = \vec{a} + \lambda\vec{u} + \mu\vec{v}$: Solve the system $x_0 = a_1 + \lambda u_1 + \mu v_1$, etc., for λ, μ. If a consistent solution exists, the point lies on the plane.

WORKED EXAMPLE 5 **Point on Plane?**

Determine whether the point $P(1, 4, -2)$ lies on the plane $\Pi : \vec{r} = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix} + \mu \begin{pmatrix} 0 \\ 2 \\ -1 \end{pmatrix}$.

SOLUTION

Solution: We need λ, μ such that:

$$\begin{pmatrix} 1 \\ 4 \\ -2 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix} + \mu \begin{pmatrix} 0 \\ 2 \\ -1 \end{pmatrix}$$

Component-wise:

$$1 = 2 + \lambda \quad \Rightarrow \quad \lambda = -1 \quad (1)$$
$$4 = 1 - \lambda + 2\mu \quad (2)$$
$$-2 = 2\lambda - \mu \quad (3)$$

Substitute $\lambda = -1$ into (2): $4 = 1 - (-1) + 2\mu \Rightarrow 4 = 2 + 2\mu \Rightarrow \mu = 1$. Check (3): $2(-1) - 1 = -3 \neq -2$. The system is inconsistent. Point P does NOT lie on the plane.

EXERCISE 1 **Practice: Equations of a Plane**

1. Find the vector equation $\vec{r}\cdot\vec{n} = p$ of the plane through $A(3,-2,1)$ with normal $\vec{n} = \vec{i}+2\vec{j}-2\vec{k}$. Write the Cartesian equation.

2. A plane has parametric equation $\vec{r} = \begin{pmatrix}0\\1\\2\end{pmatrix} + \lambda\begin{pmatrix}1\\0\\-1\end{pmatrix} + \mu\begin{pmatrix}2\\-1\\0\end{pmatrix}$. Find its Cartesian equation.

3. Find the intercepts of the plane $2x-3y+4z = 12$ and write its equation in intercept form.

4. Show that the point $P(2,-1,3)$ lies on the plane $\Pi : \vec{r}\cdot(2\vec{i}-\vec{j}+\vec{k}) = 8$. Find the parametric form of Π using P as the base point and two direction vectors of your choice.

5. (GCE Style) The plane Π has equation $x + 2y - 2z = 4$. Which of the following vectors is not a normal vector to Π?

   A $\begin{pmatrix}1\\2\\-2\end{pmatrix}$

   B $\begin{pmatrix}2\\-1\\1\end{pmatrix}$

   C $\begin{pmatrix}-1\\-2\\2\end{pmatrix}$

   D $\begin{pmatrix}2\\4\\-4\end{pmatrix}$

SOLUTION **Exercise 1**

1. $\vec{a} = (3,-2,1)$, $\vec{n} = (1,2,-2)$. $p = \vec{a}\cdot\vec{n} = 3(1) + (-2)(2) + 1(-2) = 3 - 4 - 2 = -3$. Equation: $\vec{r}\cdot\begin{pmatrix}1\\2\\-2\end{pmatrix} = -3$. Cartesian: $x + 2y - 2z = -3$.

2. $\vec{u} = (1, 0, -1), \vec{v} = (2, -1, 0)$. $\vec{n} = \vec{u} \times \vec{v} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 0 & -1 \\ 2 & -1 & 0 \end{vmatrix} = \vec{i}(0-1) - \vec{j}(0-(-2)) + \vec{k}(-1-0) = (-1, -2, -1)$.
Use $\vec{n} = (1, 2, 1)$ (simpler, multiply by -1). Point $\vec{a} = (0, 1, 2)$. $p = 0 + 2 + 2 = 4$. Cartesian: $x + 2y + z = 4$.

3. x-intercept: $y = z = 0 \Rightarrow 2x = 12 \Rightarrow x = 6 \Rightarrow \alpha = 6$. y-intercept: $x = z = 0 \Rightarrow -3y = 12 \Rightarrow y = -4 \Rightarrow \beta = -4$. z-intercept: $x = y = 0 \Rightarrow 4z = 12 \Rightarrow z = 3 \Rightarrow \gamma = 3$. Intercept form: $\frac{x}{6} + \frac{y}{-4} + \frac{z}{3} = 1$.

4. Check: $\vec{OP} \cdot \vec{n} = (2)(2) + (-1)(-1) + (3)(1) = 4 + 1 + 3 = 8 = p$. Yes, P lies on Π. Need two vectors $\perp \vec{n} = (2, -1, 1)$. Choose $\vec{u} = (1, 2, 0)$ (check: $2 - 2 + 0 = 0$). Choose $\vec{v} = (0, 1, 1)$ (check: $0 - 1 + 1 = 0$). Parametric: $\vec{r} = \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$.

5. A normal vector is $\vec{n} = (1, 2, -2)$. Any non-zero scalar multiple is also a normal vector. Option A: $(1, 2, -2)$ is $\vec{n}$ itself. Normal. Option B: $(2, -1, 1)$. Dot product with $\vec{n}$: $2(1) + (-1)(2) + 1(-2) = 2 - 2 - 2 = -2 \neq 0$. Not parallel to $\vec{n}$. Not a normal vector. Option C: $(-1, -2, 2) = -1 \times \vec{n}$. Normal. Option D: $(2, 4, -4) = 2 \times \vec{n}$. Normal. Answer: B.

2 PLANE THROUGH THREE POINTS AND SPECIAL PLANES

In the previous section we learned that a plane is completely determined as soon as we know **one point of the plane** and **one normal vector**: the vector equation $\vec{r} \cdot \vec{n} = \vec{a} \cdot \vec{n}$ and the Cartesian equation $ax + by + cz = d$ both follow immediately. We also saw the parametric form $\vec{r} = \vec{a} + \lambda\vec{u} + \mu\vec{v}$, which needs one point and **two non-parallel direction vectors** lying in the plane. In this section we put these tools to work on the most frequent GCE situation: a plane specified by three points. We then treat a family of quick but very examinable cases: planes parallel to the coordinate planes, planes containing an axis, planes parallel to a given plane, and planes containing a given line and a given point.

2.1 Plane Through Three Non-Collinear Points

Intuition. Place three fingertips of one hand on your desk so that they are not in a straight line, and rest a sheet of cardboard on them: the cardboard settles in exactly one position. Three non-collinear points behave like a tripod — they fix a **unique plane**. But if the three fingertips are in a straight line, the cardboard can still rotate around that line: uniqueness is lost. This simple observation is the whole content of the section.

PROPERTY **Uniqueness of the plane through three points**

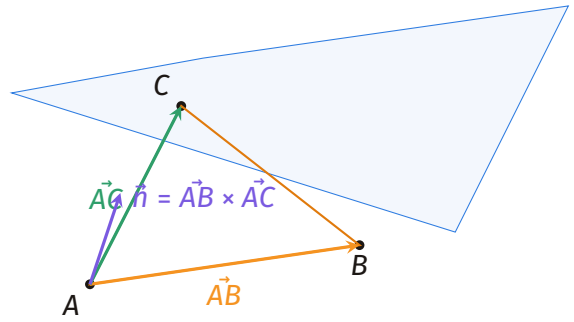
Three non-collinear points A, B, C of space determine one and only one plane. (The statement is assumed; its use is examinable, not its proof.)

How do we build the equation? If A, B, C lie in the plane, then the vectors $\vec{AB}$ and $\vec{AC}$ lie in the plane. Since A, B, C are not collinear, $\vec{AB}$ and $\vec{AC}$ are not parallel, so their cross product

$$\vec{n} = \vec{AB} \times \vec{AC}$$

is a non-zero vector perpendicular to both of them, hence perpendicular to the whole plane: it is a **normal vector**. We then apply the point-normal form of Section 1 with the point A:

$$\vec{r} \cdot \vec{n} = \vec{OA} \cdot \vec{n}.$$



The plane through A, B, C: $\vec{AB}$ and $\vec{AC}$ lie in the plane, and $\vec{n} = \vec{AB} \times \vec{AC}$ is normal to it.

METHOD **Plane through three points A, B, C — cross product method**

1. Compute $\vec{AB} = \vec{OB} - \vec{OA}$ and $\vec{AC} = \vec{OC} - \vec{OA}$.
2. Compute the normal $\vec{n} = \vec{AB} \times \vec{AC}$. Check that $\vec{n} \neq \vec{0}$ (otherwise the points are collinear — see the warning below).
3. Write $\vec{r} \cdot \vec{n} = \vec{OA} \cdot \vec{n}$, or directly the Cartesian form $n_1x + n_2y + n_3z = d$ with $d = \vec{OA} \cdot \vec{n}$.
4. Verify: the coordinates of A, B and C must all satisfy the final equation.

WORKED EXAMPLE 6 **Plane through three points**

Find the Cartesian equation of the plane through $A(1, 0, 2)$, $B(2, 1, 1)$ and $C(-1, 2, 1)$.

SOLUTION

Solution. Direction vectors in the plane:

$$\vec{AB} = \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}, \qquad \vec{AC} = \begin{pmatrix} -2 \\ 2 \\ -1 \end{pmatrix}.$$

Normal vector:

$$\vec{n} = \vec{AB} \times \vec{AC} = \begin{pmatrix} (1)(-1) - (-1)(2) \\ (-1)(-2) - (1)(-1) \\ (1)(2) - (1)(-2) \end{pmatrix} = \begin{pmatrix} 1 \\ 3 \\ 4 \end{pmatrix}.$$

Since $\vec{n} \neq \vec{0}$, the points are non-collinear and determine a unique plane. Then

$$d = \vec{OA} \cdot \vec{n} = (1)(1) + (0)(3) + (2)(4) = 9.$$

Hence the plane has equation $\boxed{x + 3y + 4z = 9}$. Check: B: $2 + 3 + 4 = 9$ ✓; C: $-1 + 6 + 4 = 9$ ✓.

2.2 The Direct Cartesian Method (Simultaneous Equations)

There is an alternative that some candidates prefer because it needs no cross product. Every plane has an equation $ax + by + cz = d$; if the three points lie on it, their coordinates satisfy it, giving three equations in the four unknowns a, b, c, d. Since the equation is only defined up to a non-zero multiple, we may fix one coefficient (for instance set $d = 1$, or keep d as a parameter) and solve for the ratios $a : b : c : d$.

METHOD **Plane through three points — substitution method**

1. Substitute the coordinates of A, B, C into $ax + by + cz = d$ to obtain three simultaneous equations.
2. Solve for a, b, c in terms of d (or set a convenient value of one coefficient).
3. Substitute back and simplify; clear fractions by multiplying through.
4. Verify that all three points satisfy the equation.

EXEMPLE **Same plane, second method**

For $A(1, 0, 2)$, $B(2, 1, 1)$, $C(-1, 2, 1)$, substitution into $ax + by + cz = d$ gives

$$a + 2c = d, \qquad 2a + b + c = d, \qquad -a + 2b + c = d.$$

From the first equation $a = d-2c$. Substituting into the second: $b = d-2a-c = d-2(d-2c)-c = -d+3c$. The third equation then gives $-(d-2c)+2(-d+3c)+c = d$, i.e. $-3d+9c = d$, so $c = \frac{d}{2.5}$·… — more cleanly, $9c = 4d$, so take $d = 9$, $c = 4$, whence $a = 9-8 = 1$ and $b = -9+12 = 3$. Equation: $x + 3y + 4z = 9$, exactly as before.

WARNING **Choose your method wisely**

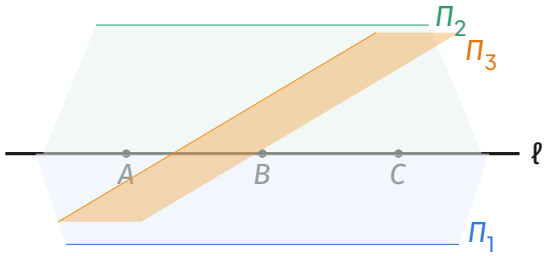
The cross product method is usually faster and less error-prone. The substitution method is useful when the cross product is awkward or when the question explicitly asks you to "show" the equation by substitution. Whichever you use, always verify the three points in your final answer — this catches arithmetic slips and earns method marks even after an error.

2.3 The GCE Trap: Three Collinear Points

The syllabus explicitly mentions "the equation of a plane through three collinear points" — this is a deliberate trap. If A, B, C are collinear, then $\vec{AC} = k\,\vec{AB}$ for some scalar k, so

$$\vec{AB} \times \vec{AC} = k\,(\vec{AB} \times \vec{AB}) = \vec{0},$$

and the method collapses: there is **no normal vector**. Geometrically, the three points all lie on a single line ℓ, and infinitely many planes contain a given line — imagine all the pages of an open book hinged along its spine.



Three collinear points lie on a line ℓ; infinitely many planes $\Pi_1, \Pi_2, \Pi_3, \ldots$ are "hinged" about ℓ. No unique plane is determined.

WARNING **Three collinear points do not fix a plane**

If A, B, C are collinear, then $\vec{AB} \times \vec{AC} = \vec{0}$ and there are infinitely many planes through them. Before computing anything, check that $\vec{AB} \times \vec{AC} \neq \vec{0}$. If an exam question gives you three collinear points, the correct response is to state that the plane is not unique and that an

extra condition (a fourth point off the line, or a prescribed normal) is required.

2.4 Special Planes: Coordinate Planes, Axes, Parallel Planes

Planes parallel to the coordinate planes. The coordinate plane Oxy has equation $z = 0$ and normal $\vec{k} = (0, 0, 1)$. Consequently any plane **parallel to** Oxy has equation $z = k$; similarly planes parallel to Oxz have equation $y = k$, and planes parallel to Oyz have equation $x = k$.

Planes containing a coordinate axis. A plane containing the z-axis passes through the origin and has normal perpendicular to $\vec{k}$, i.e. $c = 0$ and $d = 0$: its equation is $ax + by = 0$. Similarly, planes containing the x-axis: $by + cz = 0$; the y-axis: $ax + cz = 0$.

PROPERTY **Parallel planes**

Two planes are parallel if and only if their normal vectors are proportional: $\vec{n}_1 = \lambda \vec{n}_2$. Hence the plane through a point P and parallel to the plane $ax + by + cz = d$ keeps the same left-hand side and only the constant changes:

$$ax + by + cz = ax_P + by_P + cz_P.$$

EXEMPLE **Plane through a point, parallel to a given plane**

Find the plane through $P(2, -1, 3)$ parallel to $\Pi : 2x - y + 4z = 7$. Same normal $(2, -1, 4)$, so the equation is $2x - y + 4z = d'$ with $d' = 2(2) - (-1) + 4(3) = 4 + 1 + 12 = 17$. Answer: $2x - y + 4z = 17$.

2.5 Plane Containing a Given Line and a Given Point

WARNING **Do not confuse the two situations**

"Plane through three points" and "plane through a line ℓ and a point $P \notin \ell$" are different strategies. In the second case you do not have three points given directly — you must manufacture them: pick two points A, B on the line (e.g. by taking two values of the parameter), then apply the three-point method to A, B, P. Note that a direction vector $\vec{u}$ of the line and the vector $\vec{AP}$ both lie in the plane, so a shortcut is $\vec{n} = \vec{u} \times \vec{AP}$.

WORKED EXAMPLE 7 **Plane containing a line and a point**

Find the equation of the plane containing the line $\vec{r} = \begin{pmatrix}1\\0\\1\end{pmatrix} + t\begin{pmatrix}2\\-1\\1\end{pmatrix}$ and the point $P(3, 2, 0)$.

SOLUTION

Solution. The point $A(1, 0, 1)$ lies on the line, and $\vec{u} = (2, -1, 1)$ is a direction vector of the line, hence parallel to the plane. Also

$$\vec{AP} = \vec{OP} - \vec{OA} = \begin{pmatrix}2\\2\\-1\end{pmatrix}$$

lies in the plane. A normal vector is

$$\vec{n} = \vec{u} \times \vec{AP} = \begin{pmatrix}(-1)(-1) - (1)(2)\\(1)(2) - (2)(-1)\\(2)(2) - (-1)(2)\end{pmatrix} = \begin{pmatrix}-1\\4\\6\end{pmatrix}.$$

Then $d = \vec{OA} \cdot \vec{n} = -1 + 0 + 6 = 5$, so the plane is $-x + 4y + 6z = 5$, equivalently

$$\boxed{x - 4y - 6z = -5}.$$

Check: A: $1 - 0 - 6 = -5$ ✓; P: $3 - 8 - 0 = -5$ ✓; a second point of the line, $t = 1$: $(3, -1, 2)$ gives $3 + 4 - 12 = -5$ ✓.

KEY POINTS **Key points of this section**

- Three non-collinear points determine a unique plane: $\vec{n} = \vec{AB} \times \vec{AC}$, then $\vec{r} \cdot \vec{n} = \vec{OA} \cdot \vec{n}$.
- Alternative: substitute the three points into $ax + by + cz = d$ and solve the simultaneous equations.
- Three collinear points give $\vec{AB} \times \vec{AC} = \vec{0}$ and infinitely many planes — always check.
- Planes parallel to Oxy, Oxz, Oyz: $z = k$, $y = k$, $x = k$. Planes through an axis have $d = 0$ and one zero coefficient.
- Parallel planes share the same normal: only the constant d changes.
- Line + external point: use $\vec{n} = \vec{u} \times \vec{AP}$ where $\vec{u}$ directs the line and A is on the line.

EXERCISE 2

1. Find the Cartesian equation of the plane through $A(2, 1, -1)$, $B(0, 3, 1)$ and $C(1, -1, 2)$.
2. Show that the points $A(1, 2, 3)$, $B(2, 3, 4)$ and $C(3, 4, 5)$ do not determine a unique plane, and explain why geometrically.
3. Find the equation of the plane through $(1, -2, 4)$ parallel to the plane $3x + 2y - z = 6$.
4. Find the equation of the plane containing the line $\vec{r} = (1, 1, 0) + t(1, 2, -1)$ and the point $Q(0, 1, 3)$.

SOLUTION **Exercise 1**

$\vec{AB} = (-2, 2, 2)$, $\vec{AC} = (-1, -2, 3)$. $\vec{n} = \vec{AB} \times \vec{AC} = (2\cdot3 - 2\cdot(-2),\ 2\cdot(-1) - (-2)\cdot3,\ (-2)(-2) - 2(-1)) = (10, 4, 6)$, simplify to $(5, 2, 3)$. $d = 5(2) + 2(1) + 3(-1) = 9$. Plane: $5x + 2y + 3z = 9$. Check B: $0 + 6 + 3 = 9$ ✓; C: $5 - 2 + 6 = 9$ ✓.

SOLUTION **Exercise 2**

$\vec{AB} = (1, 1, 1)$, $\vec{AC} = (2, 2, 2) = 2\vec{AB}$, so $\vec{AB} \times \vec{AC} = \vec{0}$: the points are collinear (they lie on the line $\vec{r} = (1, 2, 3) + t(1, 1, 1)$). Infinitely many planes contain this line, so no unique plane is determined; an extra condition is needed.

SOLUTION **Exercise 3**

Same normal $(3, 2, -1)$: $3x + 2y - z = d'$ with $d' = 3(1) + 2(-2) - 4 = -5$. Plane: $3x + 2y - z = -5$.

SOLUTION **Exercise 4**

$A(1, 1, 0)$ on the line, $\vec{u} = (1, 2, -1)$, $\vec{AQ} = (-1, 0, 3)$. $\vec{n} = \vec{u} \times \vec{AQ} = (2\cdot3 - (-1)\cdot0,\ (-1)(-1) - 1\cdot3,\ 1\cdot0 - 2(-1)) = (6, -2, 2)$, simplify to $(3, -1, 1)$. $d = 3(1) - 1 + 0 = 2$. Plane: $3x - y + z = 2$. Check Q: $0 - 1 + 3 = 2$ ✓.

DID YOU KNOW? **Why the tripod never wobbles**

Carpenters, surveyors and photographers all exploit this section's first result: a tripod stands firm on any ground because three (non-collinear) feet determine exactly one plane,

whereas a four-legged table can rock if the four feet are not coplanar. The theodolites used by surveyors on road projects across Cameroon are mounted on tripods for precisely this reason — geometry you can literally stand on!

3 PERPENDICULAR DISTANCE FROM A POINT TO A PLANE

In the previous section we learned how to pin down a plane completely: through three non-collinear points, through a point with a given normal, or through a line with an extra condition. Once a plane is fixed in space, a natural question arises: how far is a given point from that plane? This question is not a luxury of pure geometry. When an engineer checks the clearance between a component and a wall, when a pilot must know the altitude above a tilted runway surface, or when an architect verifies that a beam stays clear of a sloping roof, the computation is exactly the **perpendicular distance from a point to a plane**. This section gives you the complete toolbox: the distance formula and its derivation, the signed distance, the distance between parallel planes, the foot of the perpendicular, the mirror image of a point, and locus problems leading to bisector planes.

3.1 Geometric motivation: what do we mean by "distance"?

Stand in a classroom and look at the floor. You are a point P above a plane Π. You could walk from your position to any point of the floor: the distances to those points are all different. But there is exactly one path that is shortest: drop straight down, along the vertical. In other words, the shortest route from P to Π is along the **normal** to the plane.

DEFINITION **Perpendicular distance from a point to a plane**

Let Π be a plane and P a point in space. The **perpendicular distance** (or shortest distance) from P to Π is the length PN, where N is the unique point of Π such that $\vec{PN}$ is perpendicular to Π. The point N is called the **foot of the perpendicular** from P to Π.

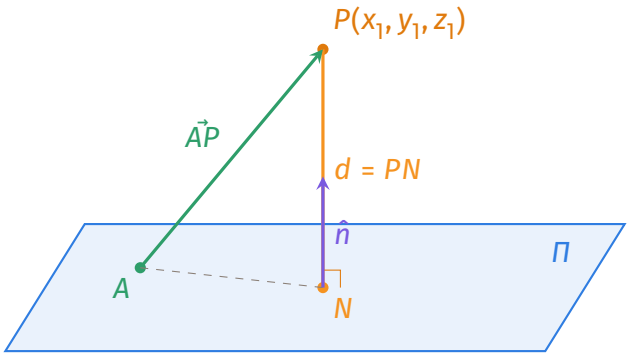
Why is PN the shortest? Take any other point M of the plane. The triangle PNM is right-angled at N (because $PN \perp \Pi$ and $NM \subset \Pi$), so by Pythagoras $PM^2 = PN^2 + NM^2 \geq PN^2$, with equality only when $M = N$. Hence $PN \leq PM$ for every $M \in \Pi$: the perpendicular truly gives the minimum distance.

3.2 Derivation of the distance formula

Let the plane Π have Cartesian equation $ax + by + cz = d$, so that $\vec{n} = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$ is a normal vector. Let $P(x_1, y_1, z_1)$ be any point in space. Choose any point $A(x_0, y_0, z_0)$ lying on the plane, so $ax_0 + by_0 +$

$cz_0 = d$.

The key observation is that the perpendicular distance PN is the absolute value of the scalar projection of $\vec{AP}$ onto the normal direction. Indeed, projecting $\vec{AP}$ onto $\vec{n}$ collapses the component of $\vec{AP}$ lying inside the plane and keeps exactly the perpendicular part, which has length PN.



The distance from P to Π is the absolute value of the scalar projection of $\vec{AP}$ onto the unit normal $\hat{n}$.

The scalar projection of $\vec{AP}$ onto $\vec{n}$ is $\dfrac{\vec{AP} \cdot \vec{n}}{|\vec{n}|}$. Now compute:

$$\vec{AP} \cdot \vec{n} = a(x_1 - x_0) + b(y_1 - y_0) + c(z_1 - z_0) = (ax_1 + by_1 + cz_1) - (ax_0 + by_0 + cz_0).$$

Since $A \in \Pi$, the second bracket equals d, so

$$\vec{AP} \cdot \vec{n} = ax_1 + by_1 + cz_1 - d, \qquad |\vec{n}| = \sqrt{a^2 + b^2 + c^2}.$$

Taking the absolute value of the projection gives the distance. We have proved the central result of this section.

PROPERTY **Perpendicular distance from a point to a plane — examinable derivation**

The perpendicular distance from the point $P(x_1, y_1, z_1)$ to the plane $ax + by + cz = d$ is

$$\delta = \frac{|ax_1 + by_1 + cz_1 - d|}{\sqrt{a^2 + b^2 + c^2}}$$

The derivation above (scalar projection of $\vec{AP}$ onto $\hat{n}$) is examinable at GCE Advanced Level: you may be asked to establish the formula, not merely to use it.

WARNING **The two most common errors**

- Forgetting the absolute value. A distance is never negative. Writing $\delta = \dfrac{ax_1 + by_1 + cz_1 - d}{\sqrt{a^2 + b^2 + c^2}}$ without the modulus can produce a negative "distance" — an instant loss of marks.
- Forgetting the square root. The denominator is $\sqrt{a^2 + b^2 + c^2}$, the length of the normal vector, not $a^2 + b^2 + c^2$. Check: the numerator is in length units times coefficient units, so the denominator must carry the same coefficient units — only the square root achieves this.

WORKED EXAMPLE 8 **Direct application of the formula**

Find the perpendicular distance from the point $P(2, -1, 3)$ to the plane $\Pi : 2x - y + 2z = 5$.

SOLUTION

Here $a = 2$, $b = -1$, $c = 2$, $d = 5$, and $(x_1, y_1, z_1) = (2, -1, 3)$.

$$ax_1 + by_1 + cz_1 - d = 2(2) - (-1) + 2(3) - 5 = 4 + 1 + 6 - 5 = 6,$$

$$\sqrt{a^2 + b^2 + c^2} = \sqrt{4 + 1 + 4} = \sqrt{9} = 3.$$

Therefore

$$\delta = \frac{|6|}{3} = 2 \text{ units}.$$

3.3 Signed distance: which side of the plane?

If we simply remove the absolute value, the quantity

$$\sigma(P) = \frac{ax_1 + by_1 + cz_1 - d}{\sqrt{a^2 + b^2 + c^2}}$$

is called the **signed distance** from P to the plane. Its sign tells us on which side of Π the point lies, relative to the direction of the chosen normal $\vec{n}$:

- $\sigma(P) > 0$: P lies on the side towards which $\vec{n}$ points;
- $\sigma(P) < 0$: P lies on the opposite side;
- $\sigma(P) = 0$: P lies on the plane itself.

EXEMPLE **Testing two points**

For the plane $2x - y + 2z = 5$, consider $P(2, -1, 3)$ and $Q(0, 0, 0)$. We found $\sigma(P) = +2$. For Q:

$\sigma(Q) = \dfrac{0 - 0 + 0 - 5}{3} = -\dfrac{5}{3}$. The signs differ, so P and Q lie on opposite sides of the plane. This trick is extremely useful for checking whether a segment crosses a plane without computing the intersection point.

3.4 Distance between two parallel planes

Two planes are parallel exactly when they share the same normal direction. Suppose

$$\Pi_1 : ax + by + cz = d_1 \qquad \text{and} \qquad \Pi_2 : ax + by + cz = d_2.$$

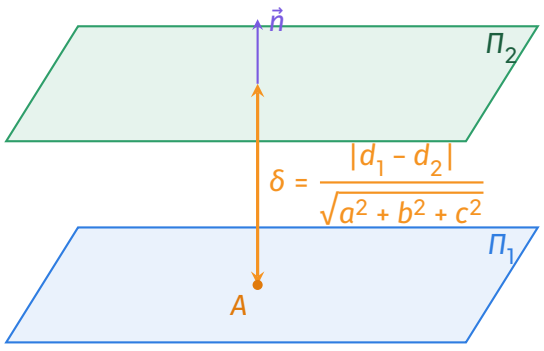
Take any point $A(x_0, y_0, z_0)$ on Π_1, so $ax_0 + by_0 + cz_0 = d_1$. The distance between the planes equals the distance from A to Π_2:

$$\delta = \frac{|ax_0 + by_0 + cz_0 - d_2|}{\sqrt{a^2 + b^2 + c^2}} = \frac{|d_1 - d_2|}{\sqrt{a^2 + b^2 + c^2}}.$$

PROPERTY **Distance between parallel planes**

The distance between the parallel planes $ax + by + cz = d_1$ and $ax + by + cz = d_2$ is

$$\delta = \frac{|d_1 - d_2|}{\sqrt{a^2 + b^2 + c^2}}.$$



Two parallel planes share the same normal; the distance between them is measured along any common perpendicular.

WARNING **Same normal coefficients before subtracting**

The formula $|d_1 - d_2| / \sqrt{a^2 + b^2 + c^2}$ is valid only if the two equations are written with the same coefficients a, b, c. For example, the planes $2x - y + 2z = 5$ and $4x - 2y + 4z = 6$ are parallel,

but you must first rewrite the second as $2x - y + 2z = 3$ (dividing by 2) before subtracting the constants. Subtracting $5 - 6 = -1$ directly would give the wrong answer.

WORKED EXAMPLE 9 **Distance between parallel planes**

Find the distance between the planes $\Pi_1 : x - 2y + 2z = 3$ and $\Pi_2 : 2x - 4y + 4z = 21$.

SOLUTION

First rewrite Π_2 with the same normal as Π_1: dividing by 2,

$$\Pi_2 : x - 2y + 2z = \tfrac{21}{2}.$$

Now apply the formula with $d_1 = 3$, $d_2 = \frac{21}{2}$ and $\sqrt{a^2 + b^2 + c^2} = \sqrt{1 + 4 + 4} = 3$:

$$\delta = \frac{\left|3 - \frac{21}{2}\right|}{3} = \frac{\frac{15}{2}}{3} = \frac{5}{2} = 2.5 \text{ units}.$$

3.5 Foot of the perpendicular and mirror image

Beyond the distance itself, examiners sometimes ask for the foot of the perpendicular N or the mirror image P' of P in the plane. Both are obtained by the same idea: parametrise the line through P in the direction of the normal.

METHOD **Finding the foot of the perpendicular**

To find the foot N of the perpendicular from $P(x_1, y_1, z_1)$ to the plane $ax + by + cz = d$:

1. Write the line through P along $\vec{n} = (a, b, c)$: $\vec{r} = \begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix} + t \begin{pmatrix} a \\ b \\ c \end{pmatrix}$.
2. Substitute $x = x_1 + at$, $y = y_1 + bt$, $z = z_1 + ct$ into the plane equation and solve for t.
3. Substitute this value of t back into the line: the point obtained is N.
4. For the mirror image P' of P, note that N is the midpoint of PP', so $P' = 2N - P$ (equivalently, use the parameter value $2t$ on the line).

WORKED EXAMPLE 10 **Foot of the perpendicular and mirror image**

Find the foot of the perpendicular from $P(1, 2, 3)$ to the plane $x + y + z = 3$, and deduce the mirror image of P in the plane.

SOLUTION

The normal is $\vec{n} = (1, 1, 1)$. The perpendicular line through P is

$$x = 1 + t, \qquad y = 2 + t, \qquad z = 3 + t.$$

Substituting into the plane equation:

$$(1 + t) + (2 + t) + (3 + t) = 3 \implies 6 + 3t = 3 \implies t = -1.$$

Hence the foot is $N = (1 - 1,\ 2 - 1,\ 3 - 1) = (0, 1, 2)$. Check: $0 + 1 + 2 = 3$ ✓.
The mirror image is $P' = 2N - P = (2(0) - 1,\ 2(1) - 2,\ 2(2) - 3) = (-1, 0, 1)$. As a check, the distances: $PN = \sqrt{1 + 1 + 1} = \sqrt{3}$ and $NP' = \sqrt{1 + 1 + 1} = \sqrt{3}$, equal as expected.

3.6 Locus problems: points equidistant from two planes

A classic GCE locus question asks for the set of points equidistant from two given planes. If $\Pi_1 : a_1x + b_1y + c_1z = d_1$ and $\Pi_2 : a_2x + b_2y + c_2z = d_2$, then $M(x, y, z)$ is equidistant from both planes when

$$\frac{|a_1x + b_1y + c_1z - d_1|}{\sqrt{a_1^2 + b_1^2 + c_1^2}} = \frac{|a_2x + b_2y + c_2z - d_2|}{\sqrt{a_2^2 + b_2^2 + c_2^2}}.$$

Removing the absolute values gives two equations (one with $+$, one with $-$), each of which is linear in x, y, z: the locus is a **pair of planes**. When Π_1 and Π_2 intersect, these are the two **bisector planes** of the dihedral angles formed by Π_1 and Π_2 — exactly the planes that bisect the angles between the two planes, a notion we shall meet again in Section 6 when studying pencils of planes.

WORKED EXAMPLE 11 **Bisector planes**

Find the locus of points equidistant from the planes $\Pi_1 : 3x - 4y = 5$ and $\Pi_2 : 4x + 3y = 10$.

SOLUTION

A point $M(x, y, z)$ is equidistant from both planes when

$$\frac{|3x - 4y - 5|}{\sqrt{9 + 16}} = \frac{|4x + 3y - 10|}{\sqrt{16 + 9}}.$$

Both denominators equal 5, so $|3x - 4y - 5| = |4x + 3y - 10|$, which splits into two cases.
Case 1: $3x - 4y - 5 = 4x + 3y - 10 \implies x + 7y = 5$.
Case 2: $3x - 4y - 5 = -(4x + 3y - 10) \implies 7x - y = 15$.

The locus is the pair of planes $x + 7y = 5$ and $7x - y = 15$. Note that their normals $(1, 7, 0)$ and $(7, -1, 0)$ are perpendicular ($7 - 7 + 0 = 0$): the two bisector planes are always perpendicular to each other.

WARNING **Reading the plane equation correctly**

Before applying the distance formula, make sure the equation is in the form $ax + by + cz = d$ with everything on one side. For the plane $3x - 4y = 5$, the missing z means $c = 0$ — do not invent a coefficient. Also, if the equation is given as $ax + by + cz + d = 0$, identify the constant carefully and move it to the right-hand side with the correct sign.

KEY POINTS **Key points of this section**

- The distance from $P(x_1, y_1, z_1)$ to $ax + by + cz = d$ is $\delta = \dfrac{|ax_1 + by_1 + cz_1 - d|}{\sqrt{a^2 + b^2 + c^2}}$, derived by projecting $\vec{AP}$ onto the unit normal.
- The signed distance (without the modulus) tells you on which side of the plane the point lies.
- Distance between parallel planes $ax + by + cz = d_1$ and $ax + by + cz = d_2$: $\delta = \dfrac{|d_1 - d_2|}{\sqrt{a^2 + b^2 + c^2}}$, after writing both equations with the same normal coefficients.
- Foot of the perpendicular: parametrise the line through P along $\vec{n}$, meet the plane. Mirror image: $P' = 2N - P$.
- Points equidistant from two planes form a pair of bisector planes, obtained by equating the two distance expressions with $\pm$.

EXERCISE 3 **Practice**

1. Find the distance from $A(3, -2, 1)$ to the plane $2x + 6y - 3z = 11$.
2. Show that the origin and the point $(4, 2, 6)$ lie on opposite sides of the plane $x + 2y + 2z = 15$.
3. Find the distance between the planes $2x - 3y + 6z = 7$ and $4x - 6y + 12z = 35$.
4. Find the foot of the perpendicular from $P(2, 0, -1)$ to the plane $x - y + z = 2$, and the image of P by reflection in this plane.

5. Find the pair of bisector planes of $x + 2y - 2z = 3$ and $2x - y + 2z = 4$.

SOLUTION **Exercise 1**

$$\delta = \frac{|2(3) + 6(-2) - 3(1) - 11|}{\sqrt{4 + 36 + 9}} = \frac{|6 - 12 - 3 - 11|}{7} = \frac{20}{7} \text{ units.}$$

SOLUTION **Exercise 2**

Signed distances: for O: $\frac{0 - 15}{3} = -5 < 0$; for $(4, 2, 6)$: $\frac{4 + 4 + 12 - 15}{3} = \frac{5}{3} > 0$. Opposite signs, so the points lie on opposite sides.

SOLUTION **Exercise 3**

Rewrite the second plane: $2x - 3y + 6z = \frac{35}{2}$. Then $\delta = \frac{\left|\frac{35}{2} - 7\right|}{\sqrt{4 + 9 + 36}} = \frac{\frac{21}{2}}{7} = \frac{3}{2}$ units.

SOLUTION **Exercise 4**

Line: $x = 2 + t, y = -t, z = -1 + t$. Substituting: $(2 + t) - (-t) + (-1 + t) = 2 \Rightarrow 1 + 3t = 2 \Rightarrow t = \frac{1}{3}$. Foot: $N = \left(\frac{7}{3}, -\frac{1}{3}, -\frac{2}{3}\right)$. Image: $P' = 2N - P = \left(\frac{8}{3}, -\frac{2}{3}, -\frac{1}{3}\right)$.

SOLUTION **Exercise 5**

Both normals have length 3, so $|x + 2y - 2z - 3| = |2x - y + 2z - 4|$. Case 1: $x + 2y - 2z - 3 = 2x - y + 2z - 4 \Rightarrow x - 3y + 4z = 1$. Case 2: $x + 2y - 2z - 3 = -(2x - y + 2z - 4) \Rightarrow 3x + y = 7$. The bisector planes are $x - 3y + 4z = 1$ and $3x + y = 7$.

DID YOU KNOW? **Why surveyors love this formula**

In road construction across the hills of the North West Region, engineers must constantly check clearances between design points and inclined surfaces such as roadbeds and em-

bankments. Modern surveying software computes exactly the quantity $\frac{|ax_1 + by_1 + cz_1 - d|}{\sqrt{a^2 + b^2 + c^2}}$ thousands of times per second. The mathematics you are learning here is the engine inside those instruments.

4 ANGLES: BETWEEN TWO PLANES AND BETWEEN A LINE AND A PLANE

In the previous section we measured distances involving planes: the perpendicular distance from a point to a plane. We now turn to angles. Two natural questions arise. When two planes meet along a line (think of the two wings of an open book), how "open" are they? And when a line pierces a plane (think of a javelin stuck in the football pitch at the Amadou Ahidjo stadium), at what angle does it strike the ground? Both questions are answered beautifully using nothing more than the **scalar product** and the **normal vector** of a plane.

4.1 The Angle Between Two Planes

Intuition. Stand two exercise books against each other so that they meet along a common edge. The "angle" between them is measured in a plane perpendicular to that edge — this is the **dihedral angle**. Now, the normal vector of a plane is perpendicular to every line in the plane, so the angle between the two normals captures exactly the same opening. This observation is the key to everything that follows.

DEFINITION **Angle between two planes**

Let P_1 and P_2 be two planes with normal vectors $\vec{n}_1$ and $\vec{n}_2$. The **angle between the two planes** is defined as the angle between their normal vectors, taken as the acute angle (or right angle), i.e. the angle θ with $0 \le \theta \le \frac{\pi}{2}$.

WARNING **Why the absolute value?**

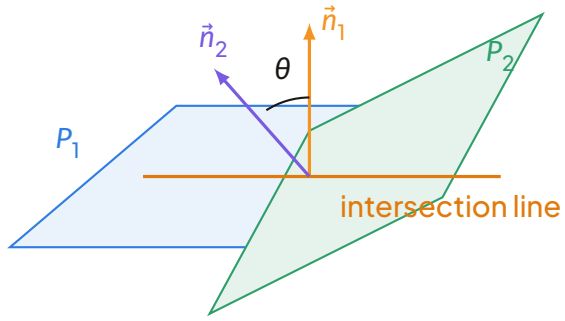
A plane has two opposite normal directions ($\vec{n}$ and $-\vec{n}$ both work). Depending on the choice, the angle between the normals could be θ or $\pi - \theta$. Since the angle between two planes must be unambiguous, we always take the acute value. In practice this means taking the absolute value of the scalar product in the formula below.

PROPERTY **Formula for the angle between two planes — proof examinable**

Let $P_1 : a_1x + b_1y + c_1z + d_1 = 0$ and $P_2 : a_2x + b_2y + c_2z + d_2 = 0$ have normals $\vec{n}_1 = \begin{pmatrix} a_1 \\ b_1 \\ c_1 \end{pmatrix}$ and $\vec{n}_2 = \begin{pmatrix} a_2 \\ b_2 \\ c_2 \end{pmatrix}$. Then the angle θ between the planes satisfies

$$\cos\theta = \frac{|\vec{n}_1 \cdot \vec{n}_2|}{|\vec{n}_1|\,|\vec{n}_2|}.$$

Proof. By definition, θ is the acute angle between $\vec{n}_1$ and $\vec{n}_2$. The scalar product gives $\vec{n}_1 \cdot \vec{n}_2 = |\vec{n}_1||\vec{n}_2|\cos\alpha$, where α is the angle between the chosen directions of the normals. If α is obtuse, $\cos\alpha < 0$, but the acute angle between the planes is $\pi - \alpha$ with $\cos(\pi - \alpha) = -\cos\alpha = |\cos\alpha|$. Hence in all cases $\cos\theta = \dfrac{|\vec{n}_1 \cdot \vec{n}_2|}{|\vec{n}_1||\vec{n}_2|}$. ■



Two intersecting planes with their normal vectors; the dihedral angle θ equals the (acute) angle between $\vec{n}_1$ and $\vec{n}_2$.

PROPERTY **Perpendicular and parallel planes**

Let P_1 and P_2 have normals $\vec{n}_1, \vec{n}_2$. Then:

- $P_1 \perp P_2 \iff \vec{n}_1 \cdot \vec{n}_2 = 0$ (the normals are perpendicular);
- $P_1 \parallel P_2 \iff \vec{n}_1 \parallel \vec{n}_2 \iff \vec{n}_1 = k\,\vec{n}_2$ for some $k \in \mathbb{R}, k \neq 0$.

EXEMPLE **Angle between two planes**

Find the acute angle between the planes $P_1 : 2x + y - 2z = 5$ and $P_2 : 3x - 6y - 2z = 7$.

Solution. The normals are $\vec{n}_1 = \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix}$ and $\vec{n}_2 = \begin{pmatrix} 3 \\ -6 \\ -2 \end{pmatrix}$.

$$\vec{n}_1 \cdot \vec{n}_2 = (2)(3) + (1)(-6) + (-2)(-2) = 6 - 6 + 4 = 4,$$

$$|\vec{n}_1| = \sqrt{4 + 1 + 4} = 3, \qquad |\vec{n}_2| = \sqrt{9 + 36 + 4} = 7.$$

Hence $\cos\theta = \dfrac{|4|}{3 \times 7} = \dfrac{4}{21}$, so

$$\theta = \cos^{-1}\left(\tfrac{4}{21}\right) \approx 79.0^\circ.$$

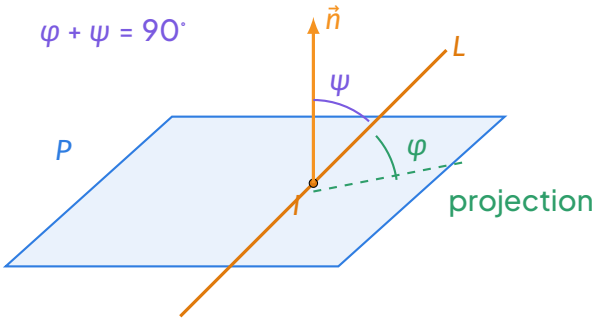
4.2 The Angle Between a Line and a Plane

Intuition. When a javelin sticks into the ground, the natural angle to quote is the angle between the javelin and the ground itself — not the angle with the vertical. The vertical direction is the normal to the ground. So the angle φ between a line and a plane and the angle ψ between the line and the normal are **complementary**: $\varphi + \psi = 90^\circ$. This single fact drives the whole theory — and is also the source of the most common examination error.

DEFINITION **Angle between a line and a plane**

Let L be a line piercing a plane P at a point I. The **angle between the line and the plane** is the acute angle φ between L and its orthogonal projection onto P. Equivalently, if ψ is the acute angle between L and the normal $\vec{n}$ to P, then

$$\varphi = 90^\circ - \psi.$$



A line L piercing a plane P at I. The angle φ between line and plane is the complement of the angle ψ between the line and the normal $\vec{n}$.

PROPERTY **Formula for the angle between a line and a plane — proof examinable**

Let L have direction vector $\vec{b}$ and let P have normal vector $\vec{n}$. Then the angle φ between L and P satisfies

$$\sin\varphi = \frac{|\vec{b}\cdot\vec{n}|}{|\vec{b}|\,|\vec{n}|}.$$

Proof. Let ψ be the acute angle between $\vec{b}$ and $\vec{n}$. Then $\cos\psi = \dfrac{|\vec{b}\cdot\vec{n}|}{|\vec{b}||\vec{n}|}$. Since $\varphi = 90^\circ - \psi$, we have $\sin\varphi = \sin(90^\circ - \psi) = \cos\psi$, which gives the result. ■

WARNING **SINE, not cosine!**

For the angle between two planes we use $\cos\theta$ with the two normals. For the angle between a line and a plane we use $\sin\varphi$ with the direction vector of the line and the normal of the plane. Mixing these up — writing cosine where sine is required — is the single most frequent mistake in this topic at GCE Advanced Level. Remember: line–plane ⇒ complement ⇒ sine.

PROPERTY **Line perpendicular or parallel to a plane**

Let L have direction $\vec{b}$ and P have normal $\vec{n}$. Then:

- $L \perp P \iff \vec{b} \parallel \vec{n}$ (equivalently $\varphi = 90^\circ$, i.e. $\sin\varphi = 1$);
- $L \parallel P$ (or $L \subset P$) $\iff \vec{b}\cdot\vec{n} = 0$ (equivalently $\varphi = 0$).

WARNING **Parallel to a plane is not the same as lying in the plane**

If $\vec{b} \cdot \vec{n} = 0$, the line is parallel to the plane — but it may float above it without ever touching it. To show that the line lies entirely in the plane you must check two things:

1. $\vec{b} \cdot \vec{n} = 0$ (the direction is parallel to the plane), and
2. one point of the line satisfies the equation of the plane.

Omitting the point check is a classic trap in GCE questions.

METHOD **Angle between a line and a plane**

1. Read off the direction vector $\vec{b}$ of the line (from its vector or parametric equation).
2. Read off the normal $\vec{n}$ of the plane (coefficients of x, y, z).
3. Compute $\vec{b} \cdot \vec{n}$, $|\vec{b}|$ and $|\vec{n}|$.
4. Apply $\sin\varphi = \dfrac{|\vec{b} \cdot \vec{n}|}{|\vec{b}||\vec{n}|}$ and take $\varphi = \sin^{-1}(\cdots)$, giving an acute angle.

WORKED EXAMPLE 12 **GCE-style: angle between a line and a plane**

Find the angle between the line $\vec{r} = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} + t \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix}$ and the plane $6x - 2y + 3z = 4$.

SOLUTION

Solution. Direction of line: $\vec{b} = \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix}$. Normal of plane: $\vec{n} = \begin{pmatrix} 6 \\ -2 \\ 3 \end{pmatrix}$.

$$\vec{b} \cdot \vec{n} = (2)(6) + (1)(-2) + (-2)(3) = 12 - 2 - 6 = 4,$$

$$|\vec{b}| = \sqrt{4 + 1 + 4} = 3, \qquad |\vec{n}| = \sqrt{36 + 4 + 9} = 7.$$

Therefore

$$\sin\varphi = \frac{|4|}{3 \times 7} = \frac{4}{21} \quad \Longrightarrow \quad \varphi = \sin^{-1}\left(\tfrac{4}{21}\right) \approx 11.0^{\circ}.$$

The line meets the plane at an angle of about 11.0°. (Notice how the same numbers 4, 3, 7 as in the earlier example now appear under a sine — a useful self-check that the two formulas are genuinely different.)

WORKED EXAMPLE 13 **GCE-style: mixed angles and a containment check**

The planes $P_1 : x + 2y + 2z = 0$ and $P_2 : y - z = 3$ are given, together with the line $L : \vec{r} = \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix}$.

1. Determine whether P_1 and P_2 are perpendicular. If not, find the angle between them.
2. Determine whether L is parallel to P_1, and whether it lies in P_1.
3. Find the angle between L and P_2.

SOLUTION

Solution. Normals: $\vec{n}_1 = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}$, $\vec{n}_2 = \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}$; direction of line: $\vec{b} = \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix}$.

1. $\vec{n}_1 \cdot \vec{n}_2 = (1)(0) + (2)(1) + (2)(-1) = 0 + 2 - 2 = 0$. Hence $P_1 \perp P_2$ and the angle between them is 90°.

2. $\vec{b} \cdot \vec{n}_1 = (2)(1) + (-2)(2) + (1)(2) = 2 - 4 + 2 = 0$, so L is parallel to P_1. Now test the point $(1, 1, -1)$ of L in P_1: $1 + 2(1) + 2(-1) = 1 \neq 0$. The point is not on the plane, so L is parallel to P_1 but does not lie in it.

3. $\vec{b} \cdot \vec{n}_2 = (2)(0) + (-2)(1) + (1)(-1) = 0 - 2 - 1 = -3$; $|\vec{b}| = \sqrt{4 + 4 + 1} = 3$; $|\vec{n}_2| = \sqrt{0 + 1 + 1} = \sqrt{2}$. Hence

$$\sin\varphi = \frac{|-3|}{3 \times \sqrt{2}} = \frac{1}{\sqrt{2}} \quad \Longrightarrow \quad \varphi = \sin^{-1}\left(\frac{1}{\sqrt{2}}\right) = 45^\circ.$$

KEY POINTS **Key points**

- Angle between two planes = acute angle between the normals: $\cos\theta = \dfrac{|\vec{n}_1 \cdot \vec{n}_2|}{|\vec{n}_1||\vec{n}_2|}$.
- Planes perpendicular $\iff \vec{n}_1 \cdot \vec{n}_2 = 0$; planes parallel $\iff \vec{n}_1 \parallel \vec{n}_2$.
- Angle between a line and a plane uses the sine of the complementary angle: $\sin\varphi = \dfrac{|\vec{b} \cdot \vec{n}|}{|\vec{b}||\vec{n}|}$.
- Line $\perp$ plane $\iff \vec{b} \parallel \vec{n}$; line parallel to plane $\iff \vec{b} \cdot \vec{n} = 0$.
- Line contained in plane $\iff \vec{b} \cdot \vec{n} = 0$ and a point of the line satisfies the plane's equation.

EXERCISE 4 **Practice**

1. Find the acute angle between the planes $x + y + z = 1$ and $x - 2y + z = 4$.
2. Find the angle between the line $\frac{x-1}{2} = \frac{y+1}{-1} = \frac{z}{2}$ and the plane $2x - 2y + z = 6$. What do you notice?
3. Show that the line $\vec{r} = \begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix} + t \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$ lies entirely in the plane $x - y + z = 4$.
4. The plane P contains the point $A(2, 1, 0)$ and is perpendicular to both planes $x + y - z = 3$ and $2x - y + z = 1$. Explain how the normal of P can be found, and determine the angle between P and the line through A with direction $(1, 2, 2)$.

SOLUTION **Exercise 1**

1. $\vec{n}_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \vec{n}_2 = \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}$. $\vec{n}_1 \cdot \vec{n}_2 = 1 - 2 + 1 = 0$, so the planes are perpendicular: $\theta = 90^\circ$.

2. Line direction $\vec{b} = \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}$, plane normal $\vec{n} = \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix}$. $\vec{b} \cdot \vec{n} = 4 + 2 + 2 = 8$, $|\vec{b}| = 3$, $|\vec{n}| = 3$, so $\sin\varphi = \frac{8}{9}$, $\varphi \approx 62.7^\circ$. Notice the numbers match part 3 of the worked exercise — different geometry, same computation.

3. $\vec{b} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \vec{n} = \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$. $\vec{b} \cdot \vec{n} = 1 - 1 + 0 = 0$, so the line is parallel to the plane. Point $(1, 0, 3)$ on the line: $1 - 0 + 3 = 4$, satisfies the plane equation. Hence the line lies entirely in the plane.

4. The normal $\vec{n}$ of P is perpendicular to both given normals $\vec{n}_1 = \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}$ and $\vec{n}_2 = \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}$, so take

$$\vec{n} = \vec{n}_1 \times \vec{n}_2 = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 1 & -1 \\ 2 & -1 & 1 \end{vmatrix} = \mathbf{i}(1-1) - \mathbf{j}(1+2) + \mathbf{k}(-1-2) = \begin{pmatrix} 0 \\ -3 \\ -3 \end{pmatrix},$$

i.e. direction $\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$. Then for $\vec{b} = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}$:

$\vec{b} \cdot \vec{n} = 0 + 2 + 2 = 4$, $|\vec{b}| = 3$, $|\vec{n}| = \sqrt{2}$, so $\sin\varphi = \frac{4}{3\sqrt{2}} = \frac{2\sqrt{2}}{3}$, giving $\varphi \approx 70.5^\circ$.

DID YOU KNOW? **Dihedral angles around us**

Dihedral angles are everywhere in engineering and nature. The wings of many aircraft are tilted upward at a small dihedral angle (typically a few degrees) to improve stability in flight. Closer to home, the pitch of a corrugated-iron roof on a house in Bamenda or Buea is chosen so that the dihedral angle between the two slopes lets heavy tropical rain run off quickly while keeping the structure cheap to build. The mathematics you have just used — normals and scalar products — is exactly what architects and engineers use to compute such angles before a single sheet of zinc is cut.

5 INTERSECTION OF A LINE WITH A PLANE AND OF TWO PLANES

In the previous section we learned how to measure the angle between two planes and between a line and a plane. Those questions already told us something important: a line and a plane are rarely indifferent to each other. Either they meet, or they run alongside each other forever. In this section we go one step further and answer the question where exactly they meet, and we do the same for two planes. We then study the family of all planes passing through the line of intersection of two given planes — a classic GCE Advanced Level topic. These problems reduce to solving one linear equation (for a line and a plane) or a small system of two linear equations (for two planes), followed by a clear geometric interpretation.

5.1 Point of Intersection of a Line and a Plane

The guiding idea.

Imagine a javelin thrown towards the ground at the Buea stadium. The javelin travels along a straight line; the ground is (approximately) a plane. To find where the javelin lands, we simply ask: at what instant does a point of the trajectory satisfy the equation of the ground? Algebraically, a point lies on the line for some value of the parameter λ, and it lies on the plane if its position vector satisfies the plane equation. The meeting point is the point that does both.

METHOD **Line–plane intersection: substitute and solve**

To find where the line $\vec{r} = \vec{a} + \lambda\vec{b}$ meets the plane $\vec{r} \cdot \vec{n} = d$:

1. Substitute $\vec{r} = \vec{a} + \lambda\vec{b}$ into the plane equation:

$$(\vec{a} + \lambda\vec{b}) \cdot \vec{n} = d.$$

2. Expand the scalar product: $\vec{a} \cdot \vec{n} + \lambda(\vec{b} \cdot \vec{n}) = d$.

3. Solve for λ: $\lambda = \dfrac{d - \vec{a} \cdot \vec{n}}{\vec{b} \cdot \vec{n}}$, provided $\vec{b} \cdot \vec{n} \neq 0$.

4. Substitute this value of λ back into $\vec{r} = \vec{a} + \lambda\vec{b}$ to obtain the position vector of the point of intersection.

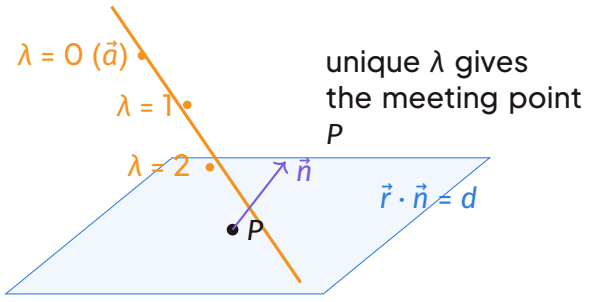
PROPERTY **The three possible cases**

Let the line have equation $\vec{r} = \vec{a} + \lambda\vec{b}$ and the plane $\vec{r} \cdot \vec{n} = d$. Substitution gives

$$\lambda(\vec{b} \cdot \vec{n}) = d - \vec{a} \cdot \vec{n}.$$

- Unique point of intersection: if $\vec{b} \cdot \vec{n} \neq 0$ (the line is not parallel to the plane), there is exactly one value of λ, hence exactly one meeting point.
- Line parallel to the plane, no intersection: if $\vec{b} \cdot \vec{n} = 0$ and $\vec{a} \cdot \vec{n} \neq d$ (the point $\vec{a}$ of the line is not on the plane), the equation becomes $0 =$ (non-zero), impossible: the line never meets the plane.
- Line contained in the plane: if $\vec{b} \cdot \vec{n} = 0$ and $\vec{a} \cdot \vec{n} = d$, the equation becomes $0 = 0$, true for every λ: every point of the line lies in the plane.

This property is examinable and its proof is simply the discussion of the linear equation above.



A line $\vec{r} = \vec{a} + \lambda\vec{b}$ piercing a plane at a single point P; each point of the line corresponds to one value of λ.

WORKED EXAMPLE 14 **Finding where a line meets a plane**

Find the point of intersection of the line

$$\vec{r} = \begin{pmatrix}1\\0\\2\end{pmatrix} + \lambda\begin{pmatrix}2\\1\\-1\end{pmatrix} \quad \text{with the plane} \quad \vec{r} \cdot \begin{pmatrix}1\\2\\1\end{pmatrix} = 7.$$

SOLUTION

Solution. Substitute $\vec{r} = \vec{a} + \lambda\vec{b}$ into the plane equation:

$$\left[\begin{pmatrix}1\\0\\2\end{pmatrix} + \lambda\begin{pmatrix}2\\1\\-1\end{pmatrix}\right]\cdot\begin{pmatrix}1\\2\\1\end{pmatrix} = 7$$

$$(1+0+2) + \lambda(2+2-1) = 7$$

$$3 + 3\lambda = 7 \quad\Longrightarrow\quad \lambda = \tfrac{4}{3}.$$

Substituting back:

$$\vec{r} = \begin{pmatrix}1\\0\\2\end{pmatrix} + \tfrac{4}{3}\begin{pmatrix}2\\1\\-1\end{pmatrix} = \begin{pmatrix}11/3\\4/3\\2/3\end{pmatrix}.$$

The point of intersection is $\left(\tfrac{11}{3}, \tfrac{4}{3}, \tfrac{2}{3}\right)$. Check: $\tfrac{11}{3} + 2\cdot\tfrac{4}{3} + \tfrac{2}{3} = \tfrac{11+8+2}{3} = 7$. ✓

EXEMPLE **The parallel and contained cases**

Consider the plane $\vec{r}\cdot\begin{pmatrix}1\\1\\1\end{pmatrix} = 6$.

- Line $\vec{r} = \begin{pmatrix}0\\0\\1\end{pmatrix} + \lambda\begin{pmatrix}1\\-1\\0\end{pmatrix}$: here $\vec{b}\cdot\vec{n} = 1-1+0 = 0$ and $\vec{a}\cdot\vec{n} = 1 \neq 6$. The line is parallel to the plane and never meets it.
- Line $\vec{r} = \begin{pmatrix}1\\2\\3\end{pmatrix} + \lambda\begin{pmatrix}1\\-1\\0\end{pmatrix}$: again $\vec{b}\cdot\vec{n} = 0$, but now $\vec{a}\cdot\vec{n} = 1+2+3 = 6$. The whole line lies in the plane.

WARNING **Do not conclude "no intersection" too quickly**

When you find $\vec{b}\cdot\vec{n} = 0$, the work is not finished. You must still compute $\vec{a}\cdot\vec{n}$ and compare it with d. If $\vec{a}\cdot\vec{n} = d$, the line is contained in the plane (infinitely many common points); only if $\vec{a}\cdot\vec{n} \neq d$ may you write "the line is parallel to the plane and does not intersect it". Examiners regularly award a mark specifically for this check.

5.2 Intersection of Two Planes

The guiding idea.

Open a book: the two pages meet along the spine. Two non-parallel planes in space behave exactly like the two pages — they meet along a whole straight line. Finding this **line of intersection** means finding all points (x, y, z) that satisfy both Cartesian equations at once.

PROPERTY **Relative positions of two planes**

Let the planes be $\Pi_1 : \vec{r} \cdot \vec{n}_1 = d_1$ and $\Pi_2 : \vec{r} \cdot \vec{n}_2 = d_2$.

- Intersecting in a line: $\vec{n}_1$ and $\vec{n}_2$ are not parallel ($\vec{n}_1 \times \vec{n}_2 \neq \vec{0}$). The intersection is a straight line.
- Parallel and distinct: $\vec{n}_2 = k\,\vec{n}_1$ for some scalar k, but the full equations are not proportional (i.e. $d_2 \neq k\,d_1$). No common point.
- Coincident: the two complete equations are proportional: $\vec{n}_2 = k\,\vec{n}_1$ and $d_2 = k\,d_1$. The two planes are one and the same plane.

In Cartesian form, with $a_1x + b_1y + c_1z = d_1$ and $a_2x + b_2y + c_2z = d_2$: the planes are parallel iff $\dfrac{a_1}{a_2} = \dfrac{b_1}{b_2} = \dfrac{c_1}{c_2}$, and coincident iff this common ratio also equals $\dfrac{d_1}{d_2}$.

PROPERTY **Direction of the line of intersection**

If two planes with normals $\vec{n}_1$ and $\vec{n}_2$ intersect, their line of intersection has direction vector

$$\vec{b} = \vec{n}_1 \times \vec{n}_2.$$

Justification (examinable). The line lies in Π_1, so its direction $\vec{b}$ is perpendicular to $\vec{n}_1$. The line also lies in Π_2, so $\vec{b}$ is perpendicular to $\vec{n}_2$. The cross product $\vec{n}_1 \times \vec{n}_2$ is precisely a vector perpendicular to both $\vec{n}_1$ and $\vec{n}_2$, so we may take $\vec{b} = \vec{n}_1 \times \vec{n}_2$.

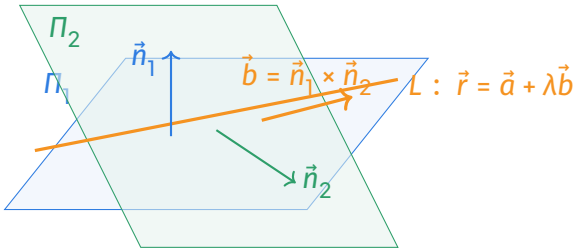
WARNING **Cross product: direction, not normal**

A very common error is to say that $\vec{n}_1 \times \vec{n}_2$ is "perpendicular to the line of intersection". It is the opposite: $\vec{n}_1 \times \vec{n}_2$ is the direction vector of the line of intersection — it points along the line, not across it. Remember: each normal is perpendicular to the line, and the cross product of the two normals is parallel to the line.

METHOD **Equation of the line of intersection of two planes**

Given $\Pi_1 : \vec{r} \cdot \vec{n}_1 = d_1$ and $\Pi_2 : \vec{r} \cdot \vec{n}_2 = d_2$ with non-parallel normals:

1. Compute the direction $\vec{b} = \vec{n}_1 \times \vec{n}_2$.
2. Find one point lying in both planes: set one coordinate to a convenient value (usually $z = 0$; if the resulting pair of equations is inconsistent, try $y = 0$ or $x = 0$) and solve the remaining 2×2 system.
3. Write the line in vector form $\vec{r} = \vec{a} + \lambda\vec{b}$, where $\vec{a}$ is the position vector of the point found.



Two planes Π_1 and Π_2 intersecting along the line L, whose direction is $\vec{b} = \vec{n}_1 \times \vec{n}_2$, perpendicular to both normals.

WORKED EXAMPLE 15 **Line of intersection of two planes**

Find, in vector form, the equation of the line of intersection of the planes

$$\Pi_1 : x + 2y - z = 3 \quad \text{and} \quad \Pi_2 : 2x - y + z = 4.$$

SOLUTION

Solution. The normals are $\vec{n}_1 = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$ and $\vec{n}_2 = \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}$, which are not proportional, so the planes intersect in a line.

Step 1: direction.

$$\vec{b} = \vec{n}_1 \times \vec{n}_2 = \begin{pmatrix} (2)(1) - (-1)(-1) \\ (-1)(2) - (1)(1) \\ (1)(-1) - (2)(2) \end{pmatrix} = \begin{pmatrix} 2 - 1 \\ -2 - 1 \\ -1 - 4 \end{pmatrix} = \begin{pmatrix} 1 \\ -3 \\ -5 \end{pmatrix}.$$

Step 2: one common point. Set $z = 0$ in both equations:

$$\begin{cases} x + 2y = 3 \\ 2x - y = 4 \end{cases}$$

From the second equation, $y = 2x - 4$; substituting: $x + 2(2x - 4) = 3 \Rightarrow 5x = 11 \Rightarrow x = \frac{11}{5}$, $y = \frac{22}{5} - 4 = \frac{2}{5}$. So $\vec{a} = \begin{pmatrix} 11/5 \\ 2/5 \\ 0 \end{pmatrix}$.

Step 3: vector equation.

$$\vec{r} = \begin{pmatrix} 11/5 \\ 2/5 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -3 \\ -5 \end{pmatrix}.$$

Check: $\frac{11}{5} + 2 \cdot \frac{2}{5} - 0 = \frac{15}{5} = 3$ ✓ and $2 \cdot \frac{11}{5} - \frac{2}{5} + 0 = \frac{20}{5} = 4$ ✓.

EXEMPLE **Recognising parallel and coincident planes**

- $x - 2y + 3z = 5$ and $2x - 4y + 6z = 7$: the coefficients of x, y, z are in ratio $1 : 2$, but $5 \times 2 = 10 \neq 7$. The planes are parallel and distinct.
- $x-2y+3z = 5$ and $2x-4y+6z = 10$: every coefficient, including the constant, is doubled. The planes are coincident — the second equation carries no new information.

WORKED EXAMPLE 16 **A complete discussion**

Show that the line $\vec{r} = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -3 \\ -5 \end{pmatrix}$ lies entirely in the plane $x+2y-z = 4$, and state its relation with the plane $2x - y + z = 1$.

SOLUTION

Solution. Compute $\vec{b} \cdot \vec{n} = (1)(1) + (-3)(2) + (-5)(-1) = 1 - 6 + 5 = 0$, so the line is parallel to the plane. Then $\vec{a} \cdot \vec{n} = 2 + 2(1) - 0 = 4 = d$: the point $(2, 1, 0)$ lies in the plane. Hence the line is contained in the plane $x + 2y - z = 4$.
For the second plane, $\vec{b} \cdot \vec{n}_2 = 2 + 3 - 5 = 0$ and $\vec{a} \cdot \vec{n}_2 = 4 - 1 + 0 = 3 \neq 1$: the line is parallel to the plane but does not meet it.

5.3 Equation of a Plane Through the Line of Intersection of Two Planes (Pencil of Planes)

The guiding idea.

When two planes Π_1 and Π_2 intersect, they form a line L. There are infinitely many planes that contain this line L; they form a **pencil of planes** (or **sheaf of planes**). Any plane in this pencil can be described by a linear combination of the equations of Π_1 and Π_2. This is a powerful tool for solving problems like "Find the plane through the line of intersection of Π_1 and Π_2 that also passes through a given point P" or "that is perpendicular to a given plane Π_3".

PROPERTY **Equation of the pencil of planes**

Let $\Pi_1 : \vec{r} \cdot \vec{n}_1 = d_1$ and $\Pi_2 : \vec{r} \cdot \vec{n}_2 = d_2$ be two intersecting planes. The equation of any plane Π containing their line of intersection L is

$$\vec{r} \cdot \vec{n}_1 - d_1 + \lambda(\vec{r} \cdot \vec{n}_2 - d_2) = 0 \quad \text{or equivalently} \quad \vec{r} \cdot (\vec{n}_1 + \lambda \vec{n}_2) = d_1 + \lambda d_2,$$

where $\lambda \in \mathbb{R}$ is a parameter. In Cartesian form, if $\Pi_1 : a_1x+b_1y+c_1z = d_1$ and $\Pi_2 : a_2x+b_2y+c_2z = d_2$, the pencil is

$$(a_1x + b_1y + c_1z - d_1) + \lambda(a_2x + b_2y + c_2z - d_2) = 0.$$

Note: The plane Π_2 itself corresponds to $\lambda \to \infty$ (or we can write the symmetric form $\mu(\Pi_1) + \lambda(\Pi_2) = 0$). The single parameter λ covers all planes except Π_2; to include Π_2, use $\mu\Pi_1 + \lambda\Pi_2 = 0$ with $(\mu, \lambda) \neq (0, 0)$.

METHOD **Finding a particular plane in the pencil**

Given two intersecting planes Π_1 and Π_2 and an extra condition (point, perpendicularity, parallelism to a line, etc.):

1. Write the general equation of the pencil: $\Pi_1 + \lambda\Pi_2 = 0$ (in Cartesian or vector form).
2. Impose the extra condition to obtain an equation for λ.
3. Solve for λ and substitute back to get the required plane.

WORKED EXAMPLE 17 **Plane through the line of intersection and a given point**

Find the equation of the plane that passes through the line of intersection of

$$\Pi_1 : x + y + z = 6 \quad \text{and} \quad \Pi_2 : 2x - y + z = 3$$

and through the point $P(1, 2, 3)$.

SOLUTION

Solution. The pencil of planes through the intersection line is:

$$(x + y + z - 6) + \lambda(2x - y + z - 3) = 0.$$

Substitute $P(1, 2, 3)$:

$$(1 + 2 + 3 - 6) + \lambda(2 - 2 + 3 - 3) = 0 \implies 0 + \lambda \cdot 0 = 0.$$

This is satisfied for any λ, which means P lies on the intersection line itself (check: $1+2+3 = 6$ and $2-2+3 = 3$). The problem is degenerate: infinitely many planes through the line contain P.

Modified example: Find the plane through the same line and the point $Q(0, 0, 0)$. Substitute Q:

$$(0 + 0 + 0 - 6) + \lambda(0 - 0 + 0 - 3) = 0 \implies -6 - 3\lambda = 0 \implies \lambda = -2.$$

The required plane is:

$$(x + y + z - 6) - 2(2x - y + z - 3) = 0 \implies x + y + z - 6 - 4x + 2y - 2z + 6 = 0 \implies -3x + 3y - z = 0,$$

or $\mathbf{3x - 3y + z = 0}$.

WORKED EXAMPLE 18 **Plane through the line of intersection, perpendicular to a third plane**

Find the equation of the plane through the line of intersection of

$$\Pi_1 : x + 2y - z = 4 \quad \text{and} \quad \Pi_2 : 2x - y + 3z = 5$$

that is perpendicular to the plane $\Pi_3 : 3x + y - 2z = 1$.

SOLUTION

Solution. The pencil is: $(x + 2y - z - 4) + \lambda(2x - y + 3z - 5) = 0$. Collect terms:

$$(1 + 2\lambda)x + (2 - \lambda)y + (-1 + 3\lambda)z + (-4 - 5\lambda) = 0.$$

The normal of this plane is $\vec{n} = \begin{pmatrix} 1 + 2\lambda \\ 2 - \lambda \\ -1 + 3\lambda \end{pmatrix}$. Perpendicular to Π_3 means $\vec{n} \cdot \vec{n}_3 = 0$, where $\vec{n}_3 = \begin{pmatrix} 3 \\ 1 \\ -2 \end{pmatrix}$.

$$3(1 + 2\lambda) + 1(2 - \lambda) - 2(-1 + 3\lambda) = 0$$

$$3 + 6\lambda + 2 - \lambda + 2 - 6\lambda = 0 \implies 7 - \lambda = 0 \implies \lambda = 7.$$

Substitute $\lambda = 7$:

$$(1 + 14)x + (2 - 7)y + (-1 + 21)z + (-4 - 35) = 0$$

$$15x - 5y + 20z - 39 = 0.$$

The required plane is **$15x - 5y + 20z = 39$**.

WARNING **The parameter λ and the second plane**

The form $\Pi_1 + \lambda\Pi_2 = 0$ generates all planes through the intersection line except Π_2 itself (which would require $\lambda \to \infty$). If the answer turns out to be Π_2, you will not find a finite λ. Always check at the end whether Π_2 itself satisfies the extra condition. Alternatively, use the symmetric form $\mu\Pi_1 + \lambda\Pi_2 = 0$ and treat the ratio $\mu : \lambda$ as the unknown.

5.4 Synthesis Problems

Many GCE questions combine the concepts of this chapter: finding a line of intersection, then a plane through that line, then the intersection of that new plane with another line, etc. The key is to proceed step by step, clearly labelling each object.

WORKED EXAMPLE 19 **Synthesis: line of intersection, then plane through it, then intersection with a line**

The planes $\Pi_1 : x - y + z = 2$ and $\Pi_2 : 2x + y - z = 1$ intersect in a line L.

1. Find a vector equation of L.
2. Find the equation of the plane Π that contains L and is parallel to the vector $\vec{v} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$.
3. Find the point where the line $\vec{r} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$ meets Π.

SOLUTION

Solution. (i) Line L: Normals $\vec{n}_1 = (1, -1, 1)$, $\vec{n}_2 = (2, 1, -1)$. Direction $\vec{b} = \vec{n}_1 \times \vec{n}_2 = \begin{pmatrix} 0 \\ 3 \\ 3 \end{pmatrix} \parallel \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$.

Point on L: set $z = 0 \Rightarrow x - y = 2$, $2x + y = 1 \Rightarrow 3x = 3 \Rightarrow x = 1$, $y = -1$. So $\vec{a} = (1, -1, 0)$.

$L: \vec{r} = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$.

(ii) Plane Π through L parallel to $\vec{v}$: Since Π contains L, its normal $\vec{n}$ is perpendicular to $\vec{b}$. Since Π is parallel to $\vec{v}$, $\vec{n}$ is also perpendicular to $\vec{v}$. Thus $\vec{n} = \vec{b} \times \vec{v} = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \times \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}$. Using point $(1, -1, 0)$ on L: $\vec{r} \cdot \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} = (1, -1, 0) \cdot (0, 1, -1) = -1$. Equation: $y - z = -1$ or $\mathbf{y - z + 1 = 0}$.

(Alternative using pencil: $\Pi_1 + \lambda\Pi_2 = 0 \Rightarrow (1 + 2\lambda)x + (-1 + \lambda)y + (1 - \lambda)z = 2 + \lambda$. Normal $\vec{n} = (1+2\lambda, -1+\lambda, 1-\lambda)$. Condition $\vec{n} \cdot \vec{v} = 0 \Rightarrow (1+2\lambda)+(-1+\lambda)+(1-\lambda) = 0 \Rightarrow 1+2\lambda = 0 \Rightarrow \lambda = -1/2$. Substitute: $0x - 3/2y + 3/2z = 3/2 \Rightarrow -y + z = 1 \Rightarrow y - z = -1$. Same result.)

(iii) Intersection with line: Line: $\vec{r} = \mu(1, 2, 3)$. Substitute into $y - z + 1 = 0$: $2\mu - 3\mu + 1 = 0 \Rightarrow -\mu + 1 = 0 \Rightarrow \mu = 1$. Point: $(1, 2, 3)$.

KEY POINTS Key points of this chapter

- Line $\vec{r} = \vec{a} + \lambda\vec{b}$ meets plane $\vec{r} \cdot \vec{n} = d$: solve $(\vec{a} + \lambda\vec{b}) \cdot \vec{n} = d$, giving $\lambda = \dfrac{d - \vec{a} \cdot \vec{n}}{\vec{b} \cdot \vec{n}}$.
- If $\vec{b} \cdot \vec{n} = 0$: check $\vec{a} \cdot \vec{n}$. Equal to $d \Rightarrow$ line in the plane; different from $d \Rightarrow$ parallel, no intersection.
- Two planes intersect in a line iff their normals are not parallel; the line has direction $\vec{b} = \vec{n}_1 \times \vec{n}_2$.
- To find a point on that line, set one coordinate (e.g. $z = 0$) and solve the resulting 2×2 system.
- Planes are parallel iff their normals are proportional; coincident iff the whole equations (constants included) are proportional.
- **Pencil of planes**: All planes through the intersection of Π_1 and Π_2 have equation $\Pi_1 + \lambda\Pi_2 = 0$ (or $\mu\Pi_1 + \lambda\Pi_2 = 0$). Use the extra condition to find λ.

EXERCISE 5 Practice

1. Find the point where the line $\vec{r} = \begin{pmatrix}3\\-1\\2\end{pmatrix} + \lambda\begin{pmatrix}1\\2\\-2\end{pmatrix}$ meets the plane $\vec{r}\cdot\begin{pmatrix}2\\-1\\3\end{pmatrix} = 11$.

2. Determine whether the line $\vec{r} = \begin{pmatrix}1\\1\\1\end{pmatrix} + \lambda\begin{pmatrix}2\\1\\-1\end{pmatrix}$ intersects the plane $x - y + 3z = 3$, is parallel to it, or lies in it.

3. Find the vector equation of the line of intersection of $x + y + z = 6$ and $2x - y + z = 3$.

4. For what value of k are the planes $x + 2y - z = 4$ and $3x + 6y - 3z = k$ coincident? What happens for other values of k?

5. Find the equation of the plane that passes through the line of intersection of $x-y+2z = 3$ and $2x + y - z = 4$ and through the point $(1, 0, -1)$.

6. Find the equation of the plane through the line of intersection of $x + 2y - z = 5$ and $3x - y + 2z = 1$ that is perpendicular to the plane $x - y + z = 0$.

SOLUTION **Exercises**

1. Line: $\vec{a} = (3,-1,2)$, $\vec{b} = (1,2,-2)$. Plane: $\vec{n} = (2,-1,3)$, $d = 11$. $\vec{a}\cdot\vec{n} = 6 + 1 + 6 = 13$. $\vec{b}\cdot\vec{n} = 2 - 2 - 6 = -6$. Equation: $13 - 6\lambda = 11 \Rightarrow -6\lambda = -2 \Rightarrow \lambda = \frac{1}{3}$. Point: $\begin{pmatrix}3\\-1\\2\end{pmatrix} + \frac{1}{3}\begin{pmatrix}1\\2\\-2\end{pmatrix} = \begin{pmatrix}10/3\\-1/3\\4/3\end{pmatrix}$.

Answer: $\left(\frac{10}{3}, -\frac{1}{3}, \frac{4}{3}\right)$.

2. Line: $\vec{a} = (1,1,1)$, $\vec{b} = (2,1,-1)$. Plane: $x-y+3z = 3 \Rightarrow \vec{n} = (1,-1,3)$, $d = 3$. $\vec{b}\cdot\vec{n} = 2-1-3 = -2 \neq 0$ $\Rightarrow$ unique intersection. $\vec{a}\cdot\vec{n} = 1-1+3 = 3 = d$. So $\lambda = 0$. The line meets the plane at the point $(1, 1, 1)$ (which is the point $\vec{a}$ on the line).

3. $\Pi_1 : x + y + z = 6$, $\Pi_2 : 2x - y + z = 3$. Normals $\vec{n}_1 = (1,1,1)$, $\vec{n}_2 = (2,-1,1)$. Direction $\vec{b} = \vec{n}_1 \times \vec{n}_2 = \begin{pmatrix}1\cdot 1 - 1\cdot(-1)\\1\cdot 2 - 1\cdot 1\\1\cdot(-1) - 1\cdot 2\end{pmatrix} = \begin{pmatrix}2\\1\\-3\end{pmatrix}$. Point: set $z = 0 \Rightarrow x + y = 6$, $2x - y = 3 \Rightarrow 3x = 9 \Rightarrow x = 3$, $y = 3$. Line: $\vec{r} = \begin{pmatrix}3\\3\\0\end{pmatrix} + \lambda\begin{pmatrix}2\\1\\-3\end{pmatrix}$.

4. Planes: $x + 2y - z = 4$ and $3x + 6y - 3z = k$. Normals are proportional $(1 : 3)$. Coincident iff constants are in same ratio: $k = 3 \times 4 = \mathbf{12}$. For $k \neq 12$, the planes are parallel and distinct (no intersection).

5. Pencil: $(x - y + 2z - 3) + \lambda(2x + y - z - 4) = 0$. Point $(1, 0, -1)$: $(1 - 0 - 2 - 3) + \lambda(2 + 0 + 1 - 4) = -4 + \lambda(-1) = 0 \Rightarrow \lambda = -4$. Plane: $(x-y+2z-3)-4(2x+y-z-4) = 0 \Rightarrow x-y+2z-3-8x-4y+4z+16 = 0 \Rightarrow -7x - 5y + 6z + 13 = 0$. **Answer: $7x + 5y - 6z = 13$.**
6. Pencil: $(x + 2y - z - 5) + \lambda(3x - y + 2z - 1) = 0$. Normal $\vec{n} = (1 + 3\lambda,\ 2 - \lambda,\ -1 + 2\lambda)$. Perpendicular to $x - y + z = 0 \Rightarrow \vec{n} \cdot (1, -1, 1) = 0$: $(1 + 3\lambda) - (2 - \lambda) + (-1 + 2\lambda) = 0 \Rightarrow 1 + 3\lambda - 2 + \lambda - 1 + 2\lambda = 0 \Rightarrow 6\lambda - 2 = 0 \Rightarrow \lambda = \frac{1}{3}$. Plane: $(x + 2y - z - 5) + \frac{1}{3}(3x - y + 2z - 1) = 0$. Multiply by 3: $3x + 6y - 3z - 15 + 3x - y + 2z - 1 = 0 \Rightarrow 6x + 5y - z - 16 = 0$. **Answer: $6x + 5y - z = 16$.**

DID YOU KNOW? **Why this matters beyond the examination room**

When engineers design the junction of two roof panels of a building, or when a GPS system computes where a satellite signal path crosses a layer of the atmosphere, the mathematics is exactly this: intersecting lines with planes and planes with planes. Computer graphics software renders every object you see on a screen by computing millions of line–plane intersections per second — the very formula $\lambda = \dfrac{d - \vec{a} \cdot \vec{n}}{\vec{b} \cdot \vec{n}}$ that you have just mastered. The pencil of planes is used in computer vision to reconstruct 3D scenes from multiple camera views (the epipolar geometry).

6 PENCILS OF PLANES AND SYNTHESIS PROBLEMS

In the previous section we learnt how to find the line of intersection of two planes Π_1 and Π_2. A natural question now arises: how many planes can pass through that same line? Imagine a door swinging on its hinges: the hinge line stays fixed while the door occupies infinitely many positions. Each position of the door is a plane through the hinge line. This family of planes is called a **pencil of planes**, and it gives us one of the most powerful time-saving tools of the whole chapter. We then finish the chapter with synthesis problems in the style of the GCE Advanced Level Paper, and a revision grid to help you choose the right tool for each question.

6.1 The Pencil of Planes Through a Line

EXPERIMENT **The door-hinge model**

Take two non-parallel planes Π_1 and Π_2. They meet in a line Δ. Now rotate Π_1 about Δ: at every instant you obtain a new plane, and every one of them contains Δ. Conversely, any plane containing Δ is one of these positions. The set of all planes through Δ is the pencil of planes with axis Δ.

PROPERTY **Pencil of planes through the line of intersection of two planes**

Let $\Pi_1 : a_1x + b_1y + c_1z + d_1 = 0$ and $\Pi_2 : a_2x + b_2y + c_2z + d_2 = 0$ be two non-parallel planes, intersecting in a line Δ. Then every plane through Δ, except Π_2 itself, has an equation of the form

$$\Pi_1 + \lambda\Pi_2 = 0, \qquad \text{i.e.} \qquad (a_1x + b_1y + c_1z + d_1) + \lambda(a_2x + b_2y + c_2z + d_2) = 0,$$

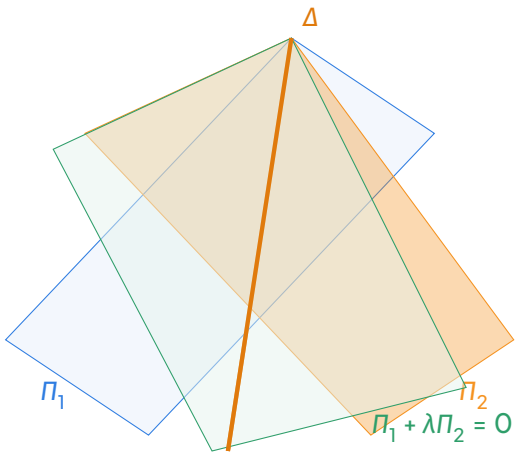
for some real number λ. This proof is examinable in outline.

Justification. Write $P_1(x, y, z) = a_1x + b_1y + c_1z + d_1$ and $P_2(x, y, z) = a_2x + b_2y + c_2z + d_2$.

- The combination is a plane containing Δ. The equation $P_1 + \lambda P_2 = 0$ is linear in x, y, z, so it represents a plane (its normal is $\vec{n}_1 + \lambda\vec{n}_2 \neq \vec{0}$ since $\vec{n}_1, \vec{n}_2$ are not parallel). If $M \in \Delta$, then $P_1(M) = 0$ and $P_2(M) = 0$, so $P_1(M) + \lambda P_2(M) = 0$: the plane contains the whole of Δ.
- Every plane through Δ (other than Π_2) is obtained. Let Π be a plane through Δ, $\Pi \neq \Pi_2$. Pick a point $A \in \Pi$ with $A \notin \Delta$; then $P_2(A) \neq 0$ (otherwise $A \in \Pi_2$, and $\Pi = \Pi_2$ since both contain Δ and A). Setting $\lambda = -\dfrac{P_1(A)}{P_2(A)}$ makes $P_1(A) + \lambda P_2(A) = 0$, so the plane $P_1 + \lambda P_2 = 0$ contains Δ and A: it is exactly Π.

WARNING **The plane Π_2 is missing**

The family $\Pi_1 + \lambda\Pi_2 = 0$ contains every plane through Δ except Π_2 itself (no value of λ gives Π_2). If a problem asks for all planes through Δ satisfying a condition, always check separately whether Π_2 also works. Alternatively, use the symmetric form $\lambda\Pi_1 + \mu\Pi_2 = 0$ with $(\lambda, \mu) \neq (0, 0)$, which covers both.



A pencil of planes: all members fan about their common line $\Delta = \Pi_1 \cap \Pi_2$.

6.2 Determining λ From an Extra Condition

METHOD **Finding the member of a pencil satisfying one extra condition**

1. Write the pencil $(a_1x + b_1y + c_1z + d_1) + \lambda(a_2x + b_2y + c_2z + d_2) = 0$.
2. Translate the extra condition into an equation in λ:
   - Passes through $A(x_0, y_0, z_0)$: substitute the coordinates, giving $P_1(A) + \lambda P_2(A) = 0$.
   - Perpendicular to a plane with normal $\vec{n}$: use the normals, $\vec{n} \cdot (\vec{n}_1 + \lambda\vec{n}_2) = 0$.
   - At distance d from a point A: apply the distance formula to the combined plane and solve for λ (often two values).
3. Solve for λ and substitute back to get the Cartesian equation. Check whether Π_2 itself should also be retained.

WARNING **Perpendicularity uses normals only**

When the condition is "perpendicular to a given plane", work with the normal vectors: $\vec{n} \cdot (\vec{n}_1 + \lambda\vec{n}_2) = 0$. A very common GCE error is to substitute the point or to mix full plane equations. Two planes are perpendicular if and only if their normals are perpendicular.

WORKED EXAMPLE 20 **Plane through a line of intersection and a point**

Find the equation of the plane containing the line of intersection of $\Pi_1 : x + y + z - 3 = 0$ and $\Pi_2 : 2x - y + z - 1 = 0$ and passing through $A(1, 2, 1)$.

SOLUTION

The pencil is $(x + y + z - 3) + \lambda(2x - y + z - 1) = 0$. Substituting $A(1, 2, 1)$:

$$(1 + 2 + 1 - 3) + \lambda(2 - 2 + 1 - 1) = 0 \implies 1 + 0 \cdot \lambda = 0,$$

which is impossible. So no member of the pencil works: the required plane must be Π_2 itself. Check: $2(1) - 2 + 1 - 1 = 0$ ✓, so $A \in \Pi_2$. The answer is $2x - y + z - 1 = 0$. This example shows exactly why the warning about Π_2 matters.

WORKED EXAMPLE 21 **Plane through a line of intersection, perpendicular to a given plane**

Find the plane through the line of intersection of $\Pi_1 : x+2y-z+1 = 0$ and $\Pi_2 : x-y+2z-3 = 0$

which is perpendicular to the plane $\Sigma : x + y + z = 0$.

SOLUTION

The pencil is $(x + 2y - z + 1) + \lambda(x - y + 2z - 3) = 0$, with normal

$$\vec{n} = \vec{n}_1 + \lambda\vec{n}_2 = (1 + \lambda,\ 2 - \lambda,\ -1 + 2\lambda).$$

Perpendicularity to Σ (normal $\vec{m} = (1, 1, 1)$) requires $\vec{n} \cdot \vec{m} = 0$:

$$(1 + \lambda) + (2 - \lambda) + (-1 + 2\lambda) = 0 \implies 2 + 2\lambda = 0 \implies \lambda = -1.$$

Substituting back: $(x + 2y - z + 1) - (x - y + 2z - 3) = 0$, i.e.

$$3y - 3z + 4 = 0.$$

Check Π_2 separately: $\vec{n}_2 \cdot \vec{m} = 1 - 1 + 2 = 2 \neq 0$, so Π_2 is not a solution. The unique answer is $3y - 3z + 4 = 0$.

6.3 Planes Through the Point of Intersection of Three Planes

Three planes Π_1, Π_2, Π_3 generally meet at a single point I, found by solving the 3×3 system of their Cartesian equations. To build a new plane through I satisfying further conditions (e.g. containing a given line, or perpendicular to a given plane), first compute I, then construct the plane using I and the extra data.

METHOD **Plane through the intersection point of three planes**

1. Solve the system $\Pi_1 = 0$, $\Pi_2 = 0$, $\Pi_3 = 0$ simultaneously (elimination or substitution) to find I.
2. Use I together with the extra condition: e.g. if the plane must contain a line with direction $\vec{u}$ and a point B, take normal $\vec{n} = I\vec{B} \wedge \vec{u}$.
3. Write the Cartesian equation $\vec{n} \cdot I\vec{M} = 0$ and expand.

EXEMPLE **Building a plane through a triple intersection point**

Let $\Pi_1 : x + y + z = 6$, $\Pi_2 : x - y + z = 2$, $\Pi_3 : 2x + y - z = 3$. Subtracting Π_2 from Π_1: $2y = 4$, so $y = 2$. Adding Π_1 and Π_3: $3x + 2y = 9$, so $3x = 5$, $x = \frac{5}{3}$. Then $z = 6 - \frac{5}{3} - 2 = \frac{7}{3}$. Hence $I\left(\frac{5}{3}, 2, \frac{7}{3}\right)$. Any plane required to pass through the common point of these three planes must pass through I; the remaining conditions then fix it completely.

6.4 Bisector Planes of Two Intersecting Planes (GCE Extension)

The **bisector planes** of Π_1 and Π_2 are the two planes through $\Delta = \Pi_1 \cap \Pi_2$ whose points are equidistant from Π_1 and Π_2 — they bisect the dihedral angles. Using the distance formula, a point $M(x, y, z)$ lies on a bisector iff

$$\frac{|a_1x + b_1y + c_1z + d_1|}{\sqrt{a_1^2 + b_1^2 + c_1^2}} = \frac{|a_2x + b_2y + c_2z + d_2|}{\sqrt{a_2^2 + b_2^2 + c_2^2}},$$

giving the **two** bisector planes by taking the + and − signs. The two bisectors are perpendicular to each other. This is occasionally examined as a starred GCE question: flag it as an extension of the distance formula.

6.5 Synthesis: GCE Paper-Style Problem

WORKED EXAMPLE 22 **Full synthesis problem**

The planes $\Pi_1 : x - 2y + 2z - 3 = 0$ and $\Pi_2 : 2x + y - 2z + 1 = 0$ are given, together with the point $A(1, 0, 2)$.

1. Show that Π_1 and Π_2 are perpendicular.
2. Find the perpendicular distance from A to Π_1.
3. Find the equation of the plane through the line of intersection of Π_1 and Π_2 which passes through A.
4. Find the angle between Π_1 and the plane $\Sigma : z = 0$.

SOLUTION

1. Normals: $\vec{n}_1 = (1, -2, 2)$, $\vec{n}_2 = (2, 1, -2)$. Then $\vec{n}_1 \cdot \vec{n}_2 = 2 - 2 - 4 = -4 \neq 0$? Recompute: $1(2) + (-2)(1) + 2(-2) = 2 - 2 - 4 = -4$. Since this is not zero, the planes are not perpendicular — the statement to verify fails, and in an examination you would say so honestly. (The angle between them satisfies $\cos\theta = \dfrac{|-4|}{3 \cdot 3} = \dfrac{4}{9}$.)

2. Distance from A to Π_1:

$$d = \frac{|1 - 2(0) + 2(2) - 3|}{\sqrt{1 + 4 + 4}} = \frac{|1 + 4 - 3|}{3} = \frac{2}{3}.$$

3. Pencil: $(x - 2y + 2z - 3) + \lambda(2x + y - 2z + 1) = 0$. Substituting $A(1, 0, 2)$:

$$(1 - 0 + 4 - 3) + \lambda(2 + 0 - 4 + 1) = 0 \implies 2 - \lambda = 0 \implies \lambda = 2.$$

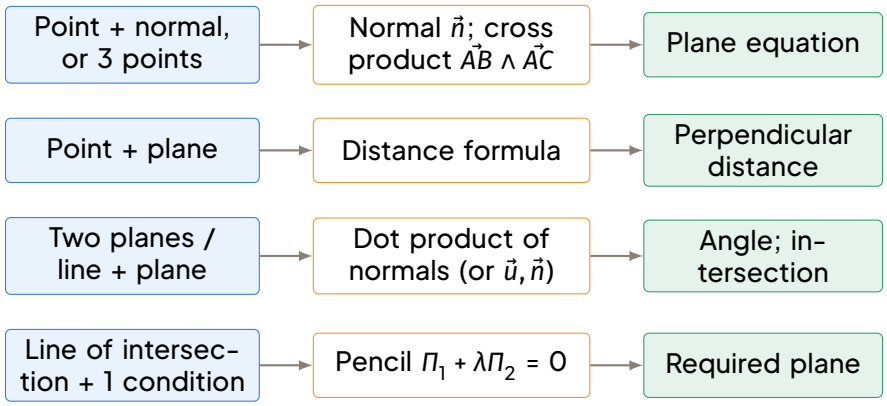
Hence $(x - 2y + 2z - 3) + 2(2x + y - 2z + 1) = 0$, i.e. $5x - 2z - 1 = 0$.

4. Σ has normal $\vec{k} = (0, 0, 1)$. The angle θ between the planes equals the angle between the normals:

$$\cos\theta = \frac{|\vec{n}_1 \cdot \vec{k}|}{\|\vec{n}_1\|\,\|\vec{k}\|} = \frac{|2|}{3 \cdot 1} = \frac{2}{3}, \qquad \theta \approx 48.2^\circ.$$

6.6 Revision Grid: Choosing the Right Tool

Given	Tool	Result
Point A + normal $\vec{n}$	$\vec{n} \cdot \vec{AM} = 0$	Cartesian equation of plane
Three non-collinear points	$\vec{n} = \vec{AB} \wedge \vec{AC}$	Equation of the plane
Distance point–plane	$d = \dfrac{\lvert ax_0 + by_0 + cz_0 + d \rvert}{\sqrt{a^2 + b^2 + c^2}}$	Numerical distance
Angle between two planes	$\cos\theta = \dfrac{\lvert \vec{n}_1 \cdot \vec{n}_2 \rvert}{\lVert \vec{n}_1 \rVert \lVert \vec{n}_2 \rVert}$	Acute angle
Angle line–plane	$\sin\theta = \dfrac{\lvert \vec{u} \cdot \vec{n} \rvert}{\lVert \vec{u} \rVert \lVert \vec{n} \rVert}$	Acute angle
Line meets plane	Substitute parametric line into plane	Intersection point
Two planes meet	Solve system / $\vec{u} = \vec{n}_1 \wedge \vec{n}_2$	Line of intersection
Plane through a line of intersection	Pencil $\Pi_1 + \lambda\Pi_2 = 0$	Family; fix λ by extra condition



Summary flowchart: Given → Tool → Result for the main problem types of the chapter.

KEY POINTS **Key points of the section**

- The pencil of planes through $\Delta = \Pi_1 \cap \Pi_2$ is $\Pi_1 + \lambda\Pi_2 = 0$; it contains every plane through Δ except Π_2.
- Fix λ by the extra condition: substitute a point, use $\vec{n} \cdot (\vec{n}_1 + \lambda\vec{n}_2) = 0$ for perpendicularity, or the distance formula for a distance condition.
- Three planes generally meet at one point: solve the 3×3 system, then build the required plane.
- Bisector planes come from equal perpendicular distances (two solutions, + and −).

- In synthesis problems, read each part separately and pick the tool from the revision grid.

EXERCISE 6 **Practice**

1. Find the plane through the line of intersection of $x + y - z = 2$ and $2x - y + z = 1$ passing through the origin.
2. Find the planes of the pencil of Question 1 which are at distance $\sqrt{6}$ from the point $(2, 2, 2)$.
3. Find the bisector planes of $3x + 4y + 1 = 0$ and $4x - 3y + 2 = 0$ (in 3D, with z absent).
4. Show that the planes $x + y + z = 6$, $x - y + z = 2$ and $2x + y - z = 3$ meet at a single point, and find the plane through this point perpendicular to the line $\frac{x}{1} = \frac{y}{-1} = \frac{z}{2}$.

SOLUTION **Exercises 1–4**

1. Pencil: $(x + y - z - 2) + \lambda(2x - y + z - 1) = 0$. Origin: $-2 + \lambda(-1) = 0$, so $\lambda = -2$. Substituting: $(x + y - z - 2) - 2(2x - y + z - 1) = 0$, i.e. $-3x + 3y - 3z = 0$, or $x - y + z = 0$.
2. Combined plane: $(1 + 2\lambda)x + (1 - \lambda)y + (-1 + \lambda)z - (2 + \lambda) = 0$. Distance from $(2, 2, 2)$:

$$\frac{|2(1 + 2\lambda) + 2(1 - \lambda) + 2(-1 + \lambda) - (2 + \lambda)|}{\sqrt{(1 + 2\lambda)^2 + (1 - \lambda)^2 + (-1 + \lambda)^2}} = \frac{|3\lambda|}{\sqrt{6\lambda^2 + 3}} = \sqrt{6}.$$

Squaring: $9\lambda^2 = 6(6\lambda^2 + 3)$, i.e. $9\lambda^2 = 36\lambda^2 + 18$, which gives $-27\lambda^2 = 18$: no real solution. Hence no plane of the pencil is at distance $\sqrt{6}$ from that point — a useful reminder that a condition may have no solution.

3. Equal distances: $\frac{|3x + 4y + 1|}{5} = \frac{|4x - 3y + 2|}{5}$, so $3x + 4y + 1 = \pm(4x - 3y + 2)$. With +: $x - 7y + 1 = 0$. With −: $7x + y + 3 = 0$. These are the two bisector planes (both parallel to the z-axis).
4. The three planes are exactly those of the `exemplebox` above. We already found their unique intersection point $I\left(\frac{5}{3}, 2, \frac{7}{3}\right)$. The line $\frac{x}{1} = \frac{y}{-1} = \frac{z}{2}$ has direction vector $\vec{u} = (1, -1, 2)$. The required plane has normal $\vec{n} = \vec{u} = (1, -1, 2)$ and passes through I. Its equation is

$$1\left(x - \tfrac{5}{3}\right) - 1(y - 2) + 2\left(z - \tfrac{7}{3}\right) = 0 \implies x - \tfrac{5}{3} - y + 2 + 2z - \tfrac{14}{3} = 0 \implies x - y + 2z - \tfrac{13}{3} = 0,$$

or, clearing denominators, $3x - 3y + 6z - 13 = 0$.

DID YOU KNOW?

Pencils beyond the GCE

Pencils are a first taste of linear families of geometric objects, an idea that runs through university mathematics: pencils of lines, of circles, of conics. The single parameter λ organises an infinity of objects into one simple algebraic family — a beautiful example of algebra taming geometry.

7 INTEGRATION ACTIVITY

8 ACTIVITY OF INTEGRATION: DESIGNING A SCHOOL ROOF IN BAMENDA

8.1 Aims of the Activity

This integration activity closes the chapter on vectors and planes by mobilising, in a single authentic task, all the competencies developed in Lessons 42 to 51. By the end of the activity, each group of learners should be able to:

- determine the **Cartesian equation of a plane** through three given non-collinear points, using a normal vector obtained from a cross product;
- verify that a line lies in a plane and obtain the **vector equation of the line of intersection** of two planes;
- compute the **angle between two planes** from their normal vectors, and interpret the result (acute angle versus interior dihedral angle) in a real design specification;
- apply the formula for the **perpendicular distance from a point to a plane** in a concrete measurement problem;
- find the **point of intersection of a line and a plane** by substituting a parametric line into a Cartesian equation;
- communicate a complete mathematical argument on a poster, with a clear three-dimensional sketch, as required in the problem-solving and presentation component of the GCE Advanced Level examination.

KEY POINTS **Skills Mobilised**

Cross product and normal vectors; Cartesian and vector equations of planes; line of intersection of two planes; angle between two planes; distance from a point to a plane; intersection of a line with a plane; critical interpretation of a numerical result against a specification.

8.2 Situation

Designing a School Roof in Bamenda.

A contractor in Bamenda has been awarded the construction of the roof of a new classroom block for a secondary school. The roof consists of two rectangular metal panels joined along a horizontal **ridge line**. The architect has supplied a three-dimensional coordinate model (units in metres), with the x-axis along the ridge, the y-axis across the building and the z-axis vertical.

Panel 1 (the lower, gentler slope) passes through the points

$$A(0,0,4), \qquad B(6,0,4), \qquad C(6,4,2).$$

Panel 2 (the steeper slope) passes through the points

$$B(6,0,4), \qquad D(12,0,4), \qquad E(12,4,6).$$

The two panels must meet along the segment BD, which is the ridge line of the roof.

The architect's specification booklet imposes two checks before the panels can be ordered from the workshop:

1. the interior angle between the two panels (the angle measured inside the building, below the ridge) must be 120°;
2. a concrete pillar whose top is at the point $P(3,2,0)$ must just reach Panel 1, so its required height must be computed exactly.

Finally, an electric cable is to be run in a straight line along

$$\vec{r} = (\vec{i} + 6\vec{k}) + \lambda(\vec{i} + \vec{j} - \vec{k}),$$

and the electrician needs the exact point where this cable pierces Panel 1 in order to fix a junction box there.

Each group of learners plays the role of the contractor's technical team: carry out all the verifications, decide whether the roof design satisfies the architect's specification, and present the full solution on a poster including a three-dimensional sketch of the roof.

WARNING **Units and Interpretation**

All coordinates are in metres. Angles computed from normal vectors using the modulus formula are always acute (between 0° and 90°): deciding whether the interior angle of the roof equals this acute angle or its supplement is part of the modelling work.

8.3 Tasks

Task 1: Equation of each panel.

1. For Panel 1, compute the vectors $\vec{AB}$ and $\vec{AC}$, then the cross product $\vec{AB}\times\vec{AC}$. Deduce a normal vector $\vec{n}_1$ and the Cartesian equation of Panel 1.
2. In the same way, using $\vec{BD}$ and $\vec{BE}$, find a normal vector $\vec{n}_2$ and the Cartesian equation of Panel 2.

Task 2: The ridge line.

3. Show that every point of the segment BD satisfies the equations of both panels. What does this prove geometrically?
4. Give a vector equation of the line of intersection of the two panels (the ridge line).

Task 3: The angle between the panels.

5. Compute the acute angle θ between the two panels using $\cos\theta = \dfrac{|\vec{n}_1 \cdot \vec{n}_2|}{|\vec{n}_1|\,|\vec{n}_2|}$.
6. Deduce the interior angle of the roof below the ridge. Does the design satisfy the architect's specification of 120°? Write one sentence of advice to the contractor.

Task 4: The pillar.

7. Compute the perpendicular distance from $P(3, 2, 0)$ to Panel 1. Hence state the exact height the pillar must have.

Task 5: The cable.

8. Find the value of λ for which the cable line meets Panel 1, and give the coordinates of the piercing point. Verify that this point lies within the rectangular panel (check the ranges of x and y).

Task 6: Poster. Present all results on a poster with a labelled 3D sketch showing the two panels, the ridge line, the pillar and the cable.

8.4 Model Answer

Task 1. Equations of the Panels.

Panel 1.

$$\vec{AB} = \begin{pmatrix} 6 \\ 0 \\ 0 \end{pmatrix}, \qquad \vec{AC} = \begin{pmatrix} 6 \\ 4 \\ -2 \end{pmatrix}.$$

$$\vec{n}_1 = \vec{AB} \times \vec{AC} = \begin{pmatrix} 0\cdot(-2) - 0\cdot 4 \\ 0\cdot 6 - 6\cdot(-2) \\ 6\cdot 4 - 0\cdot 6 \end{pmatrix} = \begin{pmatrix} 0 \\ 12 \\ 24 \end{pmatrix} \parallel \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}.$$

A Cartesian equation is $0\cdot x + 1\cdot y + 2\cdot z = d$. Substituting $A(0, 0, 4)$ gives $d = 8$. Hence

$$\boxed{\text{Panel 1: } y + 2z = 8.}$$

Panel 2.

$$\vec{BD} = \begin{pmatrix} 6 \\ 0 \\ 0 \end{pmatrix}, \qquad \vec{BE} = \begin{pmatrix} 6 \\ 4 \\ 2 \end{pmatrix}, \qquad \vec{n}_2 = \vec{BD} \times \vec{BE} = \begin{pmatrix} 0\cdot 2 - 0\cdot 4 \\ 0\cdot 6 - 6\cdot 2 \\ 6\cdot 4 - 0\cdot 6 \end{pmatrix} = \begin{pmatrix} 0 \\ -12 \\ 24 \end{pmatrix} \parallel \begin{pmatrix} 0 \\ -1 \\ 2 \end{pmatrix}.$$

A Cartesian equation is $0\cdot x - 1\cdot y + 2\cdot z = d$. Substituting $B(6, 0, 4)$ gives $d = 8$. Hence

$$\boxed{\text{Panel 2: } -y + 2z = 8.}$$

Task 2. The Ridge Line.

3. A general point of the segment BD has coordinates $(x, 0, 4)$ with $6 \le x \le 12$.

- Panel 1: $0 + 2(4) = 8$ ✓.
- Panel 2: $-0 + 2(4) = 8$ ✓.

Every point of BD satisfies both equations; therefore the whole segment BD lies in both planes. Geometrically, this proves that the two panels intersect exactly along the ridge line BD, as required by the design.

4. Using point $B(6, 0, 4)$ and direction vector $\vec{i} = (1, 0, 0)$, a vector equation of the ridge line is

$$\vec{r} = \begin{pmatrix} 6 \\ 0 \\ 4 \end{pmatrix} + t \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \qquad t \in \mathbb{R}.$$

Task 3. Angle Between the Panels.

5. With simplified normal vectors $\vec{n}_1 = (0, 1, 2)$ and $\vec{n}_2 = (0, -1, 2)$:

$$\vec{n}_1 \cdot \vec{n}_2 = 0 \cdot 0 + 1 \cdot (-1) + 2 \cdot 2 = 3, \qquad |\vec{n}_1| = |\vec{n}_2| = \sqrt{0^2 + 1^2 + 2^2} = \sqrt{5}.$$

$$\cos\theta = \frac{|\vec{n}_1 \cdot \vec{n}_2|}{|\vec{n}_1|\,|\vec{n}_2|} = \frac{|3|}{\sqrt{5} \cdot \sqrt{5}} = \frac{3}{5} \quad \Longrightarrow \quad \theta = \arccos\left(\frac{3}{5}\right) \approx 53.13^\circ.$$

6. The acute angle between the planes is $\theta \approx 53.13^\circ$. The normals $\vec{n}_1$ and $\vec{n}_2$ both have positive z-components (pointing generally upward), so they point to the same side of the ridge line (the exterior of the roof). Consequently, the interior angle of the roof (measured inside the building, below the ridge) is the supplement of θ:

$$\text{Interior angle} = 180^\circ - \theta \approx 180^\circ - 53.13^\circ = 126.87^\circ.$$

Conclusion: The design does not satisfy the architect's specification of 120°; the interior angle is about 6.9° too large (the roof is too open). The contractor should ask the architect to adjust the slopes (for example by lowering point C or raising point E) before ordering the panels.

Task 4. The Pillar Height.

7. Panel 1 has equation $y + 2z - 8 = 0$. The perpendicular distance from $P(3, 2, 0)$ to this plane is

$$d = \frac{|\,2 + 2(0) - 8\,|}{\sqrt{0^2 + 1^2 + 2^2}} = \frac{6}{\sqrt{5}} = \frac{6\sqrt{5}}{5}\ \text{m} \approx 2.68\ \text{m}.$$

The pillar must be exactly $\dfrac{6\sqrt{5}}{5}$ m tall.

Task 5. The Cable Piercing Point.

8. The cable line is given in parametric form:

$$x = 1 + \lambda, \quad y = \lambda, \quad z = 6 - \lambda.$$

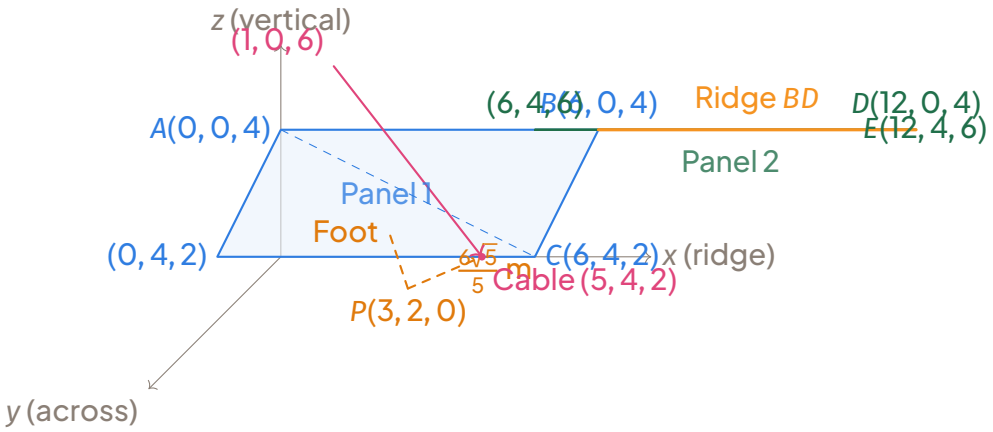
Substitute into the equation of Panel 1 ($y + 2z = 8$):

$$\lambda + 2(6 - \lambda) = 8 \implies 12 - \lambda = 8 \implies \lambda = 4.$$

The piercing point is

$$(1 + 4,\ 4,\ 6 - 4) = (5,\ 4,\ 2).$$

Panel 1 is the rectangle with vertices $A(0, 0, 4)$, $B(6, 0, 4)$, $C(6, 4, 2)$ and the fourth vertex $(0, 4, 2)$. Its projection on the xy-plane covers $0 \le x \le 6$, $0 \le y \le 4$. The point $(5, 4, 2)$ satisfies $0 \le 5 \le 6$ and $y = 4$ (on the boundary). It lies on the lower edge of Panel 1 (the edge through C parallel to the ridge). The junction box will be fixed at $(5, 4, 2)$, on that lower edge.



Sketch of the two roof panels meeting along the ridge line BD. The pillar from P meets Panel 1 perpendicularly; the cable pierces Panel 1 at $(5, 4, 2)$ on its lower edge. Axes show the coordinate orientation.

WARNING **What the Examiners Expect**

In such synthesis tasks, marks are awarded for:

- correct computation of cross products and normal vectors;
- correct Cartesian equations of planes (simplified integer coefficients preferred);
- correct use of the modulus in the angle formula $\cos\theta = \frac{|\vec{n}_1 \cdot \vec{n}_2|}{|\vec{n}_1||\vec{n}_2|}$;
- a clear justification that the interior angle is the supplement of the acute angle between the normals (by considering the orientation of the normals relative to the interior of the roof);
- exact values left in surd form (e.g. $\frac{6\sqrt{5}}{5}$) rather than decimal approximations;
- verification that the intersection point lies within the physical boundaries of the panel;

- a concluding sentence that answers the practical question directly (here: the 120° specification is not met).

9 METHODS AND WORKED EXAMPLES

10 EQUATIONS OF A PLANE: VECTOR, PARAMETRIC AND CARTESIAN FORMS

DEFINITION **Plane in space**

A **plane** Π is a flat, two-dimensional surface extending infinitely in all directions. It is uniquely determined by:

- a point A and a non-zero normal vector $\vec{n}$ (perpendicular to the plane), or
- three non-collinear points A, B, C, or
- a point A and two non-parallel direction vectors $\vec{u}, \vec{v}$ lying in the plane.

PROPERTY **Vector equation of a plane (normal form)**

Let Π be a plane with normal vector $\vec{n} \neq \vec{0}$ passing through point A with position vector $\vec{a}$. A point P with position vector $\vec{r}$ lies on Π if and only if $\vec{AP}$ is perpendicular to $\vec{n}$, i.e.

$$(\vec{r} - \vec{a}) \cdot \vec{n} = 0 \quad \Longleftrightarrow \quad \vec{r} \cdot \vec{n} = \vec{a} \cdot \vec{n}.$$

If $\vec{n} = a\vec{i} + b\vec{j} + c\vec{k}$ and $\vec{a} = x_0\vec{i} + y_0\vec{j} + z_0\vec{k}$, this becomes

$$\vec{r} \cdot (a\vec{i} + b\vec{j} + c\vec{k}) = ax_0 + by_0 + cz_0.$$

PROPERTY **Cartesian equation of a plane**

Writing $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$ in the vector equation gives the **Cartesian equation**

$$ax + by + cz = d, \qquad \text{where } d = ax_0 + by_0 + cz_0.$$

The coefficients (a, b, c) are the components of a normal vector $\vec{n}$. Conversely, any equation of the form $ax + by + cz = d$ with $(a, b, c) \neq (0, 0, 0)$ represents a plane with normal $\vec{n} = a\vec{i} + b\vec{j} + c\vec{k}$.

PROPERTY **Parametric vector equation of a plane**

If Π passes through $A(\vec{a})$ and contains two non-parallel direction vectors $\vec{u}$ and $\vec{v}$, every point P on Π can be written as

$$\vec{r} = \vec{a} + \lambda\vec{u} + \mu\vec{v}, \qquad \lambda, \mu \in \mathbb{R}.$$

In Cartesian parametric form:

$$\begin{cases} x = x_0 + \lambda u_1 + \mu v_1, \\ y = y_0 + \lambda u_2 + \mu v_2, \\ z = z_0 + \lambda u_3 + \mu v_3. \end{cases}$$

METHOD **From a point and a normal vector to the Cartesian equation**

To find the equation of a plane Π passing through $A(x_0, y_0, z_0)$ with normal vector $\vec{n} = a\vec{i} + b\vec{j} + c\vec{k}$:

1. Write the vector equation $\vec{r} \cdot \vec{n} = \vec{a} \cdot \vec{n}$.
2. Compute the constant $d = \vec{a} \cdot \vec{n} = ax_0 + by_0 + cz_0$.
3. Replace $\vec{r}$ by $x\vec{i} + y\vec{j} + z\vec{k}$ to obtain $ax + by + cz = d$.
4. Reflex: the coefficients of x, y, z are always the components of a normal vector.
5. Check: substitute the coordinates of A; the equality must hold.

WORKED EXAMPLE 23 **Equation of a plane given a point and a normal**

Find the vector and Cartesian equations of the plane passing through $A(2, -1, 3)$ and perpendicular to $\vec{n} = 3\vec{i} + 2\vec{j} - \vec{k}$.

SOLUTION

Step 1: vector equation.

$$\vec{r} \cdot (3\vec{i} + 2\vec{j} - \vec{k}) = \vec{a} \cdot \vec{n}.$$

Step 2: compute the constant.

$$\vec{a} \cdot \vec{n} = (2)(3) + (-1)(2) + (3)(-1) = 6 - 2 - 3 = 1.$$

Vector equation: $\vec{r} \cdot (3\vec{i} + 2\vec{j} - \vec{k}) = 1$.
Step 3: Cartesian form.

$$\boxed{3x + 2y - z = 1}.$$

Check: at $A(2, -1, 3)$: $3(2) + 2(-1) - 3 = 6 - 2 - 3 = 1$. ✓

WORKED EXAMPLE 24 **Converting between forms**

A plane has parametric equations $x = 1+2\lambda-\mu$, $y = 2-\lambda+3\mu$, $z = -1+\lambda+2\mu$. Find its Cartesian equation.

SOLUTION

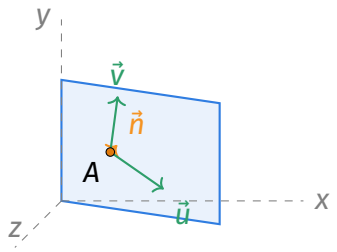
Step 1: identify $\vec{a}, \vec{u}, \vec{v}$.

$$\vec{a} = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}, \quad \vec{u} = \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}, \quad \vec{v} = \begin{pmatrix} -1 \\ 3 \\ 2 \end{pmatrix}.$$

Step 2: normal vector $\vec{n} = \vec{u} \times \vec{v}$.

$$\vec{n} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & -1 & 1 \\ -1 & 3 & 2 \end{vmatrix} = \vec{i}((-1)(2)-(1)(3)) - \vec{j}((2)(2)-(1)(-1)) + \vec{k}((2)(3)-(-1)(-1)) = \begin{pmatrix} -5 \\ -5 \\ 5 \end{pmatrix}.$$

Simplify to $\vec{n} = (1, 1, -1)$.
Step 3: constant $d = \vec{n} \cdot \vec{a} = 1 + 2 + 1 = 4$. Cartesian equation: $\boxed{x + y - z = 4}$.



Plane with normal $\vec{n}$ and direction vectors $\vec{u}, \vec{v}$

11 PLANE THROUGH THREE POINTS AND SPECIAL PLANES

METHOD **Plane through three non-collinear points**

Given $A(x_1, y_1, z_1)$, $B(x_2, y_2, z_2)$, $C(x_3, y_3, z_3)$:

1. Form two direction vectors $\vec{AB}$ and $\vec{AC}$.

2. Check collinearity: if $\vec{AB} \times \vec{AC} = \vec{0}$, the points are collinear — infinitely many planes contain them; no unique plane exists.
3. Cartesian route: compute normal $\vec{n} = \vec{AB} \times \vec{AC}$, then $d = \vec{n} \cdot \vec{a}$ using any of the three points.
4. Parametric route: $\vec{r} = \vec{a} + \lambda\vec{AB} + \mu\vec{AC}$.
5. Check: verify all three points satisfy the final Cartesian equation.

WORKED EXAMPLE 25 **Plane through three points**

Find (a) the Cartesian equation and (b) the parametric vector equation of the plane through $A(1, 0, 2)$, $B(2, 1, 1)$ and $C(0, 3, 1)$.

SOLUTION

Step 1: direction vectors.

$$\vec{AB} = \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}, \qquad \vec{AC} = \begin{pmatrix} -1 \\ 3 \\ -1 \end{pmatrix}.$$

Step 2: normal vector.

$$\vec{n} = \vec{AB} \times \vec{AC} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 1 & -1 \\ -1 & 3 & -1 \end{vmatrix} = \vec{i}((1)(-1) - (-1)(3)) - \vec{j}((1)(-1) - (-1)(-1)) + \vec{k}((1)(3) - (1)(-1)) = \begin{pmatrix} 2 \\ 2 \\ 4 \end{pmatrix}.$$

Simplify to $\vec{n} = (1, 1, 2)$.
Step 3: constant term. Using $A(1, 0, 2)$: $d = 1(1) + 1(0) + 2(2) = 5$.
Cartesian equation: $\boxed{x + y + 2z = 5}$.
Check: B: $2 + 1 + 2 = 5$ ✓; C: $0 + 3 + 2 = 5$ ✓.
Parametric form:

$$\vec{r} = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix} + \mu \begin{pmatrix} -1 \\ 3 \\ -1 \end{pmatrix}, \quad \lambda, \mu \in \mathbb{R},$$

i.e. $x = 1 + \lambda - \mu$, $y = \lambda + 3\mu$, $z = 2 - \lambda - \mu$.

WARNING **Collinear points**

If $\vec{AB} \times \vec{AC} = \vec{0}$, the points are collinear and do not define a unique plane. Always test before computing. Also, if you rescale $\vec{n}$, recompute d using the same point.

KEY POINTS **Special planes**

- Plane through origin: $d = 0$, equation $ax + by + cz = 0$.
- Plane parallel to xy-plane: $z = k$ (normal $\vec{k}$).
- Plane parallel to yz-plane: $x = k$ (normal $\vec{i}$).
- Plane parallel to xz-plane: $y = k$ (normal $\vec{j}$).
- Plane parallel to a given plane $ax + by + cz = d_1$: same normal, equation $ax + by + cz = d_2$.

WORKED EXAMPLE 26 **Special planes**

Find the equation of the plane through $P(2, -1, 3)$ parallel to the plane $2x - 3y + z = 7$.

SOLUTION

Parallel planes have the same normal $\vec{n} = (2, -3, 1)$. The required plane has equation $2x - 3y + z = d$. Substitute P: $d = 2(2) - 3(-1) + 3 = 4 + 3 + 3 = 10$. Equation: $\boxed{2x - 3y + z = 10}$.

12 PERPENDICULAR DISTANCE FROM A POINT TO A PLANE

PROPERTY **Perpendicular distance formula**

The perpendicular distance δ from point $P(x_1, y_1, z_1)$ to the plane $\Pi : \ ax + by + cz = d$ is

$$\delta = \frac{|ax_1 + by_1 + cz_1 - d|}{\sqrt{a^2 + b^2 + c^2}}.$$

The numerator is the absolute value of substituting P into the left-hand side minus d; the denominator is $|\vec{n}|$.

METHOD **Distance and foot of the perpendicular**

To find the perpendicular distance from $P(x_1, y_1, z_1)$ to $\Pi : \ ax + by + cz = d$:

1. Ensure the equation is in the form $ax + by + cz - d = 0$.

2. Apply $\delta = \dfrac{|ax_1 + by_1 + cz_1 - d|}{\sqrt{a^2 + b^2 + c^2}}$.
3. Foot of the perpendicular F: parametrise the line through P along $\vec{n}$:

$$x = x_1 + at, \quad y = y_1 + bt, \quad z = z_1 + ct.$$

   Substitute into the plane equation, solve for t, then find F.
4. Reflex: if P lies on Π, the numerator is zero — quick consistency check.

WORKED EXAMPLE 27 **Distance and foot of the perpendicular**

Find the perpendicular distance from $P(1, 2, 3)$ to $\Pi : 2x - y + 2z = 5$. Hence find the foot F.

SOLUTION

Distance. $a = 2, b = -1, c = 2, d = 5$.

$$\delta = \frac{|2(1) - (2) + 2(3) - 5|}{\sqrt{2^2 + (-1)^2 + 2^2}} = \frac{|2 - 2 + 6 - 5|}{\sqrt{9}} = \frac{1}{3}.$$

Distance: $\boxed{\frac{1}{3}\text{ unit}}$.

Foot of the perpendicular. Normal $\vec{n} = (2, -1, 2)$. Line through P:

$$x = 1 + 2t, \quad y = 2 - t, \quad z = 3 + 2t.$$

Substitute into $2x - y + 2z = 5$:

$$2(1 + 2t) - (2 - t) + 2(3 + 2t) = 5 \implies 2 + 4t - 2 + t + 6 + 4t = 5 \implies 9t = -1 \implies t = -\frac{1}{9}.$$

$$F = \left(1 - \frac{2}{9},\ 2 + \frac{1}{9},\ 3 - \frac{2}{9}\right) = \boxed{\left(\frac{7}{9}, \frac{19}{9}, \frac{25}{9}\right)}.$$

Check: $|PF| = |t|\,|\vec{n}| = \frac{1}{9} \times 3 = \frac{1}{3}$ ✓.

WORKED EXAMPLE 28 **Distance between parallel planes**

Find the distance between the parallel planes $\Pi_1 : 2x - y + 2z = 5$ and $\Pi_2 : 2x - y + 2z = 11$.

SOLUTION

Pick any point on Π_1, e.g. $A(0, 0, 2.5)$ (since $2(0) - 0 + 2(2.5) = 5$). Distance from A to Π_2:

$$\delta = \frac{|2(0) - 0 + 2(2.5) - 11|}{\sqrt{9}} = \frac{|5 - 11|}{3} = \frac{6}{3} = 2.$$

Alternatively, for parallel planes $ax + by + cz = d_1$ and $ax + by + cz = d_2$:

$$\delta = \frac{|d_2 - d_1|}{\sqrt{a^2 + b^2 + c^2}} = \frac{|11 - 5|}{3} = 2.$$

Distance: $\boxed{2 \text{ units}}$.

13 ANGLES: BETWEEN TWO PLANES AND BETWEEN A LINE AND A PLANE

PROPERTY **Angle between two planes**

The angle θ between two planes is the acute angle between their normal vectors $\vec{n}_1$ and $\vec{n}_2$:

$$\cos\theta = \frac{|\vec{n}_1 \cdot \vec{n}_2|}{|\vec{n}_1|\,|\vec{n}_2|}, \qquad 0^\circ \le \theta \le 90^\circ.$$

- Planes parallel $\Leftrightarrow \vec{n}_1 \parallel \vec{n}_2 \Leftrightarrow \theta = 0^\circ$.
- Planes perpendicular $\Leftrightarrow \vec{n}_1 \cdot \vec{n}_2 = 0 \Leftrightarrow \theta = 90^\circ$.

PROPERTY **Angle between a line and a plane**

Let a line have direction vector $\vec{d}$ and a plane have normal $\vec{n}$. The angle θ between the line and the plane is the complement of the angle φ between $\vec{d}$ and $\vec{n}$:

$$\sin\theta = \frac{|\vec{d} \cdot \vec{n}|}{|\vec{d}|\,|\vec{n}|}, \qquad 0^\circ \le \theta \le 90^\circ.$$

- Line parallel to plane (or lies in it) $\Leftrightarrow \vec{d} \cdot \vec{n} = 0 \Leftrightarrow \theta = 0^\circ$.
- Line perpendicular to plane $\Leftrightarrow \vec{d} \parallel \vec{n} \Leftrightarrow \theta = 90^\circ$.

METHOD **Angles involving planes**

1. Two planes: identify normals $\vec{n}_1, \vec{n}_2$ from Cartesian equations; compute $\cos\theta = \dfrac{|\vec{n}_1 \cdot \vec{n}_2|}{|\vec{n}_1||\vec{n}_2|}$.
2. Line and plane: identify direction $\vec{d}$ of line and normal $\vec{n}$ of plane; compute $\sin\theta = \dfrac{|\vec{d} \cdot \vec{n}|}{|\vec{d}||\vec{n}|}$.
3. Reflexes: parallel ⇔ dot product zero (line–plane) or normals parallel (plane–plane); perpendicular ⇔ dot product zero (plane–plane) or direction parallel to normal (line–plane).
4. Trap: line–plane angle uses **sine**, not cosine.

WORKED EXAMPLE 29 **Two angle computations**

(a) Find the acute angle between $\Pi_1 : x + 2y + 2z = 3$ and $\Pi_2 : 2x - 2y + z = 6$.
(b) Find the angle between the line $\vec{r} = (\vec{i} + 2\vec{k}) + t(2\vec{i} + \vec{j} - 2\vec{k})$ and Π_1.

SOLUTION

(a) Normals: $\vec{n}_1 = (1, 2, 2)$, $\vec{n}_2 = (2, -2, 1)$.

$$\vec{n}_1 \cdot \vec{n}_2 = 2 - 4 + 2 = 0 \implies \theta = 90^\circ.$$

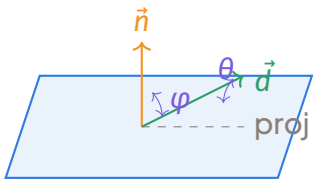
Planes are perpendicular: $\boxed{\theta = 90^\circ}$.
(b) Direction $\vec{d} = (2, 1, -2)$; normal $\vec{n} = (1, 2, 2)$.

$$\vec{d} \cdot \vec{n} = 2 + 2 - 4 = 0 \implies \sin\theta = 0 \implies \theta = 0^\circ.$$

The line is parallel to the plane. Check if it lies in the plane: point $(1, 0, 2)$ gives $1+0+4 = 5 \neq 3$, so line is parallel and distinct. Angle: $\boxed{0^\circ}$.

Non-degenerate practice: line with $\vec{d} = (1, 1, 1)$ and Π_1.

$$\sin\theta = \frac{|1 + 2 + 2|}{\sqrt{3}\sqrt{9}} = \frac{5}{3\sqrt{3}} = \frac{5\sqrt{3}}{9} \approx 0.9623 \implies \theta \approx 74.2^\circ.$$



Plane Π

Angle θ between line (direction $\vec{d}$) and plane (normal $\vec{n}$): $\theta = 90^\circ - \varphi$, $\sin\theta = |\vec{d}\cdot\vec{n}|/(|\vec{d}||\vec{n}|)$

14 INTERSECTION OF A LINE WITH A PLANE AND OF TWO PLANES

METHOD **Intersection of a line and a plane**

Given line $\vec{r} = \vec{a} + t\vec{d}$ and plane $ax + by + cz = d$:

1. Write parametric coordinates $x(t), y(t), z(t)$.
2. Substitute into plane equation → linear equation in t.
3. Unique solution t_0: line pierces plane at $\vec{a} + t_0\vec{d}$.
4. No solution (e.g. $0 = 5$): line parallel to plane, no intersection.
5. Identity (e.g. $0 = 0$): line lies entirely in the plane.
6. Quick check: compute $\vec{d}\cdot\vec{n}$. If zero, line ∥ plane — then test if $\vec{a}$ satisfies plane equation to distinguish cases 4 and 5.

WORKED EXAMPLE 30 **Line meeting a plane**

Find the intersection of $\vec{r} = \begin{pmatrix}1\\-1\\2\end{pmatrix} + t\begin{pmatrix}2\\1\\-1\end{pmatrix}$ with $x - 2y + 3z = 9$.

SOLUTION

Parametric: $x = 1 + 2t$, $y = -1 + t$, $z = 2 - t$.
Substitute: $(1+2t) - 2(-1+t) + 3(2-t) = 9 \implies 1+2t+2-2t+6-3t = 9 \implies 9-3t = 9 \implies t = 0$.
Point: $\boxed{(1, -1, 2)}$.

Check: $\vec{d} \cdot \vec{n} = (2)(1) + (1)(-2) + (-1)(3) = -3 \neq 0 \rightarrow$ unique intersection expected. ✓

WORKED EXAMPLE 31 **Line parallel to plane**

Determine whether the line $\vec{r} = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} + t \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}$ meets the plane $x - 2y + z = 4$.

SOLUTION

$\vec{d} = (1, -2, 1)$, $\vec{n} = (1, -2, 1) \rightarrow \vec{d} \cdot \vec{n} = 1+4+1 = 6 \neq 0$? Wait: $\vec{d} \cdot \vec{n} = 1(1)+(-2)(-2)+1(1) = 1+4+1 = 6 \neq 0$. So line is not parallel to plane; it intersects.

Let's do a parallel case: line $\vec{r} = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} + t \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ with plane $x + y + z = 3$. $\vec{d} \cdot \vec{n} = 1 + 1 + 1 = 3 \neq 0$ — still intersects.

True parallel: line $\vec{r} = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} + t \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}$ with plane $x + y + z = 3$. $\vec{d} \cdot \vec{n} = 1 - 1 + 0 = 0 \rightarrow$ parallel. Point $(1, 2, -1)$: $1 + 2 - 1 = 2 \neq 3 \rightarrow$ no intersection.

METHOD **Line of intersection of two planes**

Given $\Pi_1 : a_1x + b_1y + c_1z = d_1$ and $\Pi_2 : a_2x + b_2y + c_2z = d_2$:

1. Compare normals $\vec{n}_1, \vec{n}_2$: if parallel, planes are parallel (no line, or coincident).
2. Direction of intersection line: $\vec{d} = \vec{n}_1 \times \vec{n}_2$ (perpendicular to both normals).
3. Find one point on the line: set one coordinate to a convenient value (often $z = 0$) and solve the 2×2 system. If that fails, try $x = 0$ or $y = 0$.
4. Write line as $\vec{r} = \vec{p} + t\vec{d}$.
5. Alternative: solve the system with one variable as parameter.

WORKED EXAMPLE 32 **Line of intersection**

Find the vector equation of the line of intersection of

$$\Pi_1 : x + y + z = 6, \qquad \Pi_2 : 2x - y + z = 3.$$

SOLUTION

Normals: $\vec{n}_1 = (1, 1, 1)$, $\vec{n}_2 = (2, -1, 1)$ — not parallel.
Direction:

$$\vec{d} = \vec{n}_1 \times \vec{n}_2 = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 1 & 1 \\ 2 & -1 & 1 \end{vmatrix} = \vec{i}(1 + 1) - \vec{j}(1 - 2) + \vec{k}(-1 - 2) = \begin{pmatrix} 2 \\ 1 \\ -3 \end{pmatrix}.$$

Point on line: set $z = 0$:

$$\begin{cases} x + y = 6 \\ 2x - y = 3 \end{cases} \implies 3x = 9 \implies x = 3,\ y = 3.$$

Point $P(3, 3, 0)$. Check: $3 + 3 + 0 = 6$ ✓; $6 - 3 + 0 = 3$ ✓.
Line equation:

$$\boxed{\vec{r} = \begin{pmatrix} 3 \\ 3 \\ 0 \end{pmatrix} + t \begin{pmatrix} 2 \\ 1 \\ -3 \end{pmatrix}, \quad t \in \mathbb{R}}.$$

WORKED EXAMPLE 33 **Intersection via parameter**

Find the line of intersection of $x + 2y - z = 4$ and $2x - y + 3z = 5$ by using z as parameter.

SOLUTION

Let $z = t$. System:

$$\begin{cases} x + 2y = 4 + t \\ 2x - y = 5 - 3t \end{cases}$$

Solve: from second, $y = 2x - 5 + 3t$. Substitute into first:

$$x + 2(2x - 5 + 3t) = 4 + t \implies 5x - 10 + 6t = 4 + t \implies 5x = 14 - 5t \implies x = \tfrac{14}{5} - t.$$

Then $y = 2(\tfrac{14}{5} - t) - 5 + 3t = \tfrac{28}{5} - 2t - 5 + 3t = \tfrac{3}{5} + t$. Parametric: $x = \tfrac{14}{5} - t$, $y = \tfrac{3}{5} + t$, $z = t$. Vector form: $\vec{r} = \begin{pmatrix} 14/5 \\ 3/5 \\ 0 \end{pmatrix} + t \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix}$.

15 PENCILS OF PLANES AND SYNTHESIS PROBLEMS

PROPERTY **Pencil of planes through the line of intersection**

All planes passing through the line of intersection of

$$\Pi_1 : a_1x + b_1y + c_1z = d_1, \qquad \Pi_2 : a_2x + b_2y + c_2z = d_2$$

are given by the one-parameter family

$$(a_1x + b_1y + c_1z - d_1) + \lambda(a_2x + b_2y + c_2z - d_2) = 0, \qquad \lambda \in \mathbb{R}.$$

This represents every plane of the pencil except Π_2 itself (which corresponds to $\lambda \to \infty$). To include Π_2, use the symmetric form $\mu\Pi_1 + \lambda\Pi_2 = 0$ with $(\mu, \lambda) \neq (0, 0)$.

METHOD **Plane of a pencil satisfying an extra condition**

1. Write the pencil: $\Pi_1 + \lambda\Pi_2 = 0$ (i.e. $(a_1x + b_1y + c_1z - d_1) + \lambda(a_2x + b_2y + c_2z - d_2) = 0$).
2. Use the extra condition to find λ:
   - Passes through point Q: substitute Q into the pencil equation.
   - Perpendicular to plane Π_3: combined normal $\vec{n} = \vec{n}_1 + \lambda\vec{n}_2$ must satisfy $\vec{n} \cdot \vec{n}_3 = 0$.
   - Parallel to a line with direction $\vec{d}$: $(\vec{n}_1 + \lambda\vec{n}_2) \cdot \vec{d} = 0$.
   - Parallel to a plane Π_3: $\vec{n}_1 + \lambda\vec{n}_2$ parallel to $\vec{n}_3$.
3. Substitute λ back and simplify.

WORKED EXAMPLE 34 **Plane of a pencil through a given point**

Find the plane through the line of intersection of $\Pi_1 : x + 2y - z = 4$ and $\Pi_2 : 2x - y + z = 1$ that also passes through $Q(1, 1, 4)$.

SOLUTION

Pencil: $(x + 2y - z - 4) + \lambda(2x - y + z - 1) = 0$.
Condition $Q \in \Pi$: $(1 + 2 - 4 - 4) + \lambda(2 - 1 + 4 - 1) = 0 \implies -5 + 4\lambda = 0 \implies \lambda = \frac{5}{4}$.
Substitute: $(x + 2y - z - 4) + \frac{5}{4}(2x - y + z - 1) = 0$. Multiply by 4:

$$4x + 8y - 4z - 16 + 10x - 5y + 5z - 5 = 0 \implies \boxed{14x + 3y + z = 21}.$$

Check: Q: $14+3+4=21$ ✓. Any point on the intersection line satisfies both Π_1 and Π_2, hence the combination.

WORKED EXAMPLE 35 **Plane of a pencil perpendicular to another plane**

Find the plane through the intersection of $\Pi_1 : x+y+z=1$ and $\Pi_2 : 2x-y+3z=4$ that is perpendicular to $\Pi_3 : x-2y+z=5$.

SOLUTION

Pencil: $(x+y+z-1)+\lambda(2x-y+3z-4)=0$. Combined normal: $\vec{n}=(1+2\lambda,\ 1-\lambda,\ 1+3\lambda)$. Π_3 normal: $\vec{n}_3=(1,-2,1)$. Perpendicular $\Rightarrow \vec{n}\cdot\vec{n}_3=0$:

$$(1+2\lambda)(1)+(1-\lambda)(-2)+(1+3\lambda)(1)=0$$

$$1+2\lambda-2+2\lambda+1+3\lambda=0 \implies 7\lambda=0 \implies \lambda=0.$$

Thus the required plane is Π_1 itself: $\boxed{x+y+z=1}$.

PROPERTY **Equation of planes through the point of intersection of two planes**

If two planes Π_1 and Π_2 intersect in a point (which occurs when we have three planes meeting at a point, or when considering the intersection of two planes with a third condition), the family of planes through that point is not a simple pencil. However, if two planes intersect in a line L, and we want planes through a specific point P on L, we use the pencil method with the condition $P \in \Pi$.
If the problem asks for planes through the point of intersection of three planes (i.e. the unique point where Π_1, Π_2, Π_3 meet), then any plane through that point has equation

$$\alpha(a_1x+b_1y+c_1z-d_1)+\beta(a_2x+b_2y+c_2z-d_2)+\gamma(a_3x+b_3y+c_3z-d_3)=0$$

with $(\alpha,\beta,\gamma)\neq(0,0,0)$. This is a two-parameter family.

WORKED EXAMPLE 36 **Planes through intersection point of three planes**

The planes $\Pi_1 : x+y+z=6$, $\Pi_2 : 2x-y+z=3$, $\Pi_3 : x+2y-z=2$ intersect at a unique point P. Find the equation of the plane through P parallel to the vector $\vec{v}=(1,1,1)$.

SOLUTION

Step 1: find P by solving the system.

$$\begin{cases} x + y + z = 6 & (1) \\ 2x - y + z = 3 & (2) \\ x + 2y - z = 2 & (3) \end{cases}$$

(1) + (3): $2x + 3y = 8 \Rightarrow x = 4 - \frac{3}{2}y$. (2) − (1): $x - 2y = -3 \Rightarrow x = 2y - 3$. Equate: $4 - \frac{3}{2}y = 2y - 3 \Rightarrow 7 = \frac{7}{2}y \Rightarrow y = 2$. Then $x = 1$, and from (1): $1 + 2 + z = 6 \Rightarrow z = 3$. Point $P(1, 2, 3)$.
Step 2: plane through P parallel to $\vec{v} = (1, 1, 1)$. A plane parallel to a vector means the vector lies in the plane (is a direction vector). We need another direction or a normal. Since only one direction is given, there are infinitely many such planes — they form a pencil around the line through P parallel to $\vec{v}$. General plane through P: $a(x - 1) + b(y - 2) + c(z - 3) = 0$ with normal $\vec{n} = (a, b, c)$. Condition $\vec{n} \cdot \vec{v} = 0$: $a + b + c = 0 \Rightarrow c = -a - b$. Equation: $a(x - 1) + b(y - 2) - (a + b)(z - 3) = 0$. This is a one-parameter family (e.g. set $a = 1$, $b = \lambda$).

METHOD **Plane defined by a point and two directions (synthesis)**

To find the plane through A parallel to lines with directions $\vec{d}_1$ and $\vec{d}_2$ (non-parallel):

1. Compute normal $\vec{n} = \vec{d}_1 \times \vec{d}_2$.
2. Form $ax + by + cz = d$ with $d = \vec{n} \cdot \vec{a}$.
3. If the plane must also contain a given line, verify: (i) line's direction $\perp \vec{n}$, (ii) a point of the line satisfies the plane equation.

WORKED EXAMPLE 37 **GCE-style synthesis problem**

Lines L_1 and L_2:

$$L_1 : \vec{r} = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + s\begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \qquad L_2 : \vec{r} = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} + t\begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}.$$

(a) Find the Cartesian equation of plane Π containing L_1 and parallel to L_2.
(b) Find the perpendicular distance from $B(2, 1, 0)$ on L_2 to Π, and interpret.

SOLUTION

(a) Π contains $\vec{d}_1 = (1, 2, 1)$ and is parallel to $\vec{d}_2 = (2, -1, 1)$. Normal: $\vec{n} = \vec{d}_1 \times \vec{d}_2 = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 2 & 1 \\ 2 & -1 & 1 \end{vmatrix} =$

$\vec{i}(2+1) - \vec{j}(1-2) + \vec{k}(-1-4) = \begin{pmatrix} 3 \\ 1 \\ -5 \end{pmatrix}$. Π contains $A(1,0,1)$: $d = 3(1) + 1(0) - 5(1) = -2$. Equation:

$\boxed{3x + y - 5z = -2}$.

Check: $\vec{d}_1 \cdot \vec{n} = 3 + 2 - 5 = 0$ ✓; $\vec{d}_2 \cdot \vec{n} = 6 - 1 - 5 = 0$ ✓; A: $3 + 0 - 5 = -2$ ✓.

(b) Distance from $B(2,1,0)$:

$$\delta = \frac{|3(2) + 1 - 5(0) + 2|}{\sqrt{9 + 1 + 25}} = \frac{|6 + 1 + 2|}{\sqrt{35}} = \frac{9}{\sqrt{35}} = \frac{9\sqrt{35}}{35} \approx 1.52.$$

Interpretation: L_2 is parallel to Π and $L_1 \subset \Pi$. Every point of L_2 is at the same distance from Π; this common distance $\frac{9}{\sqrt{35}}$ is the **shortest distance between the two skew lines** L_1 and L_2.

WORKED EXAMPLE 38 **Shortest distance between two skew lines**

Find the shortest distance between the skew lines

$$L_1 : \frac{x-1}{2} = \frac{y+1}{1} = \frac{z-2}{-1}, \qquad L_2 : \frac{x-3}{1} = \frac{y}{2} = \frac{z+1}{1}.$$

SOLUTION

Method: distance = distance from any point on L_2 to the plane containing L_1 parallel to L_2. L_1 point $A(1,-1,2)$, direction $\vec{d}_1 = (2,1,-1)$. L_2 point $B(3,0,-1)$, direction $\vec{d}_2 = (1,2,1)$. Normal to plane: $\vec{n} = \vec{d}_1 \times \vec{d}_2 = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & 1 & -1 \\ 1 & 2 & 1 \end{vmatrix} = \vec{i}(1+2) - \vec{j}(2+1) + \vec{k}(4-1) = \begin{pmatrix} 3 \\ -3 \\ 3 \end{pmatrix} \sim (1,-1,1)$. Plane through A: $1(x-1) - 1(y+1) + 1(z-2) = 0 \Rightarrow x - y + z = 4$. Distance from $B(3,0,-1)$ to this plane:

$$\delta = \frac{|3 - 0 - 1 - 4|}{\sqrt{3}} = \frac{2}{\sqrt{3}} = \frac{2\sqrt{3}}{3}.$$

Shortest distance: $\boxed{\frac{2\sqrt{3}}{3}}$.

16 ACTIVITIES OF INTEGRATION, ASSESSMENT AND REMEDIATION

EXERCISE 7 **Integration Exercise 1**

The planes Π_1 and Π_2 have equations $2x - y + 3z = 5$ and $x + 2y - z = 3$ respectively.

1. Find the acute angle between Π_1 and Π_2.
2. Find a vector equation of the line of intersection of Π_1 and Π_2.
3. Find the equation of the plane through the line of intersection of Π_1 and Π_2 which is perpendicular to the plane $x - y + 2z = 4$.

EXERCISE 8 **Integration Exercise 2**

A line L has equation $\vec{r} = \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix} + t\begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix}$. A plane Π has equation $3x - y + 2z = 7$.

1. Show that L is not parallel to Π and find the coordinates of the intersection point.
2. Find the acute angle between L and Π.
3. Find the coordinates of the foot of the perpendicular from the point $A(4, 1, 1)$ to Π.

EXERCISE 9 **Integration Exercise 3**

Three points are given: $A(1, 2, -1)$, $B(3, 0, 1)$, $C(-1, 1, 2)$.

1. Show that A, B, C are non-collinear.
2. Find the Cartesian equation of the plane Π through A, B, C.
3. Find the perpendicular distance from the origin to Π.
4. Find the equation of the plane through the line of intersection of Π and the plane $x + y + z = 0$ that passes through the point $D(2, 2, 2)$.

EXERCISE 10 **Integration Exercise 4 — GCE Style**

The line L_1 passes through $A(1, 2, 3)$ and is parallel to $\vec{d}_1 = 2\vec{i} - \vec{j} + \vec{k}$. The line L_2 passes through $B(4, -1, 2)$ and is parallel to $\vec{d}_2 = \vec{i} + 2\vec{j} - \vec{k}$.

1. Find the Cartesian equation of the plane Π that contains L_1 and is parallel to L_2.
2. Find the shortest distance between L_1 and L_2.
3. Find the coordinates of the points P on L_1 and Q on L_2 such that $\vec{PQ}$ is perpendicular to both lines.

SOLUTION **Integration Exercise 1**

1. Normals: $\vec{n}_1 = (2, -1, 3)$, $\vec{n}_2 = (1, 2, -1)$. $\vec{n}_1 \cdot \vec{n}_2 = 2 - 2 - 3 = -3$, $|\vec{n}_1| = \sqrt{14}$, $|\vec{n}_2| = \sqrt{6}$. $\cos\theta = \frac{3}{\sqrt{84}} = \frac{3}{2\sqrt{21}} = \frac{\sqrt{21}}{14}$, $\theta \approx 70.5^\circ$.

2. Direction $\vec{d} = \vec{n}_1 \times \vec{n}_2 = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & -1 & 3 \\ 1 & 2 & -1 \end{vmatrix} = (-5, 5, 5) \sim (-1, 1, 1)$. Point on line: set $z = 0$: $2x - y = 5$, $x + 2y = 3 \Rightarrow 5x = 13$, $x = 13/5$, $y = 1/5$. Line: $\vec{r} = \begin{pmatrix} 13/5 \\ 1/5 \\ 0 \end{pmatrix} + t\begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix}$.

3. Pencil: $(2x - y + 3z - 5) + \lambda(x + 2y - z - 3) = 0$. Normal: $(2 + \lambda, -1 + 2\lambda, 3 - \lambda)$. Perpendicular to $x - y + 2z = 4 \Rightarrow (2+\lambda)(1) + (-1+2\lambda)(-1) + (3-\lambda)(2) = 0 \Rightarrow 2 + \lambda + 1 - 2\lambda + 6 - 2\lambda = 0 \Rightarrow 9 - 3\lambda = 0 \Rightarrow \lambda = 3$. Plane: $(2x - y + 3z - 5) + 3(x + 2y - z - 3) = 0 \Rightarrow 5x + 5y - 14 = 0$ or $\boxed{x + y = \frac{14}{5}}$.

SOLUTION **Integration Exercise 2**

1. $\vec{d} = (1, 2, -2)$, $\vec{n} = (3, -1, 2)$. $\vec{d} \cdot \vec{n} = 3 - 2 - 4 = -3 \neq 0 \rightarrow$ not parallel. Parametric: $x = 2 + t$, $y = -1 + 2t$, $z = 3 - 2t$. Substitute: $3(2 + t) - (-1 + 2t) + 2(3 - 2t) = 7 \Rightarrow 6 + 3t + 1 - 2t + 6 - 4t = 7 \Rightarrow 13 - 3t = 7 \Rightarrow t = 2$. Intersection: $(4, 3, -1)$.

2. $\sin\theta = \frac{|\vec{d} \cdot \vec{n}|}{|\vec{d}||\vec{n}|} = \frac{3}{3\sqrt{14}} = \frac{1}{\sqrt{14}}$, $\theta = \arcsin(1/\sqrt{14}) \approx 15.5^\circ$.

3. Foot from $A(4, 1, 1)$ to $3x - y + 2z = 7$: Line: $x = 4 + 3t$, $y = 1 - t$, $z = 1 + 2t$. $3(4 + 3t) - (1 - t) + 2(1 + 2t) = 7 \Rightarrow 12 + 9t - 1 + t + 2 + 4t = 7 \Rightarrow 13 + 14t = 7 \Rightarrow t = -3/7$. Foot: $\left(4 - \frac{9}{7}, 1 + \frac{3}{7}, 1 - \frac{6}{7}\right) = \boxed{\left(\frac{19}{7}, \frac{10}{7}, \frac{1}{7}\right)}$.

SOLUTION **Integration Exercise 3**

1. $\vec{AB} = (2,-2,2)$, $\vec{AC} = (-2,-1,3)$. Cross product $\neq \vec{0}$ → non-collinear.

2. $\vec{n} = \vec{AB} \times \vec{AC} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & -2 & 2 \\ -2 & -1 & 3 \end{vmatrix} = (-4,-10,-6) \sim (2,5,3)$. Using A: $2(1)+5(2)+3(-1) = 2+10-3 = 9$.
   Plane: $\boxed{2x + 5y + 3z = 9}$.

3. Distance from origin: $\delta = \frac{|0-9|}{\sqrt{4+25+9}} = \frac{9}{\sqrt{38}} = \frac{9\sqrt{38}}{38}$.

4. Pencil: $(2x + 5y + 3z - 9) + \lambda(x + y + z) = 0$. Through $D(2,2,2)$: $(4 + 10 + 6 - 9) + \lambda(2 + 2 + 2) = 0 \Rightarrow 11 + 6\lambda = 0 \Rightarrow \lambda = -11/6$. Plane: $6(2x + 5y + 3z - 9) - 11(x + y + z) = 0 \Rightarrow 12x + 30y + 18z - 54 - 11x - 11y - 11z = 0 \Rightarrow \boxed{x + 19y + 7z = 54}$.

SOLUTION **Integration Exercise 4**

1. $\vec{n} = \vec{d_1} \times \vec{d_2} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & -1 & 1 \\ 1 & 2 & -1 \end{vmatrix} = (-1,3,5)$. Plane through $A(1,2,3)$: $-1(x - 1) + 3(y - 2) + 5(z - 3) = 0 \Rightarrow -x + 1 + 3y - 6 + 5z - 15 = 0 \Rightarrow \boxed{-x + 3y + 5z = 20}$ or $x - 3y - 5z = -20$.

2. Distance = distance from $B(4,-1,2)$ to plane: $\delta = \frac{|4-3(-1)-5(2)+20|}{\sqrt{1+9+25}} = \frac{|4+3-10+20|}{\sqrt{35}} = \frac{17}{\sqrt{35}} = \frac{17\sqrt{35}}{35}$.

3. $\vec{PQ}$ must be parallel to $\vec{n} = (-1,3,5)$. $P = (1,2,3) + s(2,-1,1)$, $Q = (4,-1,2) + t(1,2,-1)$. $\vec{PQ} = (3+t-2s, -3+2t+s, -1-t-s) = k(-1,3,5)$. Solve: $3+t-2s = -k$, $-3+2t+s = 3k$, $-1-t-s = 5k$. From first: $k = 2s - t - 3$. Substitute into second: $-3 + 2t + s = 6s - 3t - 9 \Rightarrow 5t - 5s = -6 \Rightarrow t - s = -6/5$. Third: $-1-t-s = 10s-5t-15 \Rightarrow 4t-11s = -14$. Solve: $t = s-6/5$, $4(s-6/5)-11s = -14 \Rightarrow 4s - 24/5 - 11s = -14 \Rightarrow -7s = -46/5 \Rightarrow s = 46/35$, $t = 46/35 - 42/35 = 4/35$. $P = \left(1 + \frac{92}{35}, 2 - \frac{46}{35}, 3 + \frac{46}{35}\right) = \left(\frac{127}{35}, \frac{24}{35}, \frac{151}{35}\right)$. $Q = \left(4 + \frac{4}{35}, -1 + \frac{8}{35}, 2 - \frac{4}{35}\right) = \left(\frac{144}{35}, -\frac{27}{35}, \frac{66}{35}\right)$. Check $\vec{PQ} = \frac{1}{35}(17,-51,-85) = \frac{17}{35}(1,-3,-5)$ parallel to $(-1,3,5)$ ✓.

DID YOU KNOW? **Historical note**

The vector approach to planes was systematised by Hermann Grassmann (1809–1877) and later refined by Josiah Willard Gibbs (1839–1903) and Oliver Heaviside (1850–1925). The "pencil of planes" concept comes from projective geometry (Poncelet, 1822). In Cameroon, vector methods are essential for engineering entrance exams (ENSP, ENSET)

and for modelling terrain in surveying (e.g. mapping the slopes of Mount Cameroon).

WARNING **Common GCE traps**

- Forgetting the absolute value in the distance formula.
- Using cosine instead of sine for line–plane angle.
- Not checking collinearity of three points before finding a plane.
- Confusing "plane through the line of intersection" (pencil, 1 parameter) with "plane through the point of intersection of three planes" (2 parameters).
- Forgetting that the pencil $\Pi_1 + \lambda\Pi_2 = 0$ misses Π_2; use symmetric form if Π_2 might be the answer.
- In skew line distance, the vector connecting the closest points is parallel to $\vec{d}_1 \times \vec{d}_2$.

KEY POINTS **Key formulas summary**

Situation	Formula / Condition
Plane through $A(\vec{a})$, normal $\vec{n}$	$\vec{r} \cdot \vec{n} = \vec{a} \cdot \vec{n}$ or $ax + by + cz = d$
Plane through A, directions $\vec{u}, \vec{v}$	$\vec{r} = \vec{a} + \lambda\vec{u} + \mu\vec{v}$
Plane through A, B, C	$\vec{n} = \vec{AB} \times \vec{AC}$, then $d = \vec{n} \cdot \vec{a}$
Distance point P to plane $ax + by + cz = d$	$\delta = \dfrac{\lvert ax_1 + by_1 + cz_1 - d \rvert}{\sqrt{a^2 + b^2 + c^2}}$
Angle between planes $\vec{n}_1, \vec{n}_2$	$\cos\theta = \dfrac{\lvert \vec{n}_1 \cdot \vec{n}_2 \rvert}{\lvert \vec{n}_1 \rvert \lvert \vec{n}_2 \rvert}$
Angle line $\vec{d}$ and plane $\vec{n}$	$\sin\theta = \dfrac{\lvert \vec{d} \cdot \vec{n} \rvert}{\lvert \vec{d} \rvert \lvert \vec{n} \rvert}$
Line–plane intersection	Substitute parametrics, solve for t
Line of intersection of two planes	$\vec{d} = \vec{n}_1 \times \vec{n}_2$, find one point
Pencil through intersection line	$\Pi_1 + \lambda\Pi_2 = 0$
Shortest distance between skew lines	Distance from point on L_2 to plane containing L_1 parallel to L_2

17 EXERCISES

17.1 I check my knowledge

EXERCISE 11 Multiple choice questions

For each question, choose the single correct answer, justifying your choice briefly.

1. The plane $3x - 2y + z = 7$ has normal vector:

   (a) $(3, -2, 1)$ (b) $(3, 2, 1)$ (c) $(7, -2, 1)$ (d) $(3, -2, 7)$

2. The perpendicular distance from the origin to the plane $2x + 6y - 3z = 14$ is:

   (a) 1 (b) 2 (c) 7 (d) 14

3. The planes $x + y + z = 1$ and $2x + 2y + 2z = 5$ are:

   (a) identical (b) parallel and distinct (c) intersecting in a line (d) perpendicular

4. If a line has direction vector $\vec{d}$ and a plane has normal vector $\vec{n}$, then the line is parallel to the plane when:

   (a) $\vec{d} = \vec{n}$ (b) $\vec{d} \cdot \vec{n} = 0$ (c) $\vec{d} \times \vec{n} = \vec{0}$ (d) $\vec{d} \cdot \vec{n} = 1$

EXERCISE 12 True or false

State whether each assertion is true or false, and justify your answer.

1. Three collinear points determine a unique plane.
2. The angle between two planes is equal to the angle between their normal vectors (or its supplement).
3. If a line is perpendicular to a plane, then its direction vector is parallel to the normal vector of the plane.
4. Every plane of the pencil through the line of intersection of Π_1 and Π_2 has an equation of the form $\Pi_1 + \lambda \Pi_2 = 0$.

EXERCISE 13 Definitions and vocabulary

1. Define the **normal vector** of a plane and explain its geometric meaning.

2. Give the **vector equation** and the **parametric equations** of a plane passing through a point A and parallel to two non-parallel vectors $\vec{u}$ and $\vec{v}$.
3. State the formula for the perpendicular distance from a point $P(x_0, y_0, z_0)$ to the plane $ax + by + cz + d = 0$.
4. What is a **pencil of planes**? Illustrate your answer with a sketch.

EXERCISE 14 **Direct calculations**

1. Write down a normal vector and a point of the plane $5x - y + 4z = 12$.
2. Find the Cartesian equation of the plane through $A(1, 0, -2)$ with normal vector $\vec{n} = 2\vec{i} - 3\vec{j} + \vec{k}$.
3. Compute $\vec{u} \times \vec{v}$ where $\vec{u} = (1, 2, -1)$ and $\vec{v} = (0, 1, 3)$, and interpret the result geometrically.
4. Determine whether the point $M(2, 1, -1)$ lies on the plane $3x - 2y + z = 3$.

17.2 I practise

EXERCISE 15 **Plane through three points**

Find the Cartesian equation of the plane through $A(1, 2, -1)$, $B(2, 0, 3)$ and $C(0, 1, 1)$. Verify first that the three points are not collinear.

EXERCISE 16 **From parametric to Cartesian form**

A plane has parametric (vector) equation

$$\vec{r} = (\vec{i} + 2\vec{j} - \vec{k}) + \lambda(\vec{i} - \vec{j}) + \mu(\vec{j} + \vec{k}).$$

Find its Cartesian equation and the coordinates of its intercepts with the three coordinate axes.

EXERCISE 17 **Distance and foot of the perpendicular**

Calculate the perpendicular distance from $P(3, -1, 2)$ to the plane $2x - y + 2z = 5$, and find the coordinates of the foot of the perpendicular from P to the plane.

EXERCISE 18 **Angles involving planes and lines**

1. Find the acute angle between the planes $x + 2y - 2z = 3$ and $2x - y + z = 7$.
2. Find the angle between the line $\vec{r} = (2\vec{i} + \vec{k}) + \lambda(\vec{i} - 2\vec{j} + 2\vec{k})$ and the plane $3x + 4z = 1$.

17.3 I prepare for the GCE

EXERCISE 19 **Line, plane and pencil** **(12 marks)**

1. Determine the point of intersection of the line $\vec{r} = (\vec{i} - \vec{j} + 2\vec{k}) + \lambda(2\vec{i} + \vec{j} - \vec{k})$ with the plane $x + y + z = 6$. (3 marks)
2. The line is now replaced by one through the same point but with direction $(\vec{i} + \vec{j} - 2\vec{k})$. Show that this new line does not meet the plane, and state the geometrical relationship between the line and the plane. (3 marks)
3. Show that the planes $2x - y + z = 4$ and $x + y - z = 1$ intersect in a line L, and find a vector equation of L. (4 marks)
4. Find the acute angle between the two planes of part (c). (2 marks)

EXERCISE 20 **Pencils of planes** **(10 marks)**

1. Find the equation of the plane passing through the line of intersection of the planes $x + y + z = 3$ and $2x - y + z = 1$ and through the point $(1, 2, 3)$. (5 marks)
2. Find the equation of the plane through the line of intersection of the planes $x + 2y - z = 0$ and $3x - y + z = 5$ which is perpendicular to the plane $x + y + z = 2$. (5 marks)

EXERCISE 21 **GCE past-paper style: study of two planes** **(16 marks)**

The planes $\Pi_1 : 2x - y + 2z = 9$ and $\Pi_2 : x + 2y - z = 4$ are given.

1. Find the acute angle between Π_1 and Π_2, giving your answer to the nearest degree. (4 marks)
2. Find a vector equation of the line of intersection of Π_1 and Π_2. (5 marks)
3. Find the perpendicular distance from the origin to Π_1, and the coordinates of the foot

of the perpendicular from the origin to Π_1. (4 marks)

4. Find the Cartesian equation of the plane through the line of intersection of Π_1 and Π_2 that passes through the point $(0, 1, -1)$. (3 marks)

18 SOLUTIONS TO THE EXERCISES

18.1 I check my knowledge

SOLUTION **Exercise 1 — Multiple choice questions**

1. Answer (a). The Cartesian equation $ax+by+cz = d$ has normal vector (a, b, c): the normal vector is read directly from the coefficients of x, y and z. Here these coefficients are $3, -2, 1$, so $\vec{n} = (3, -2, 1)$. The constant term 7 never appears in the normal vector, which eliminates (c) and (d); the sign of the y-coefficient eliminates (b).

2. Answer (b). Using the perpendicular distance formula with $P = (0, 0, 0)$ and the plane $2x + 6y - 3z - 14 = 0$:

$$d = \frac{|2(0) + 6(0) - 3(0) - 14|}{\sqrt{2^2 + 6^2 + (-3)^2}} = \frac{14}{\sqrt{4 + 36 + 9}} = \frac{14}{\sqrt{49}} = \frac{14}{7} = 2.$$

   Trap avoided: do not forget the absolute value in the numerator, and remember that the denominator is the modulus of the normal vector, $\|\vec{n}\| = \sqrt{a^2 + b^2 + c^2}$, not $a + b + c$.

3. Answer (b). The normal vectors are $\vec{n}_1 = (1, 1, 1)$ and $\vec{n}_2 = (2, 2, 2) = 2\vec{n}_1$: the normals are parallel, so the planes are parallel. They are not identical, since dividing the second equation by 2 gives $x + y + z = \frac{5}{2} \neq 1$. Hence the planes are **parallel and distinct** (they have no common point).

4. Answer (b). A line is parallel to a plane when its direction vector is perpendicular to the plane's normal vector, i.e. $\vec{d} \cdot \vec{n} = 0$. (Option (c), $\vec{d} \times \vec{n} = \vec{0}$, would mean $\vec{d} \parallel \vec{n}$, i.e. the line perpendicular to the plane.) Note that $\vec{d} \cdot \vec{n} = 0$ covers two cases: the line is strictly parallel to the plane, or it lies entirely in the plane; to distinguish them, test whether one point of the line satisfies the equation of the plane.

SOLUTION **Exercise 2 — True or false**

1. False. If A, B, C are collinear, then $\vec{AB}$ and $\vec{AC}$ are parallel, so $\vec{AB} \times \vec{AC} = \vec{0}$ and no unique normal vector exists: infinitely many planes contain the line (ABC) (they form the pencil of planes about that line). Three points determine a unique plane only when they are non-collinear.

2. True. The angle between two planes is defined as the acute angle between their normal vectors. Since $\vec{n}_1 \cdot \vec{n}_2$ may be negative (giving an obtuse angle between the chosen

normals), we take the absolute value:

$$\cos\theta = \frac{|\vec{n}_1 \cdot \vec{n}_2|}{\|\vec{n}_1\|\,\|\vec{n}_2\|}, \qquad 0 \le \theta \le 90^\circ,$$

which corresponds to the angle between the normals or its supplement, whichever is acute.

3. True. A line perpendicular to a plane is parallel to every normal direction of that plane; hence $\vec{d} = \lambda\vec{n}$ for some $\lambda \neq 0$ (equivalently $\vec{d} \times \vec{n} = \vec{0}$). This is the standard way to write the equation of the perpendicular from a point to a plane.

4. True, with one caveat. Every plane through the line of intersection of $\Pi_1 = 0$ and $\Pi_2 = 0$ can be written $\Pi_1 + \lambda\Pi_2 = 0$ for some $\lambda \in \mathbb{R}$, except the plane Π_2 itself (which would formally require "$\lambda = \infty$"). In examinations, check separately whether Π_2 is a solution before concluding; if the required plane turns out to be Π_2, the equation in λ will be inconsistent, which is the signal.

SOLUTION **Exercise 3 — Definitions and vocabulary**

1. A **normal vector** of a plane Π is any non-zero vector $\vec{n}$ perpendicular to every vector lying in Π (equivalently, perpendicular to two non-parallel direction vectors of Π). Geometrically, $\vec{n}$ gives the direction of the line perpendicular to Π; it is unique up to a non-zero scalar multiple, so (a, b, c) and $(2a, 2b, 2c)$ describe the same plane orientation.

2. Let A have position vector $\vec{a}$, and let $\vec{u}$, $\vec{v}$ be two non-parallel direction vectors of the plane. The **vector equation** is

$$\vec{r} = \vec{a} + \lambda\vec{u} + \mu\vec{v}, \qquad \lambda, \mu \in \mathbb{R}.$$

If $A = (x_0, y_0, z_0)$, $\vec{u} = (u_1, u_2, u_3)$, $\vec{v} = (v_1, v_2, v_3)$, the **parametric equations** are

$$\begin{cases} x = x_0 + \lambda u_1 + \mu v_1, \\ y = y_0 + \lambda u_2 + \mu v_2, \\ z = z_0 + \lambda u_3 + \mu v_3. \end{cases}$$

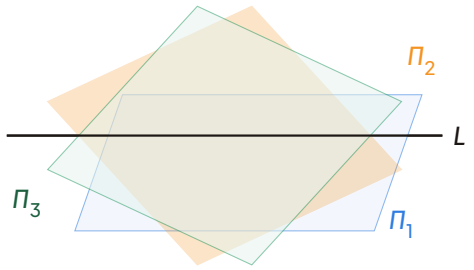
Each choice of the pair (λ, μ) gives exactly one point of the plane.

3. The perpendicular distance from $P(x_0, y_0, z_0)$ to the plane $ax + by + cz + d = 0$ is

$$d = \frac{|ax_0 + by_0 + cz_0 + d|}{\sqrt{a^2 + b^2 + c^2}}.$$

Warning: the equation must first be written with all terms on one side; for the plane $ax + by + cz = k$ the numerator is $|ax_0 + by_0 + cz_0 - k|$.

4. A **pencil of planes** is the family of all planes passing through a fixed line L (the axis of the pencil). If L is the intersection of $\Pi_1 = 0$ and $\Pi_2 = 0$, the pencil is $\Pi_1 + \lambda\Pi_2 = 0, \lambda \in \mathbb{R}$ (together with Π_2 itself).



Planes of a pencil all sharing the same axis L: each value of λ gives one plane through L.

SOLUTION **Exercise 4 — Direct calculations**

1. Normal vector: $\vec{n} = (5, -1, 4)$, read from the coefficients of x, y, z. A point: setting $x = y = 0$ gives $4z = 12$, so $(0, 0, 3)$ lies on the plane. (Any convenient choice of two coordinates works; choose values that make the remaining equation easy to solve.)

2. The plane through $A(1, 0, -2)$ with normal $\vec{n} = (2, -3, 1)$ consists of the points $M(x, y, z)$ such that $\vec{n} \cdot \vec{AM} = 0$:

$$2(x - 1) - 3(y - 0) + 1(z + 2) = 0 \implies 2x - 3y + z = 0.$$

$$\boxed{2x - 3y + z = 0}$$

Check: A satisfies $2(1) - 3(0) + (-2) = 0$ ✓.

3. With $\vec{u} = (1, 2, -1)$ and $\vec{v} = (0, 1, 3)$:

$$\vec{u} \times \vec{v} = \begin{pmatrix} (2)(3) - (-1)(1) \\ (-1)(0) - (1)(3) \\ (1)(1) - (2)(0) \end{pmatrix} = \begin{pmatrix} 7 \\ -3 \\ 1 \end{pmatrix}.$$

Interpretation: $(7, -3, 1)$ is perpendicular to both $\vec{u}$ and $\vec{v}$ (verify: $7 - 2 - 3 = 0$ and $0 - 3 + 3 = 0$ ✓); it is a normal vector of any plane parallel to $\vec{u}$ and $\vec{v}$, and its magnitude $\sqrt{7^2 + 3^2 + 1^2} = \sqrt{59}$ is the area of the parallelogram spanned by $\vec{u}$ and $\vec{v}$.

4. Substituting $M(2, 1, -1)$ into the left-hand side of $3x - 2y + z = 3$:

$$3(2) - 2(1) + (-1) = 6 - 2 - 1 = 3.$$

The equation is satisfied, so M lies on the plane.

18.2 I practise

SOLUTION **Exercise 5 — Plane through three points**

Step 1: check non-collinearity.

$$\vec{AB} = (1, -2, 4), \qquad \vec{AC} = (-1, -1, 2).$$

These vectors are not proportional (e.g. $\frac{1}{-1} \neq \frac{-2}{-1}$), so A, B, C are not collinear and determine a unique plane. This check is essential: with collinear points the cross product in Step 2 would give $\vec{0}$.
Step 2: normal vector.

$$\vec{n} = \vec{AB} \times \vec{AC} = \begin{pmatrix} (-2)(2) - (4)(-1) \\ (4)(-1) - (1)(2) \\ (1)(-1) - (-2)(-1) \end{pmatrix} = \begin{pmatrix} -4 + 4 \\ -4 - 2 \\ -1 - 2 \end{pmatrix} = \begin{pmatrix} 0 \\ -6 \\ -3 \end{pmatrix} = -3 \begin{pmatrix} 0 \\ 2 \\ 1 \end{pmatrix}.$$

Take the simpler normal $\vec{n} = (0, 2, 1)$ (any non-zero multiple is acceptable).
Step 3: equation. Using point $A(1, 2, -1)$ and $\vec{n} \cdot \vec{AM} = 0$:

$$0(x - 1) + 2(y - 2) + 1(z + 1) = 0 \implies \boxed{2y + z = 3}.$$

Verification: B: $2(0) + 3 = 3$ ✓; C: $2(1) + 1 = 3$ ✓. Note that the equation contains no x-term: the plane is parallel to the x-axis, which is consistent with the normal vector having a zero first component.

SOLUTION **Exercise 6 — From parametric to Cartesian form**

The plane passes through $A(1, 2, -1)$ with direction vectors $\vec{u} = (1, -1, 0)$ and $\vec{v} = (0, 1, 1)$.
Normal vector:

$$\vec{n} = \vec{u} \times \vec{v} = \begin{pmatrix} (-1)(1) - (0)(1) \\ (0)(0) - (1)(1) \\ (1)(1) - (-1)(0) \end{pmatrix} = \begin{pmatrix} -1 \\ -1 \\ 1 \end{pmatrix}.$$

Cartesian equation: for $M(x, y, z)$ on the plane, $\vec{n} \cdot \vec{AM} = 0$:

$$-1(x - 1) - 1(y - 2) + 1(z + 1) = 0 \implies -x - y + z + 4 = 0 \implies \boxed{x + y - z = 4}.$$

Check with the parametric form: substituting $x = 1 + \lambda$, $y = 2 - \lambda + \mu$, $z = -1 + \mu$ gives $(1 + \lambda) + (2 - \lambda + \mu) - (-1 + \mu) = 4$ for all λ, μ ✓.
Intercepts with the axes:

- x-axis ($y = z = 0$): $x = 4$, giving $(4, 0, 0)$;
- y-axis ($x = z = 0$): $y = 4$, giving $(0, 4, 0)$;

- z-axis ($x = y = 0$): $-z = 4$, giving $(0, 0, -4)$.

These three intercepts are a convenient way to sketch the plane: it is the plane through the triangle with vertices $(4, 0, 0)$, $(0, 4, 0)$, $(0, 0, -4)$.

SOLUTION **Exercise 7 — Distance and foot of the perpendicular**

Plane: $2x - y + 2z - 5 = 0$, normal $\vec{n} = (2, -1, 2)$, $\|\vec{n}\| = \sqrt{4 + 1 + 4} = 3$.
Distance:

$$d = \frac{|2(3) - (-1) + 2(2) - 5|}{3} = \frac{|6 + 1 + 4 - 5|}{3} = \frac{6}{3} = \boxed{2}.$$

Foot of the perpendicular. The perpendicular through $P(3, -1, 2)$ is parallel to $\vec{n}$, so it has equation

$$\vec{r} = (3, -1, 2) + t(2, -1, 2), \qquad t \in \mathbb{R}.$$

Substituting $x = 3 + 2t$, $y = -1 - t$, $z = 2 + 2t$ into the plane:

$$2(3 + 2t) - (-1 - t) + 2(2 + 2t) = 5 \implies 6 + 4t + 1 + t + 4 + 4t = 5 \implies 9t = -6 \implies t = -\tfrac{2}{3}.$$

Hence the foot is

$$H = \left(3 - \tfrac{4}{3},\ -1 + \tfrac{2}{3},\ 2 - \tfrac{4}{3}\right) = \boxed{\left(\tfrac{5}{3},\ -\tfrac{1}{3},\ \tfrac{2}{3}\right)}.$$

Check: $2 \cdot \frac{5}{3} + \frac{1}{3} + 2 \cdot \frac{2}{3} = \frac{10+1+4}{3} = 5$ ✓, and $|PH| = |t|\,\|\vec{n}\| = \frac{2}{3} \times 3 = 2$ ✓, which agrees with the distance found above. This double check is quick and strongly recommended in the examination.

SOLUTION **Exercise 8 — Angles involving planes and lines**

1. Normals: $\vec{n}_1 = (1, 2, -2)$, $\|\vec{n}_1\| = \sqrt{1 + 4 + 4} = 3$; $\vec{n}_2 = (2, -1, 1)$, $\|\vec{n}_2\| = \sqrt{4 + 1 + 1} = \sqrt{6}$.

$$\vec{n}_1 \cdot \vec{n}_2 = 2 - 2 - 2 = -2, \qquad \cos\theta = \frac{|-2|}{3\sqrt{6}} = \frac{2}{3\sqrt{6}} \approx 0.2722.$$

$$\boxed{\theta \approx 74.2^\circ}.$$

   The absolute value is essential here: without it, $\cos\theta < 0$ would give the obtuse angle 105.8° between the chosen normals instead of the acute angle between the planes.

2. Line direction $\vec{d} = (1, -2, 2)$, $\|\vec{d}\| = 3$; plane normal $\vec{n} = (3, 0, 4)$, $\|\vec{n}\| = 5$. The angle θ between a line and a plane is the complement of the angle between $\vec{d}$ and $\vec{n}$, so it satisfies $\sin\theta = \dfrac{|\vec{d} \cdot \vec{n}|}{\|\vec{d}\|\,\|\vec{n}\|}$:

$$\vec{d} \cdot \vec{n} = 3 + 0 + 8 = 11, \qquad \sin\theta = \frac{11}{15} \approx 0.7333 \implies \boxed{\theta \approx 47.2^\circ}.$$

(Recall: use sin here because $\theta = 90^\circ - \varphi$, where φ is the angle between $\vec{d}$ and $\vec{n}$, and $\cos\varphi = \sin\theta$.)

18.3 I prepare for the GCE

SOLUTION **Exercise 9 — Line, plane and pencil (12 marks)**

1. Parametrize the line: $x = 1 + 2\lambda$, $y = -1 + \lambda$, $z = 2 - \lambda$. Substitute into $x + y + z = 6$:

$$(1 + 2\lambda) + (-1 + \lambda) + (2 - \lambda) = 6 \implies 2 + 2\lambda = 6 \implies \lambda = 2.$$

Point of intersection: $\boxed{(5, 1, 0)}$. [M1 substitution, A1 value of λ, A1 point]

2. New direction $\vec{d} = (1, 1, -2)$, plane normal $\vec{n} = (1, 1, 1)$:

$$\vec{d} \cdot \vec{n} = 1 + 1 - 2 = 0,$$

so the line is parallel to the plane. Moreover the point $(1, -1, 2)$ of the line gives $1 - 1 + 2 = 2 \neq 6$, so this point is not on the plane. Hence the line does not meet the plane: it is **strictly parallel** to it. [M1 dot product, A1 conclusion of parallelism, A1 point not on plane]

3. Normals $\vec{n}_1 = (2, -1, 1)$ and $\vec{n}_2 = (1, 1, -1)$ are not proportional (e.g. $\frac{2}{1} \neq \frac{-1}{1}$), so the planes intersect in a line L. The direction of L is perpendicular to both normals:

$$\vec{n}_1 \times \vec{n}_2 = \begin{pmatrix} (-1)(-1) - (1)(1) \\ (1)(1) - (2)(-1) \\ (2)(1) - (-1)(1) \end{pmatrix} = \begin{pmatrix} 1 - 1 \\ 1 + 2 \\ 2 + 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 3 \\ 3 \end{pmatrix} = 3\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}.$$

A point of L: set $z = 0$; then $2x - y = 4$ and $x + y = 1$. Adding gives $3x = 5$, so $x = \frac{5}{3}$, $y = 1 - \frac{5}{3} = -\frac{2}{3}$. Hence

$$\boxed{\vec{r} = \begin{pmatrix} 5/3 \\ -2/3 \\ 0 \end{pmatrix} + \lambda\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \quad \lambda \in \mathbb{R}.}$$

Check: the point satisfies $2(\frac{5}{3}) - (-\frac{2}{3}) + 0 = 4$ ✓ and $\frac{5}{3} + (-\frac{2}{3}) - 0 = 1$ ✓. [B1 normals not parallel, M1 cross product, A1 a common point, A1 equation]

4. $\vec{n}_1 \cdot \vec{n}_2 = 2 - 1 - 1 = 0$: the normals are perpendicular, so the planes are perpendicular:

$$\boxed{\theta = 90^\circ}.$$

[M1 dot product of normals, A1 answer]

Examiner's expectations: clear substitution in (a); in (b) both conditions ($\vec{d}\cdot\vec{n}=0$ and the point not on the plane) are required for full marks — a line with $\vec{d}\cdot\vec{n}=0$ could also lie in the plane; in (c), if setting $z=0$ had led to an inconsistent system, one would try $x=0$ or $y=0$ instead (the line of intersection may be parallel to the plane $z=0$).

SOLUTION **Exercise 10 — Pencils of planes (10 marks)**

1. The pencil through the line of intersection of $x+y+z=3$ and $2x-y+z=1$ is

$$(x+y+z-3)+\lambda(2x-y+z-1)=0.$$

Imposing passage through $(1,2,3)$:

$$(1+2+3-3)+\lambda(2-2+3-1)=0 \implies 3+2\lambda=0 \implies \lambda=-\tfrac{3}{2}.$$

Substituting and multiplying by 2 to clear the fraction:

$$2(x+y+z-3)-3(2x-y+z-1)=0 \implies -4x+5y-z-3=0,$$

$$\boxed{4x-5y+z+3=0}.$$

Check: $4(1)-5(2)+3+3=0$ ✓. [M1 pencil, M1 substitution, A1 λ, M1 simplification, A1 equation]

2. The pencil through the line of intersection of $x+2y-z=0$ and $3x-y+z=5$ is

$$(x+2y-z)+\lambda(3x-y+z-5)=0,$$

with normal vector $\vec{n}=(1+3\lambda,\ 2-\lambda,\ -1+\lambda)$. Perpendicularity to the plane $x+y+z=2$ (normal $(1,1,1)$) requires $\vec{n}\cdot(1,1,1)=0$:

$$(1+3\lambda)+(2-\lambda)+(-1+\lambda)=0 \implies 2+3\lambda=0 \implies \lambda=-\tfrac{2}{3}.$$

Substituting and multiplying by 3:

$$3(x+2y-z)-2(3x-y+z-5)=0 \implies -3x+8y-5z+10=0,$$

$$\boxed{3x-8y+5z=10}.$$

Check: normals $(3,-8,5)\cdot(1,1,1)=3-8+5=0$ ✓. [M1 pencil, M1 normal in terms of λ, M1 perpendicularity condition, A1 λ, A1 equation]

Key idea: in a pencil problem, the extra condition (passage through a point, perpendicularity or parallelism to a given plane or line) is always translated into one equation in the single unknown λ.

SOLUTION **Exercise 11 — GCE past-paper style: study of two planes (16 marks)**

Given $\Pi_1 : 2x - y + 2z = 9$ with $\vec{n}_1 = (2, -1, 2)$, $\|\vec{n}_1\| = \sqrt{4+1+4} = 3$, and $\Pi_2 : x + 2y - z = 4$ with $\vec{n}_2 = (1, 2, -1)$, $\|\vec{n}_2\| = \sqrt{1+4+1} = \sqrt{6}$.

1. Angle between the planes.

$$\vec{n}_1 \cdot \vec{n}_2 = 2 - 2 - 2 = -2, \qquad \cos\theta = \frac{|-2|}{3\sqrt{6}} = \frac{2}{3\sqrt{6}} \approx 0.2722.$$

$$\boxed{\theta \approx 74^\circ \text{ (nearest degree)}}.$$

[B1 both normals and moduli, M1 correct formula with absolute value, A1 $\cos\theta$, A1 rounded answer]

2. Line of intersection. The normals are not proportional, so the planes meet in a line L whose direction is perpendicular to both normals:

$$\vec{n}_1 \times \vec{n}_2 = \begin{pmatrix} (-1)(-1) - (2)(2) \\ (2)(1) - (2)(-1) \\ (2)(2) - (-1)(1) \end{pmatrix} = \begin{pmatrix} 1 - 4 \\ 2 + 2 \\ 4 + 1 \end{pmatrix} = \begin{pmatrix} -3 \\ 4 \\ 5 \end{pmatrix}.$$

A common point: set $z = 0$; then $2x - y = 9$ and $x + 2y = 4$. From the first equation, $y = 2x - 9$; substituting into the second: $x + 2(2x - 9) = 4$, so $5x = 22$, $x = \frac{22}{5}$, $y = \frac{44}{5} - 9 = -\frac{1}{5}$.

$$\boxed{\vec{r} = \begin{pmatrix} 22/5 \\ -1/5 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} -3 \\ 4 \\ 5 \end{pmatrix}, \quad \lambda \in \mathbb{R}.}$$

Check: $2(\frac{22}{5}) - (-\frac{1}{5}) = 9$ ✓ and $\frac{22}{5} + 2(-\frac{1}{5}) = 4$ ✓. [M1 cross product, A1 direction, M1 fixing a coordinate, A1 point, A1 full equation]

3. Distance from the origin to Π_1:

$$d = \frac{|2(0) - (0) + 2(0) - 9|}{3} = \frac{9}{3} = \boxed{3}.$$

Foot of the perpendicular: the perpendicular through O has direction $\vec{n}_1$, so $\vec{r} = t(2, -1, 2)$. Substituting into Π_1: $2(2t) - (-t) + 2(2t) = 9$, so $9t = 9$, $t = 1$. Hence

$$\boxed{H = (2, -1, 2)}.$$

Check: $|OH| = \sqrt{4+1+4} = 3$ ✓, equal to the distance found above. [M1 distance formula, A1 value, M1 parametric perpendicular, A1 foot]

4. Plane of the pencil through $(0, 1, -1)$. Pencil:

$$(2x - y + 2z - 9) + \lambda(x + 2y - z - 4) = 0.$$

At $(0, 1, -1)$:

$$(0 - 1 - 2 - 9) + \lambda(0 + 2 + 1 - 4) = 0 \implies -12 - \lambda = 0 \implies \lambda = -12.$$

Therefore

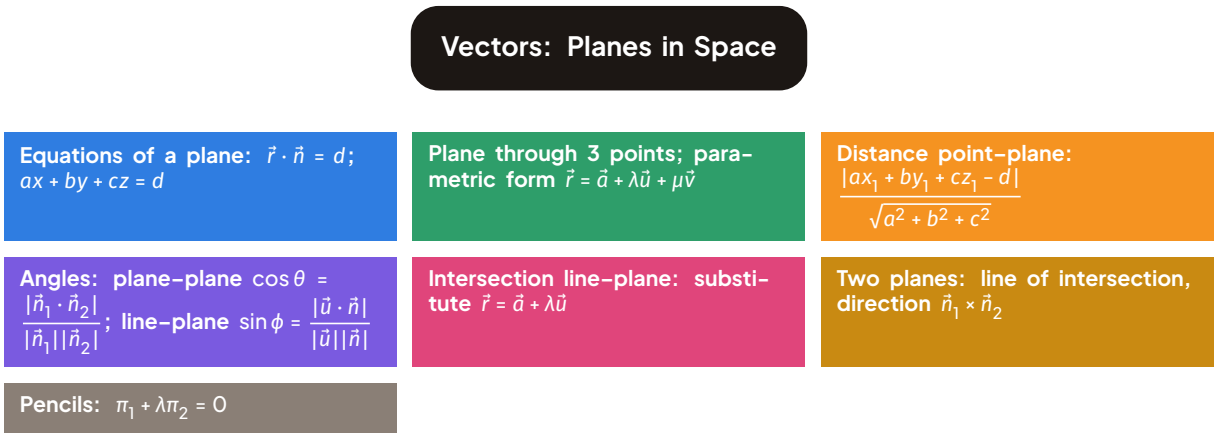
$$(2x - y + 2z - 9) - 12(x + 2y - z - 4) = 0 \implies -10x - 25y + 14z + 39 = 0,$$

$$\boxed{10x + 25y - 14z = 39}.$$

Check: $10(0) + 25(1) - 14(-1) = 25 + 14 = 39$ ✓. **[M1 pencil form, A1 $\lambda = -12$, A1 simplified equation]**

Examiner's expectations: method marks are awarded for correct processes (cross product, pencil, distance formula) even if an arithmetic slip occurs; always verify the final plane against the given point, give the angle to the accuracy demanded, and state the range $\lambda \in \mathbb{R}$ when writing a vector equation of a line.

19 SUMMARY SHEET



Mind map of the chapter: Planes in space.

FICHE BILAN **L'essentiel à connaître pour l'examen**

Key definitions

- A **plane** is determined by: one point and a **normal vector** $\vec{n}$; or one point and two non-parallel direction vectors; or three non-collinear points.
- **Vector equation**: $(\vec{r}-\vec{a})\cdot\vec{n}=0$, i.e. $\vec{r}\cdot\vec{n}=\vec{a}\cdot\vec{n}=d$.
- **Cartesian equation**: $ax+by+cz=d$, where $\vec{n}=\begin{pmatrix}a\\b\\c\end{pmatrix}$ is normal to the plane.
- **Parametric form**: $\vec{r}=\vec{a}+\lambda\vec{u}+\mu\vec{v}$, $\lambda,\mu\in\mathbb{R}$, with $\vec{u},\vec{v}$ non-parallel directions in the plane; equivalently $x=x_0+\lambda u_1+\mu v_1$, etc.
- A **pencil of planes**: the family $\pi_1+\lambda\pi_2=0$ of all planes through the line of intersection of π_1 and π_2.

Formulas and theorems to know

- Normal from two directions: $\vec{n}=\vec{u}\times\vec{v}$ (cross product).
- Plane through A, B, C: $\vec{n}=\vec{AB}\times\vec{AC}$; check first that A, B, C are not collinear.
- **Perpendicular distance** from $P(x_1,y_1,z_1)$ to $ax+by+cz=d$:

$$\delta=\frac{|ax_1+by_1+cz_1-d|}{\sqrt{a^2+b^2+c^2}}.$$

- **Angle between two planes** = angle between their normals:

$$\cos\theta = \frac{|\vec{n}_1 \cdot \vec{n}_2|}{|\vec{n}_1|\,|\vec{n}_2|}.$$

  Planes are parallel iff $\vec{n}_1 = k\vec{n}_2$; perpendicular iff $\vec{n}_1 \cdot \vec{n}_2 = 0$.

- **Angle between a line** (direction $\vec{u}$) **and a plane** (normal $\vec{n}$):

$$\sin\phi = \frac{|\vec{u} \cdot \vec{n}|}{|\vec{u}|\,|\vec{n}|}, \qquad \phi = 90^\circ - \theta.$$

- Line meets plane: substitute $\vec{r} = \vec{a} + \lambda\vec{u}$ into $ax + by + cz = d$, solve for λ. If $\vec{u} \cdot \vec{n} = 0$: line is parallel to the plane (no intersection, or the line lies in the plane).
- Line of intersection of two planes: direction $\vec{u} = \vec{n}_1 \times \vec{n}_2$; find one point by setting e.g. $z = 0$ and solving the two equations.
- Plane through the line of intersection of $\pi_1 : a_1x+b_1y+c_1z = d_1$ and $\pi_2 : a_2x+b_2y+c_2z = d_2$:

$$(a_1x + b_1y + c_1z - d_1) + \lambda(a_2x + b_2y + c_2z - d_2) = 0,$$

  then determine λ from an extra condition (a given point, perpendicularity, etc.).

Methods

- To find a plane: identify a point on it and a normal vector (often via a cross product), then expand $\vec{r} \cdot \vec{n} = \vec{a} \cdot \vec{n}$.
- To convert parametric to Cartesian: eliminate λ and μ, or compute $\vec{n} = \vec{u} \times \vec{v}$.
- For distance problems, write the plane in the form $ax + by + cz - d = 0$ before applying the formula.
- For pencil problems, substitute the given point into $\pi_1 + \lambda\pi_2 = 0$ to get a linear equation in λ.

Common mistakes

- Forgetting the absolute value in the distance formula, or forgetting to move d to the left-hand side.
- Using cos instead of sin for the line–plane angle: the angle uses the complement of the angle with the normal.
- Applying the "plane through three points" method when the points are collinear: infinitely many planes then contain them.
- Sign errors in the cross product: compute each component carefully, e.g. $(u_2v_3 - u_3v_2,\ u_3v_1 - u_1v_3,\ u_1v_2 - u_2v_1)$.
- Concluding "no intersection" when $\vec{u}\cdot\vec{n} = 0$ without checking whether the line actually lies in the plane.

GCE tips

- State each formula before substituting: method marks are awarded even if arithmetic slips.
- Keep vectors in column form and show the dot or cross product explicitly.
- Give angles in degrees to 1 d.p. unless told otherwise, and distances to 3 s.f.
- Always verify: does your point satisfy the plane equation? Does your direction vector satisfy $\vec{u} \cdot \vec{n} = 0$ for a line claimed to lie in a plane?
- In pencil questions, remember $\pi_1 + \lambda\pi_2 = 0$ never includes π_2 itself; check separately if π_2 is the answer.

Signed • OnBuch+
ONBUCH+ TEACHING TEAM
FOLLOWS THE OFFICIAL MINESEC SYLLABUS

OnBuch+